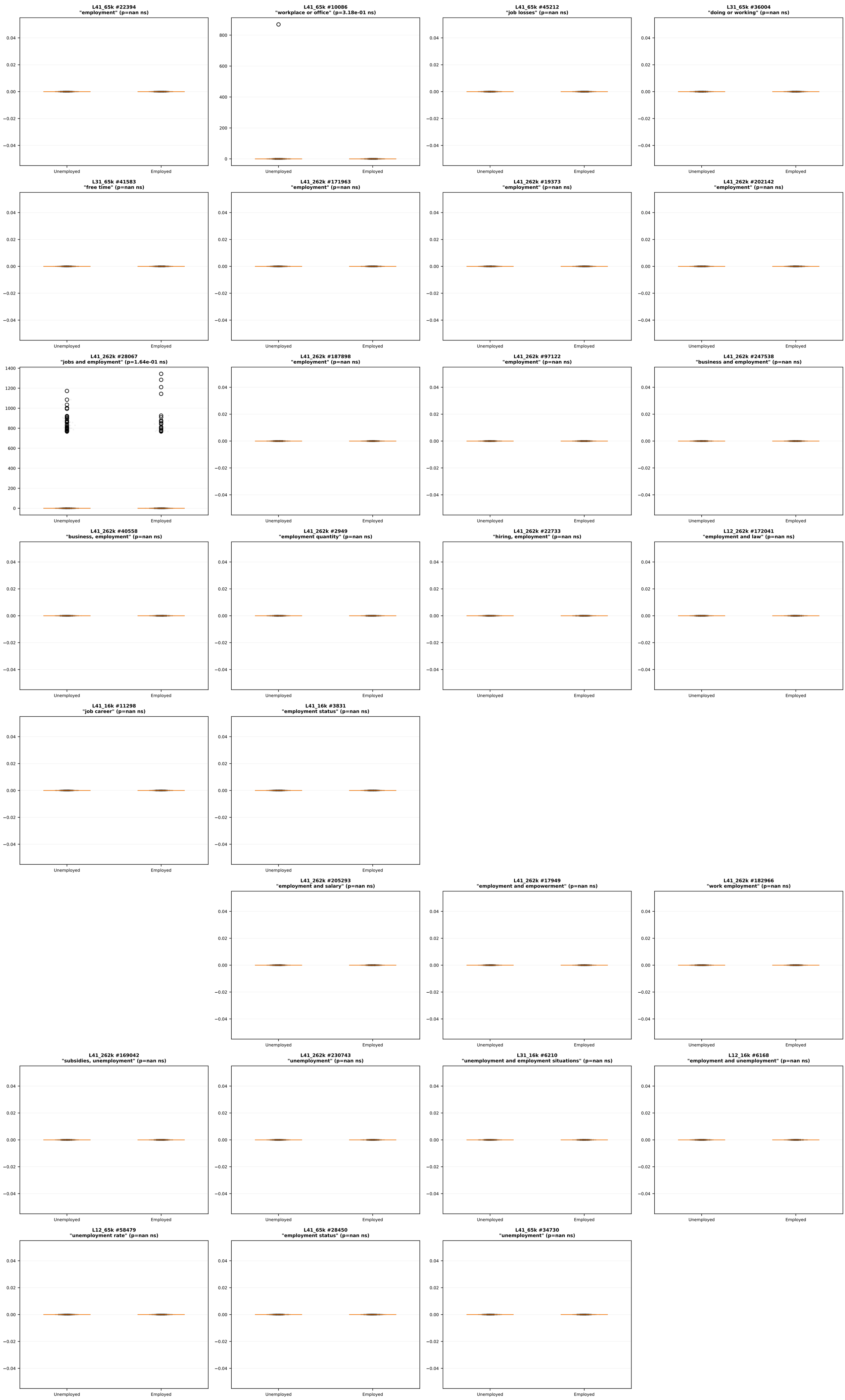
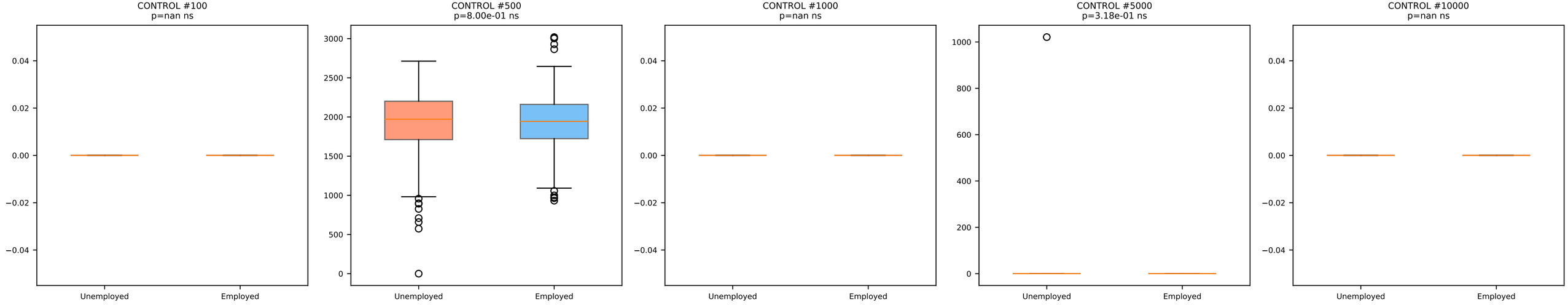
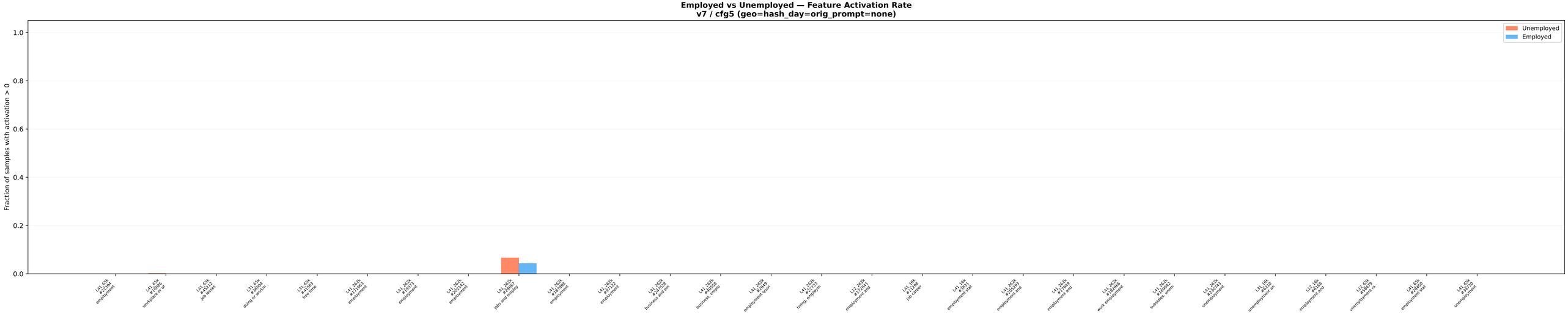
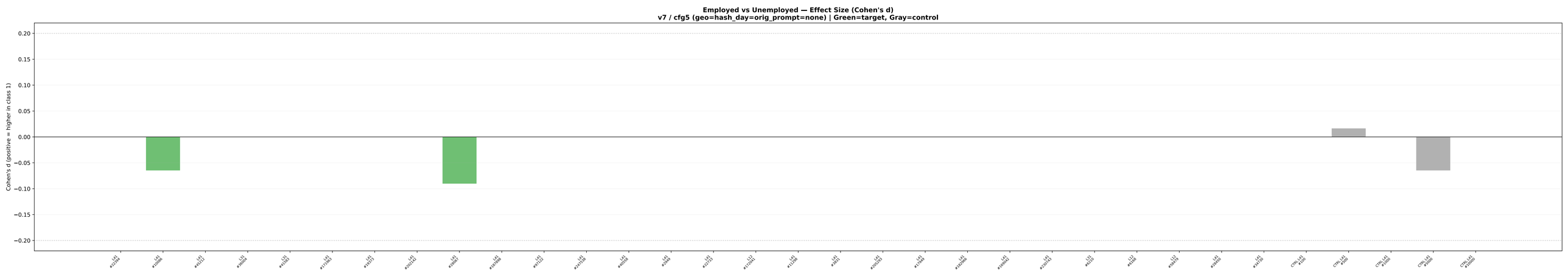
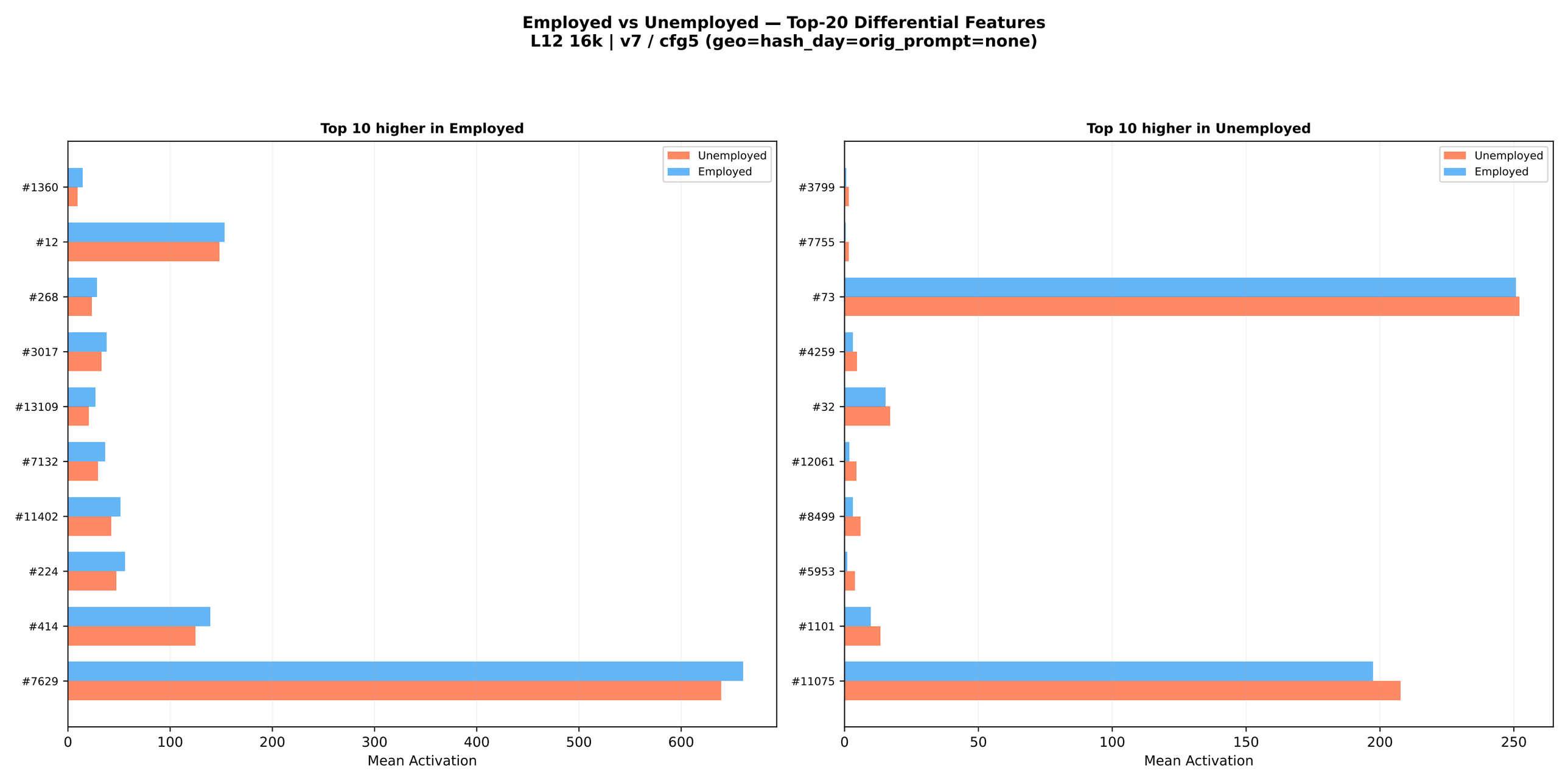
Employed vs Unemployed — SAE Feature Activations
v7 / cfg5 (geo=hash_day=orig_prompt=none) | Target features (NeuronPedia)

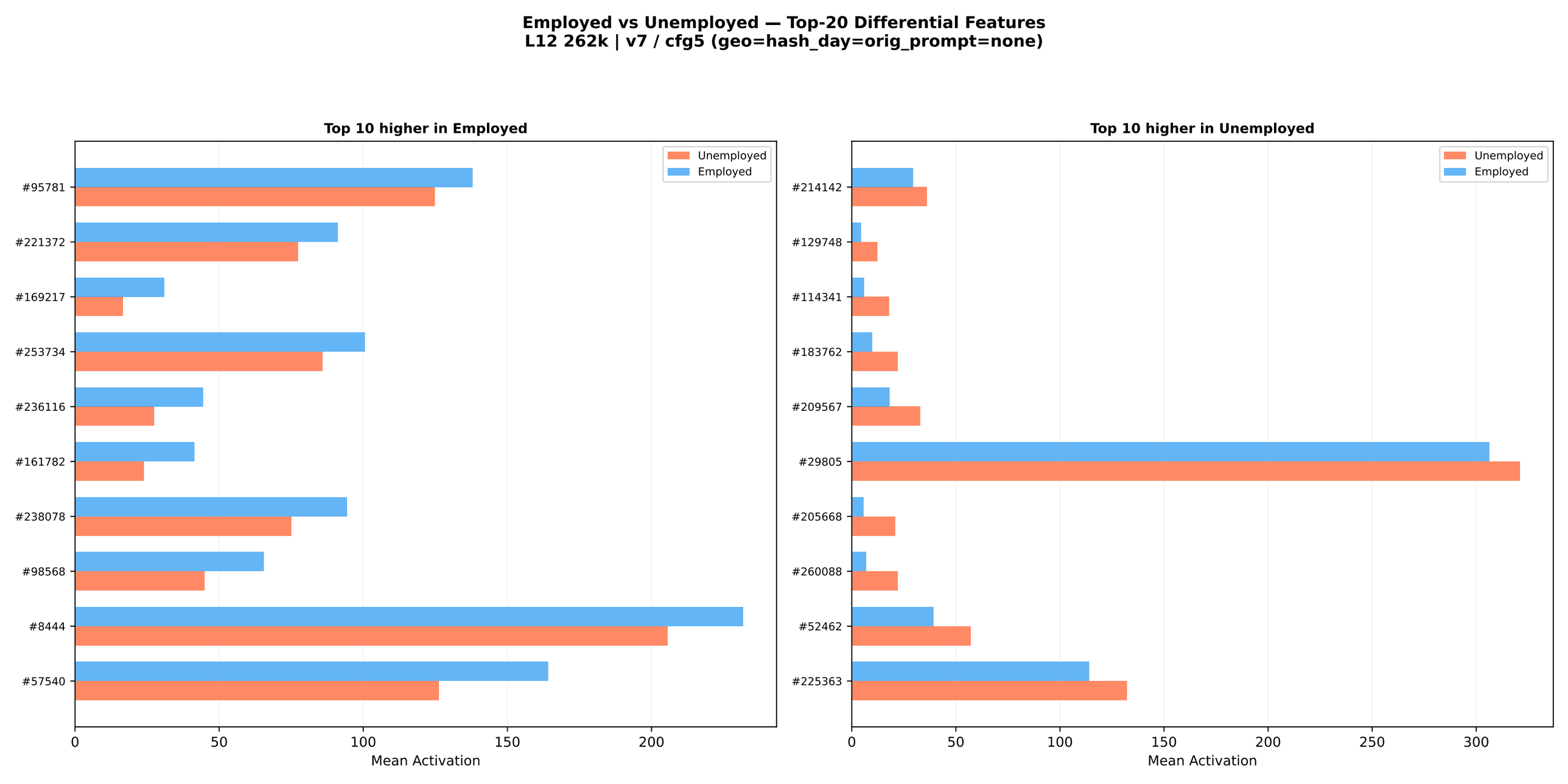


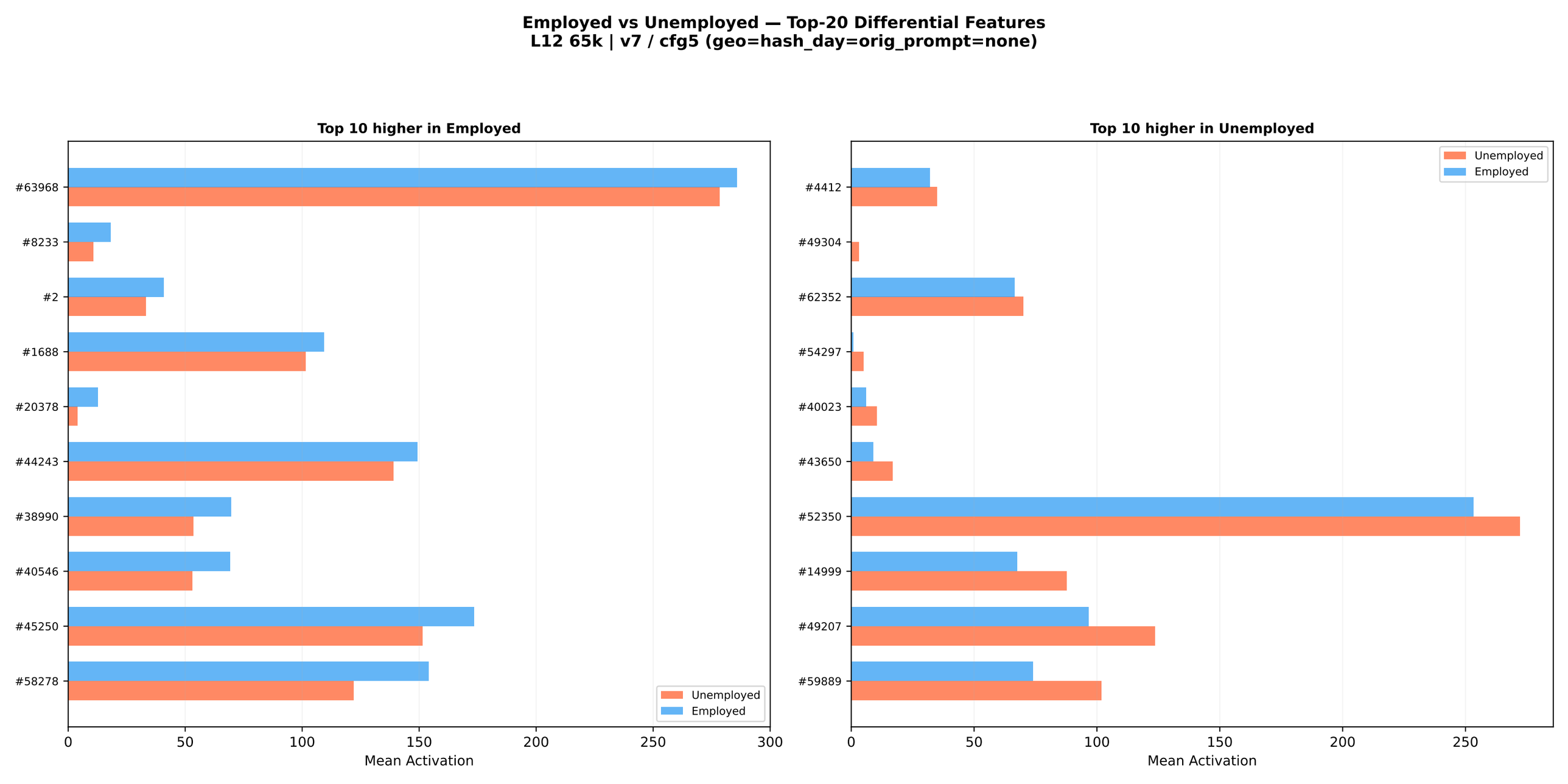

Employed vs Unemployed — Control Features (should be ns)
v7 / cfg5 (geo=hash_day=orig_prompt=none)
CONTROL #100
p=nan ns
CONTROL #500
p=8.00e-01 ns
CONTROL #1000
p=nan ns
CONTROL #5000
p=3.18e-01 ns
CONTROL #10000
p=nan ns
Unemployed
Employed

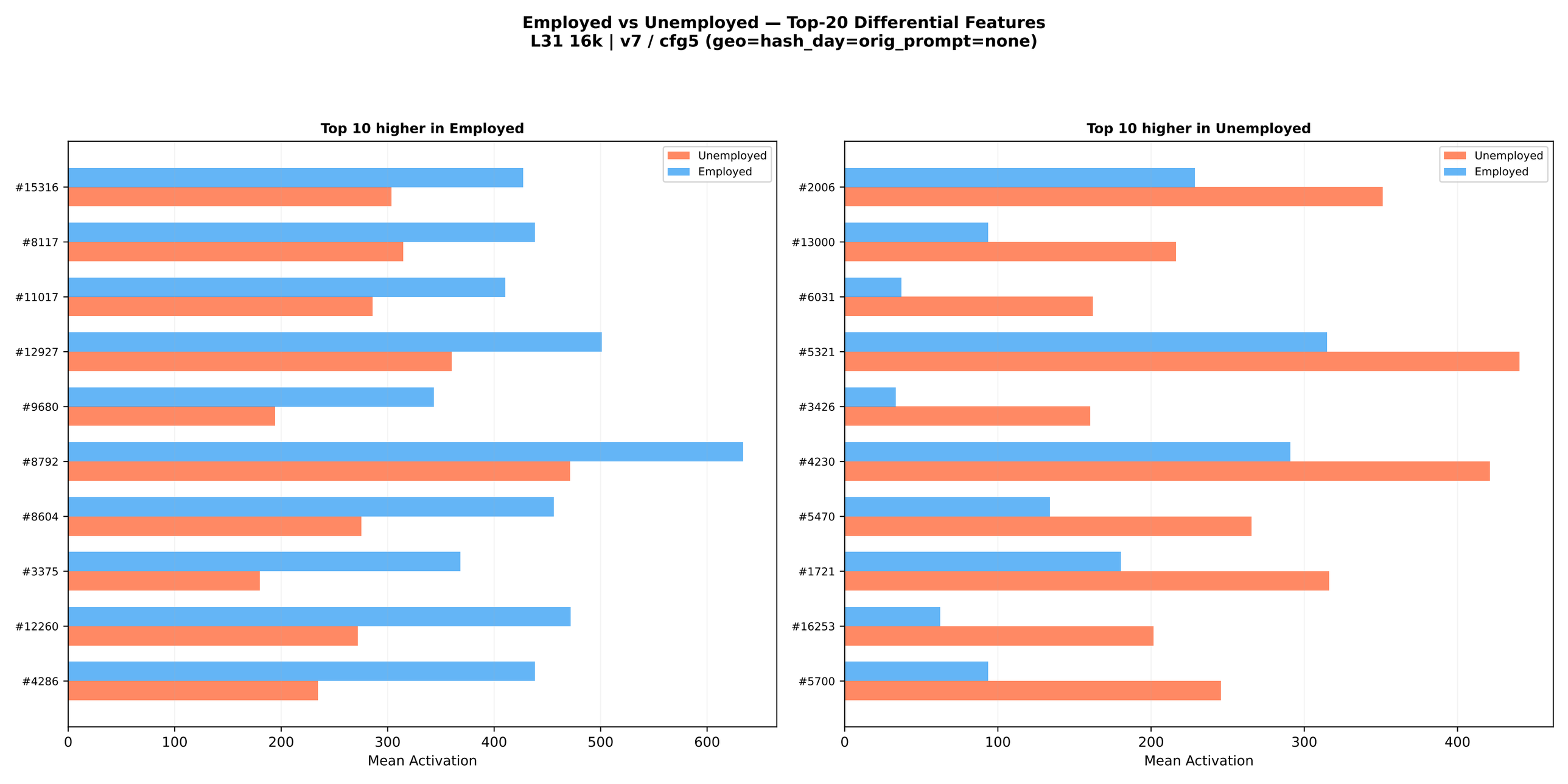

Employed vs Unemployed — Feature Activation Rate
v7 / cfg5 (geo=hash_day=orig_prompt=none)
Fraction of samples with activation > 0
0.0
0.2
0.4
0.6
0.8
1.0
Unemployed
Employed
L41 65k #22384 employment
L41 65k #10086 workplace or of
L41 65k #4523 job dates
L31 65k #4604 doing, or workin
L31 65k #1589 free time
L41 -60% #17063 employment
L41 -60% #4573 employment
L41 -60% #2014 employment
L41 -60% #2867 jobs and employ
L41 -60% #18188 employment
L41 -60% #9733 employment
L41 -60% #27538 business and em
L41 -60% #4658 business, emplo
L41 -60% #2348 employment stain
L41 -60% #2723 hiring, employm
L7 -60% #17043 employment stata
L41 -16% #1348 job centre
L41 -16% #1861 employment stat
L41 -60% #30503 employment and
L41 -60% #17949 employment and
L41 -60% #16086 work employment
L41 -60% #30642 subsidies, unem
L41 -60% #30729 unemployment
L31 -16% #4019 unemployment in
L12 -16% #3148 employment stata
L12 -65% #8478 unemployment ra
L41 -65% #14503 employment stat
L41 -65% #14710 unemployment

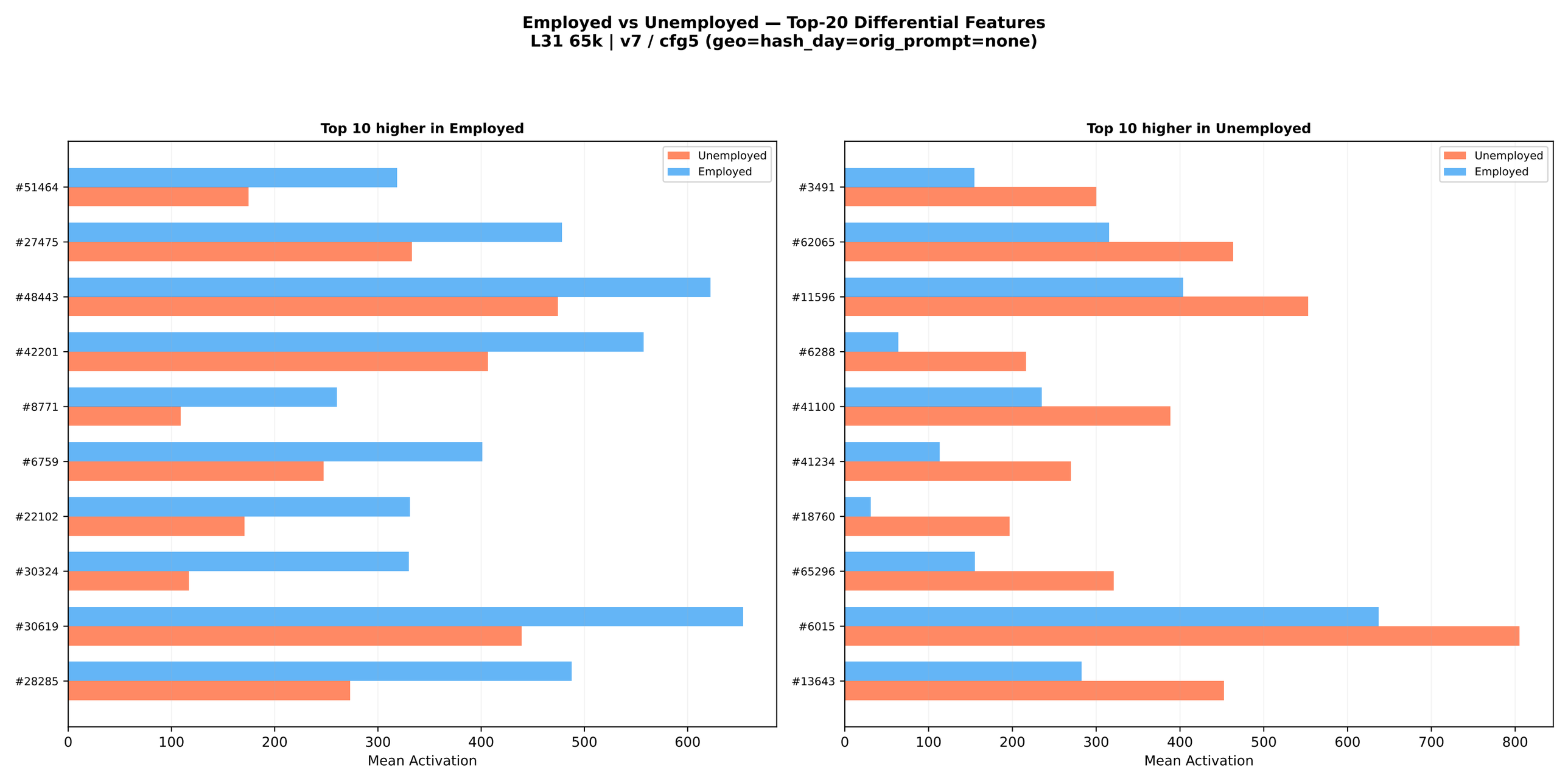

Employed vs Unemployed — Effect Size (Cohen's d)
v7 / cfg5 (geo=hash_day=orig_prompt=none) | Green=target, Gray=control
Cohen's d (positive = higher in class 1)
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
L45 #2584
L45 #1088
L45 #4513
L45 #3859
L45 #4183
L45 #7143
L45 #1673
L45 #2042
L45 #3867
L45 #1988
L45 #9122
L45 #2458
L45 #4058
L45 #848
L45 #2733
L45 #7242
L45 #1108
L45 #831
L45 #2093
L45 #1589
L45 #3088
L45 #1982
L45 #2583
L45 #610
L45 #638
L45 #3879
L45 #3930
L45 #3430
CTRL L45 #30
CTRL L45 #50
CTRL L45 #100
CTRL L45 #200
CTRL L45 #2000

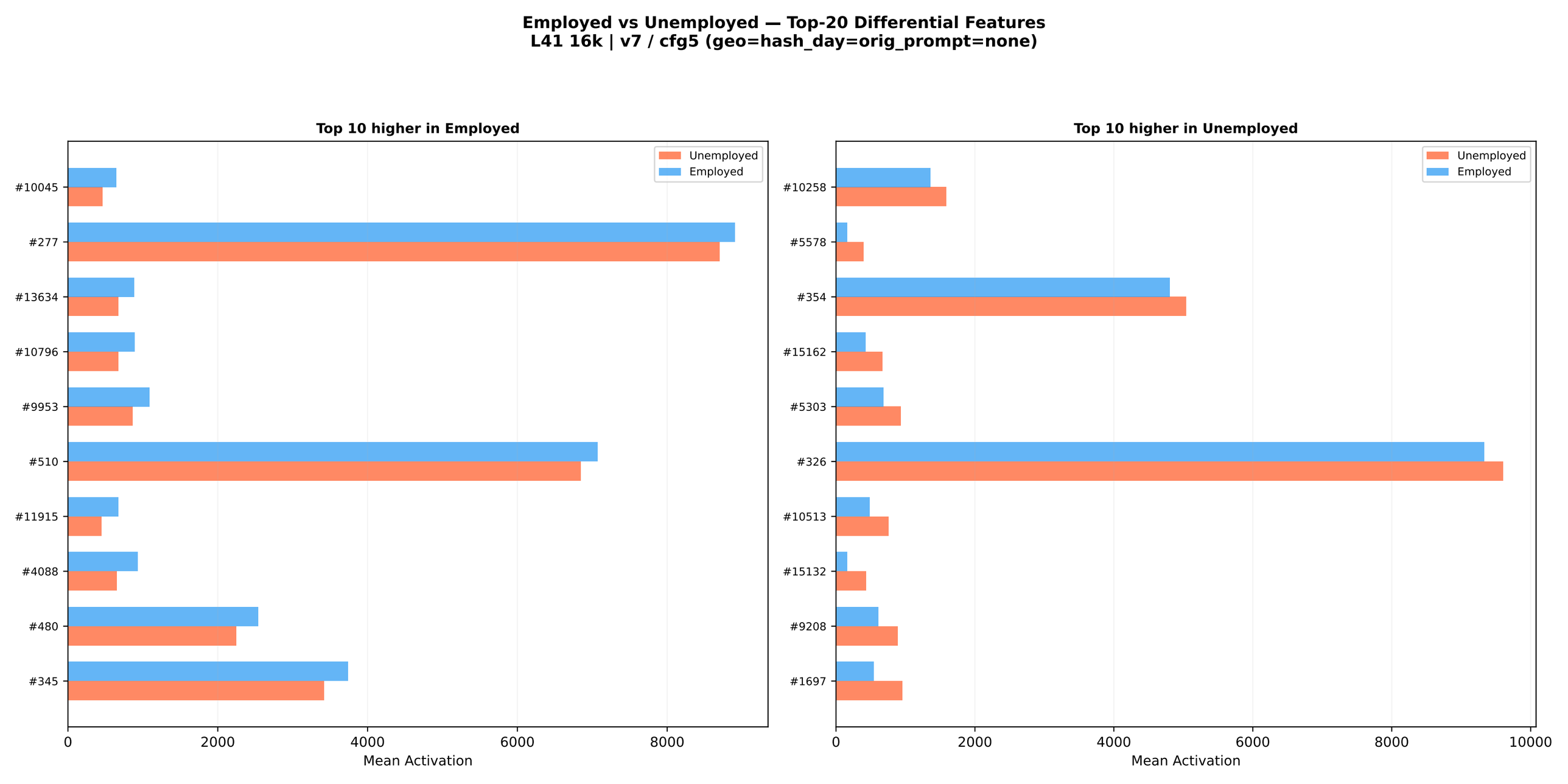

Employed vs Unemployed — Top-20 Differential Features
L12 16k | v7 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#1360
#12
#268
#3017
#13109
#7132
#11402
#224
#414
#7629
0
100
200
300
400
500
600
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#3799
#7755
#73
#4259
#32
#12061
#8499
#5953
#1101
#11075
0
50
100
150
200
250
Mean Activation

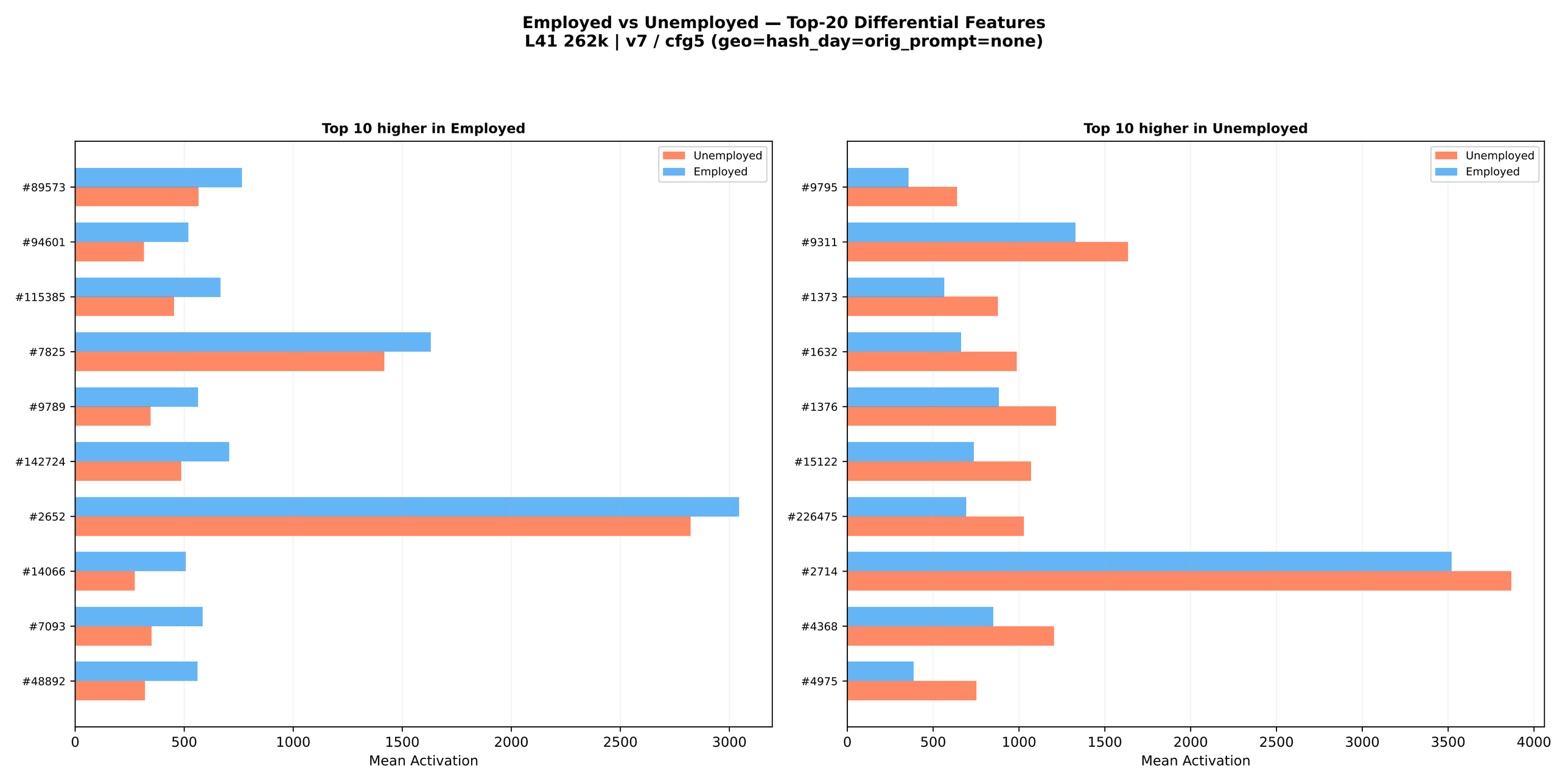

Employed vs Unemployed — Top-20 Differential Features
L12 262k | v7 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#95781
#221372
#169217
#253734
#236116
#161782
#238078
#98568
#8444
#57540
0
50
100
150
200
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#214142
#129748
#114341
#183762
#209567
#29805
#205668
#260088
#52462
#225363
0
50
100
150
200
250
300
Mean Activation

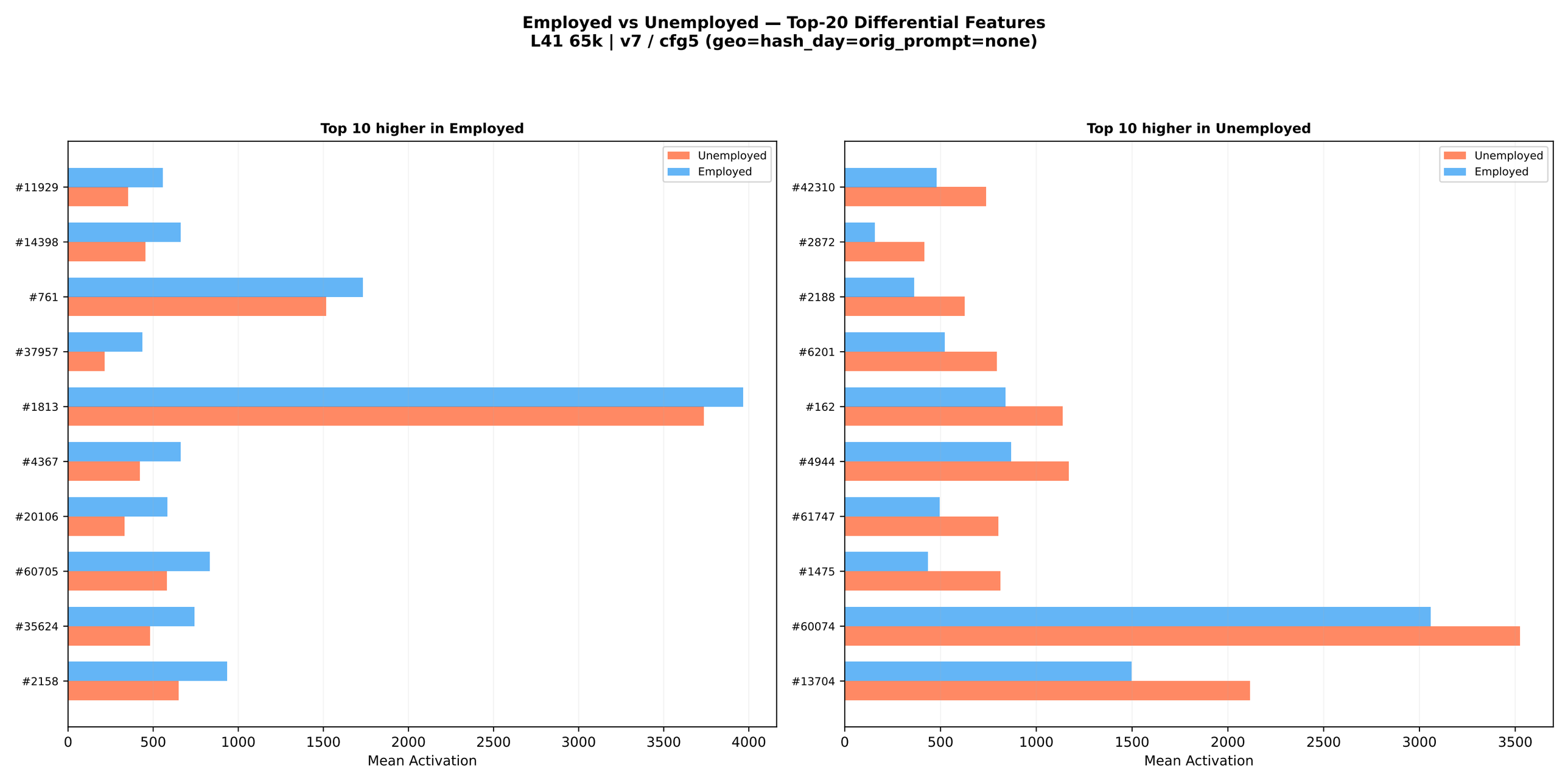

Employed vs Unemployed — Top-20 Differential Features
L12 65k | v7 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
#63968
#8233
#2
#1688
#20378
#44243
#38990
#40546
#45250
#58278
Unemployed
Employed
0
50
100
150
200
250
300
Mean Activation
Top 10 higher in Unemployed
#4412
#49304
#62352
#54297
#40023
#43650
#52350
#14999
#49207
#59889
Unemployed
Employed
0
50
100
150
200
250
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L31 16k | v7 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#15316
#8117
#11017
#12927
#9680
#8792
#8604
#3375
#12260
#4286
0
100
200
300
400
500
600
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#2006
#13000
#6031
#5321
#3426
#4230
#5470
#1721
#16253
#5700
0
100
200
300
400
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L31 65k | v7 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#51464
#27475
#48443
#42201
#8771
#6759
#22102
#30324
#30619
#28285
0
100
200
300
400
500
600
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#3491
#62065
#11596
#6288
#41100
#41234
#18760
#65296
#6015
#13643
0
100
200
300
400
500
600
700
800
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 16k | v7 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#10045
#277
#13634
#10796
#9953
#510
#11915
#4088
#480
#345
0
2000
4000
6000
8000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#10258
#5578
#354
#15162
#5303
#326
#10513
#15132
#9208
#1697
0
2000
4000
6000
8000
10000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 262k | v7 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#89573
#94601
#115385
#7825
#9789
#142724
#2652
#14066
#7093
#48892
0
500
1000
1500
2000
2500
3000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#9795
#9311
#1373
#1632
#1376
#15122
#226475
#2714
#4368
#4975
0
500
1000
1500
2000
2500
3000
3500
4000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 65k | v7 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#11929
#14398
#761
#37957
#1813
#4367
#20106
#60705
#35624
#2158
0
500
1000
1500
2000
2500
3000
3500
4000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#42310
#2872
#2188
#6201
#162
#4944
#61747
#1475
#60074
#13704
0
500
1000
1500
2000
2500
3000
3500
Mean Activation

Summary

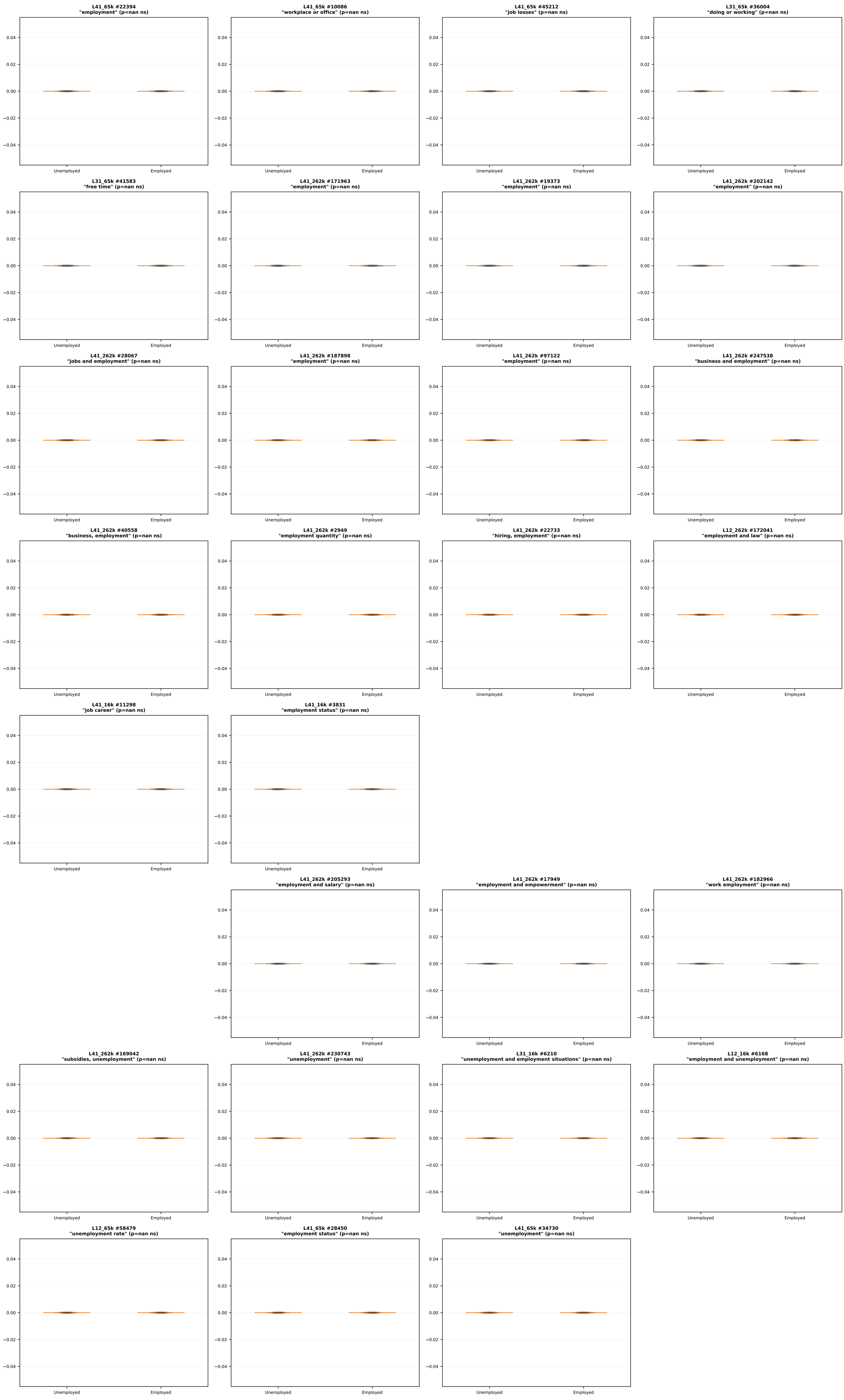
Employed vs Unemployed — SAE Feature Summary | v7 / cfg5 (geo=hash_day=orig_prompt=none)

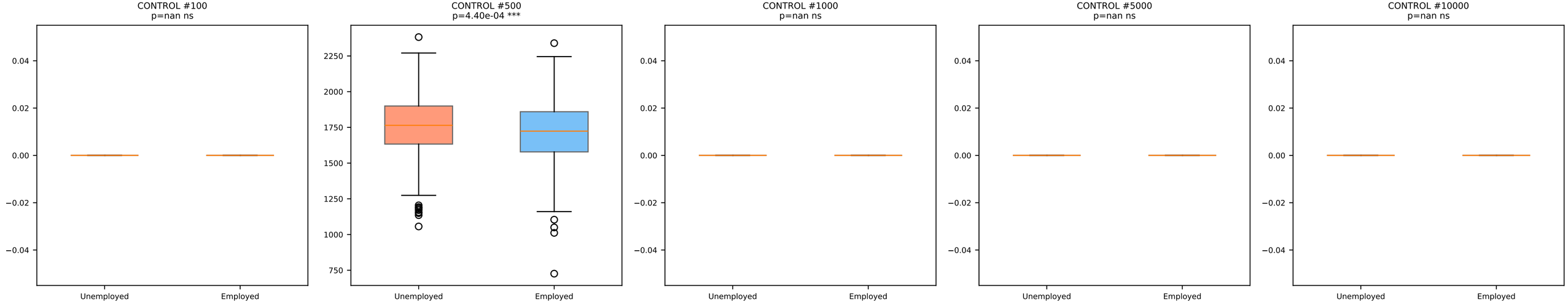
Layer	Width	Feat#	Label	mean_c0	mean_c1	d	p-val	sig	act%_c0	act%_c1
L 41	65k	22394	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10086	workplace or office	1.8107	0.0000	-0.065	3.18e-01	ns	0.2%	0.0%
L 41	65k	45212	job losses	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	36004	doing or working	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	41583	free time	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	171963	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	19373	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	202142	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	28067	jobs and employment	57.9010	39.5583	-0.090	1.64e-01	ns	6.7%	4.4%
L 41	262k	187898	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	97122	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	247538	business and employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	40558	business, employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	2949	employment quantity	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	22733	hiring, employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	262k	172041	employment and law	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	16k	11298	job career	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	16k	3831	employment status	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	205293	employment and salary	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	17949	employment and empowerment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	182966	work employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	169042	subsidies, unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	230743	unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	16k	6210	unemployment and employment situations	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	16k	6168	employment and unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	65k	58479	unemployment rate	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	28450	employment status	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	34730	unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	100	CONTROL_100	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	500	CONTROL_500	1934.3932	1940.2030	+0.016	8.00e-01	ns	99.8%	100.0%
L 41	65k	1000	CONTROL_1000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	5000	CONTROL_5000	2.1274	0.0000	-0.065	3.18e-01	ns	0.2%	0.0%
L 41	65k	10000	CONTROL_10000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%

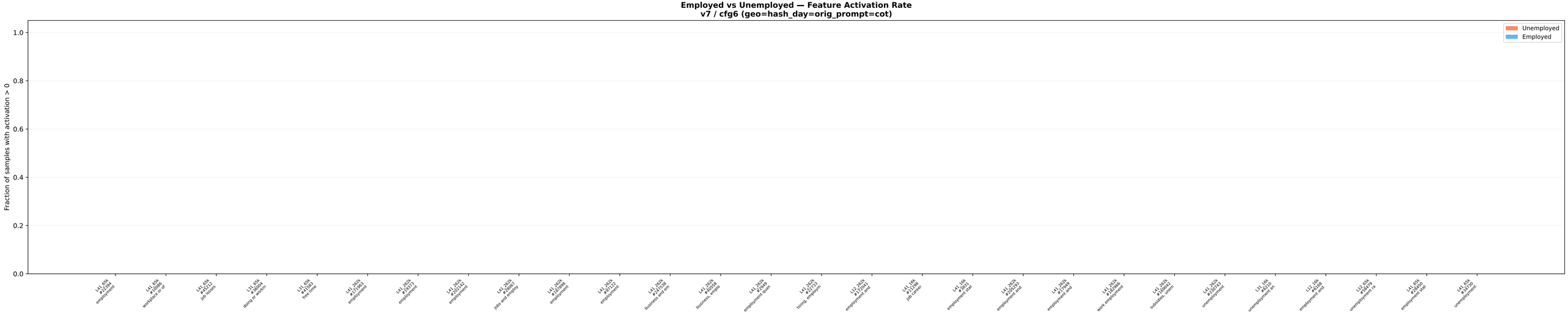
c0 = Unemployed, c1 = Employed
Cohen's d: |d|>0.2 small, |d|>0.5 medium, |d|>0.8 large
act% = fraction of samples where feature activation > 0

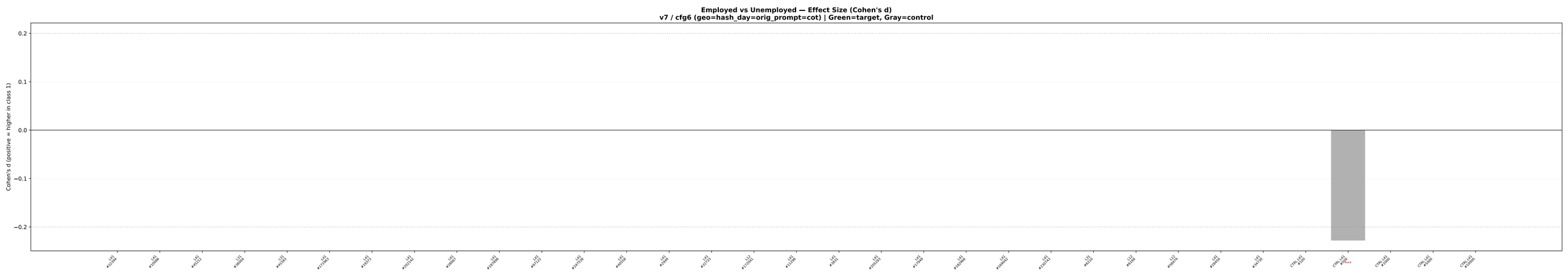
Significant target features: 0/28
Significant control features: 0/5 (should be ~0)

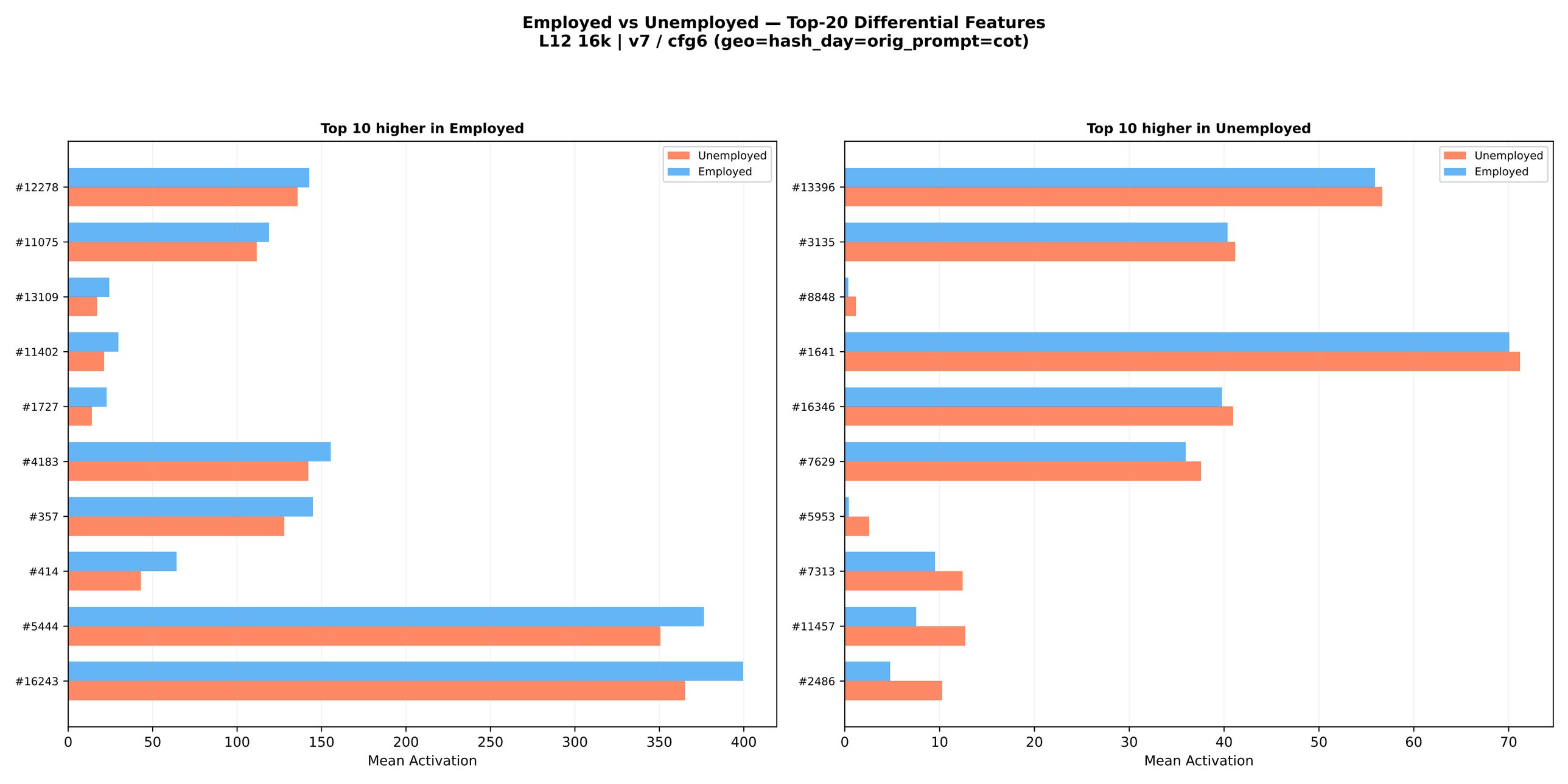
Employed vs Unemployed — SAE Feature Activations
v7 / cfg6 (geo=hash_day=orig_prompt=cot) | Target features (NeuronPedia)

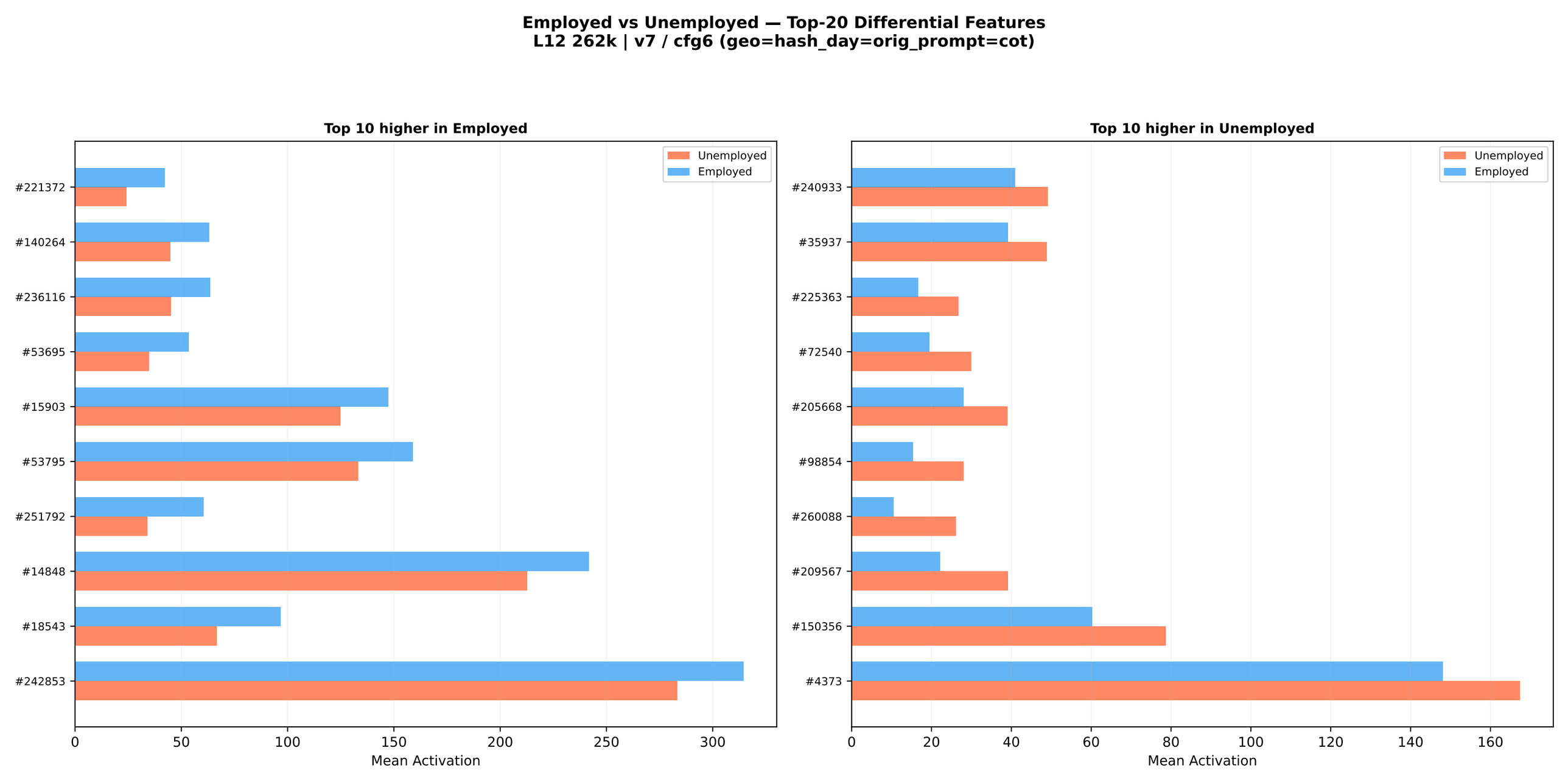


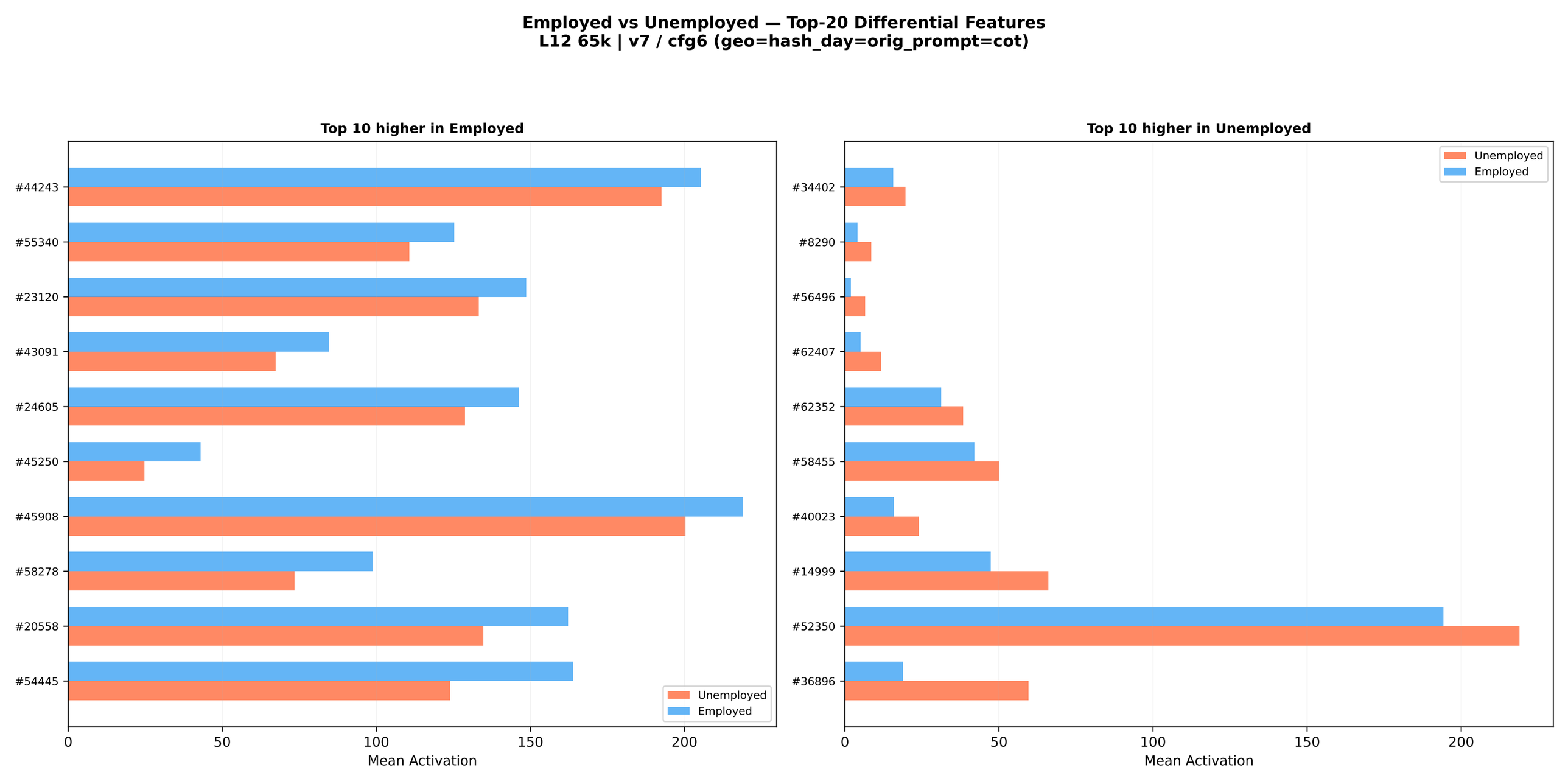

Employed vs Unemployed — Control Features (should be ns)
v7 / cfg6 (geo=hash_day=orig_prompt=cot)
CONTROL #100
p=nan ns
CONTROL #500
p=4.40e-04 ***
CONTROL #1000
p=nan ns
CONTROL #5000
p=nan ns
CONTROL #10000
p=nan ns
Unemployed
Employed

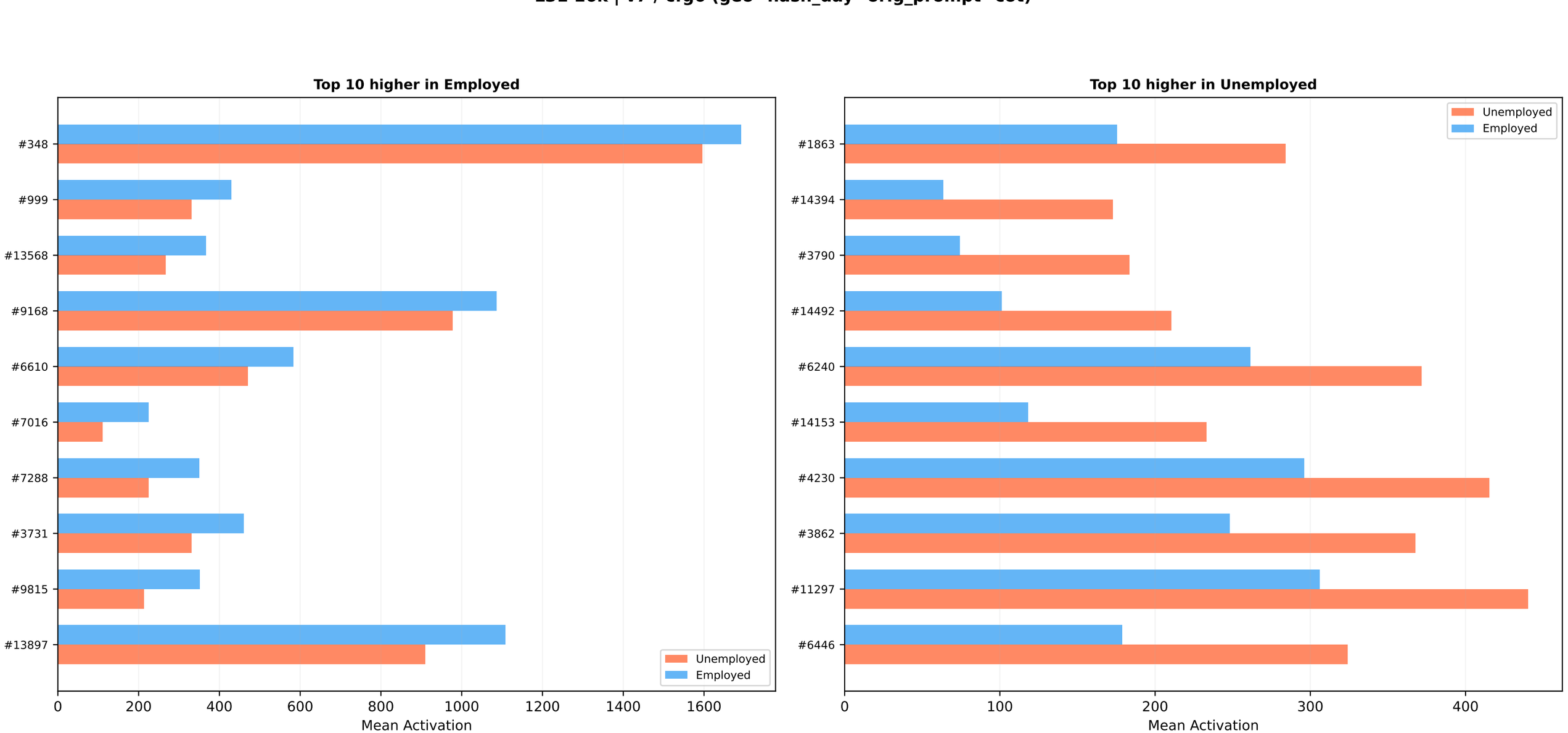

Employed vs Unemployed — Feature Activation Rate
v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Fraction of samples with activation > 0
0.0
0.2
0.4
0.6
0.8
1.0
Unemployed
Employed
L41 65k #22384 employment
L41 65k #10088 workplace or of
L41 65k #4523 job dates
L31 65k #4604 doing, or workin
L31 65k #1589 free time
L41 262k #17063 employment
L41 262k #4373 employment
L41 262k #2673 employments
L41 262k #2867 jobs and employ
L41 262k #18788 employment
L41 262k #97123 employment
L41 262k #27538 business asia em
L41 262k #4658 business, emplo
L41 262k #2549 employment stain
L41 262k #2723 hiring, employm
L17 262k #17043 employment asia
L41 16k #1348 job career
L41 16k #1861 employment stat
L41 262k #30503 employment and
L41 262k #2769 employment and
L41 262k #18286 work employment
L41 262k #30643 subsidiaries, employ
L41 262k #30729 unemployment
L31 16k #4019 unemployment in
L12 16k #3148 employment asia
L12 65k #8439 unemployment in
L41 65k #4503 employment stat
L41 65k #14710 unemployment

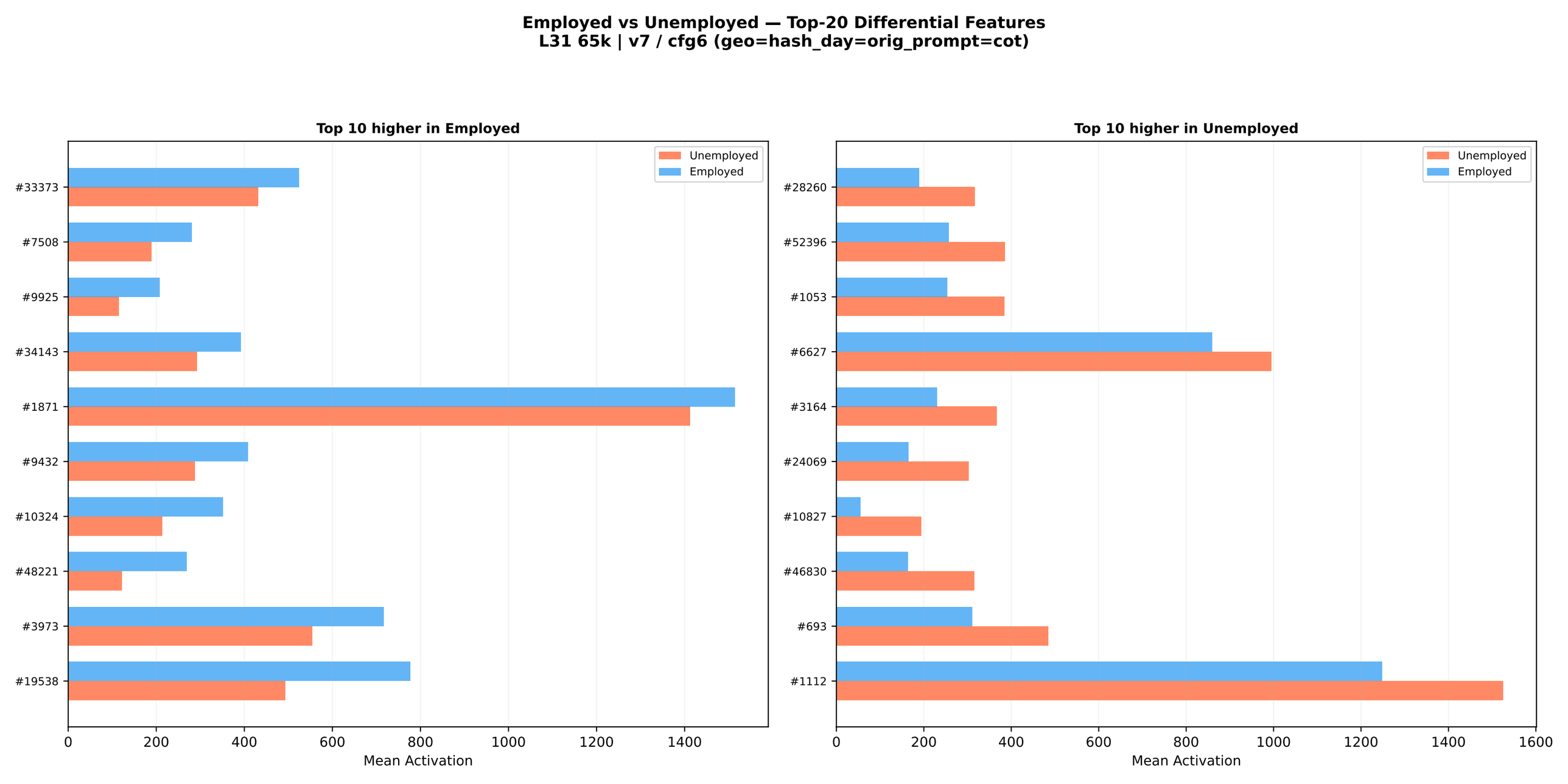

Employed vs Unemployed — Effect Size (Cohen's d)
v7 / cfg6 (geo=hash_day=orig_prompt=cot) | Green=target, Gray=control
Cohen's d (positive = higher in class 1)
0.2
0.1
0.0
−0.1
−0.2
L0 #2084
L0 #3088
L0 #4012
L0 #3004
L0 #4083
L0 #3083
L0 #3973
L0 #2042
L0 #2067
L0 #3088
L0 #3122
L0 #2038
L0 #4058
L0 #3 48
L0 #2133
L0 #3041
L0 #1148
L0 #81
L0 #2093
L0 #1189
L0 #3048
L0 #3082
L0 #2093
L0 #630
L0 #638
L0 #3049
L0 #2040
L0 #3 40
CTR L0 #40
CTR L0 #38 ***
CTR L0 #3 08
CTR L0 #2 08
CTR L0 #2080

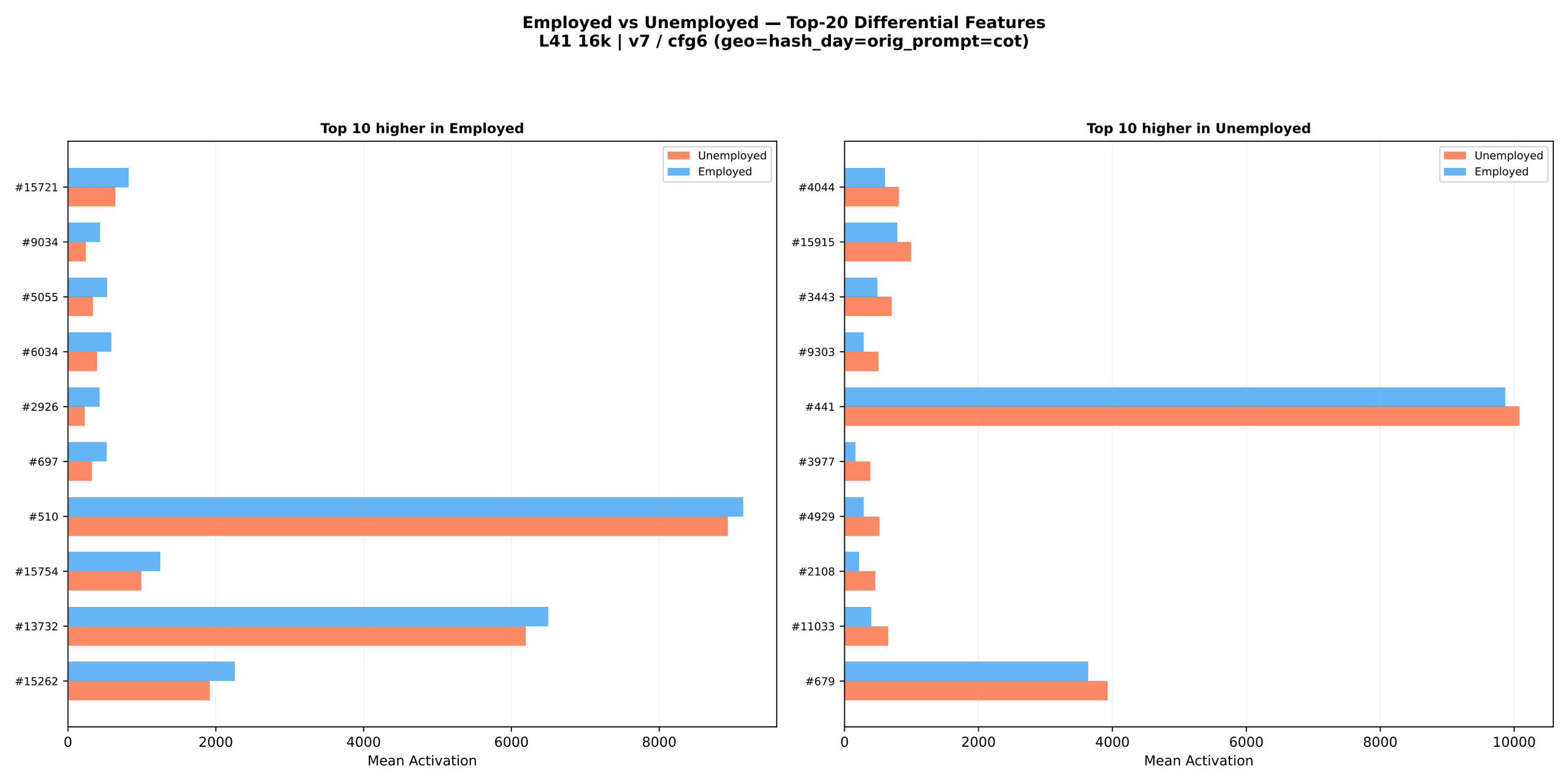

Employed vs Unemployed — Top-20 Differential Features
L12 16k | v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#12278
#11075
#13109
#11402
#1727
#4183
#357
#414
#5444
#16243
0
50
100
150
200
250
300
350
400
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#13396
#3135
#8848
#1641
#16346
#7629
#5953
#7313
#11457
#2486
0
10
20
30
40
50
60
70
Mean Activation

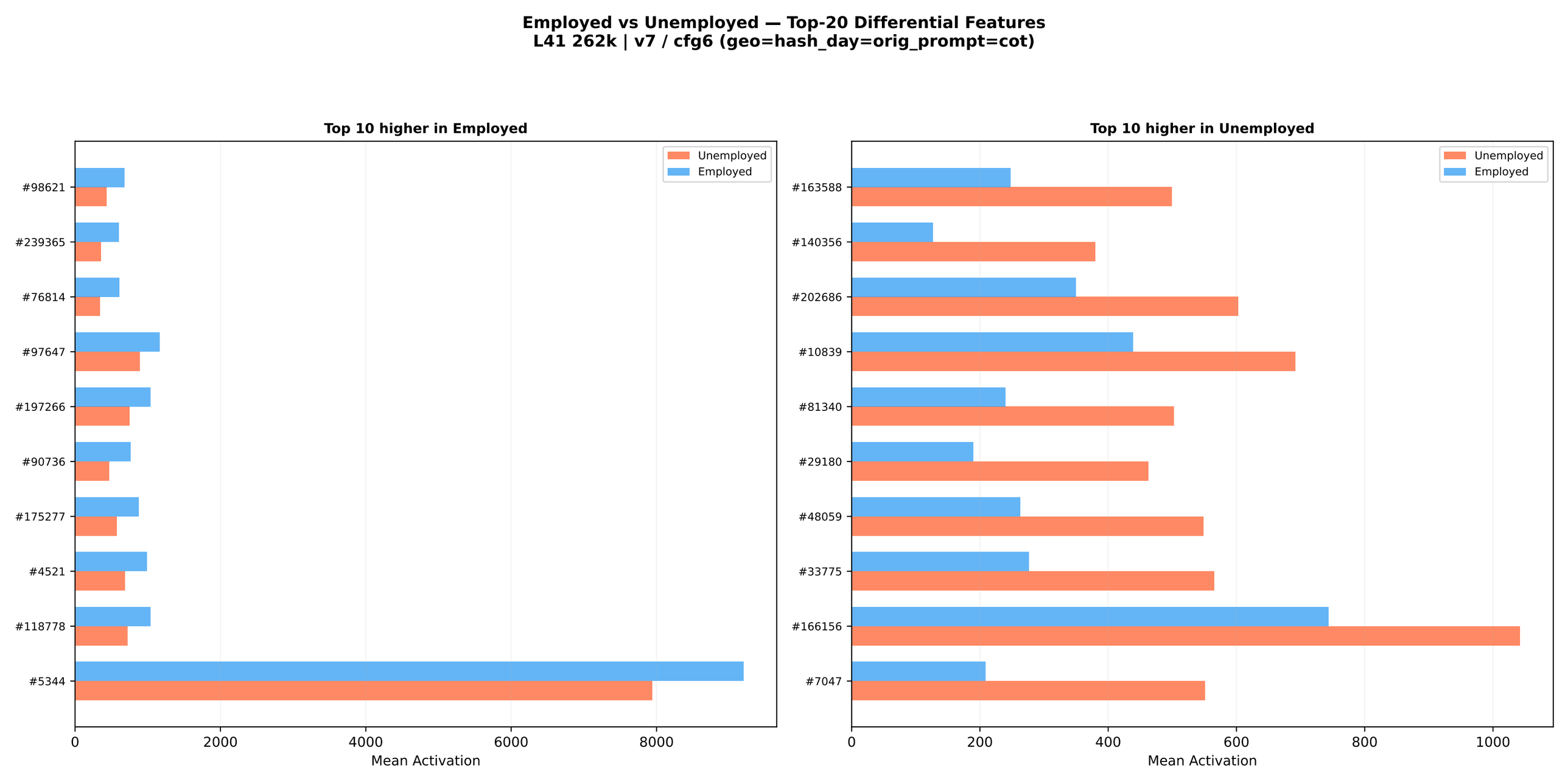

Employed vs Unemployed — Top-20 Differential Features
L12 262k | v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#221372
#140264
#236116
#53695
#15903
#53795
#251792
#14848
#18543
#242853
0
50
100
150
200
250
300
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#240933
#35937
#225363
#72540
#205668
#98854
#260088
#209567
#150356
#4373
0
20
40
60
80
100
120
140
160
Mean Activation

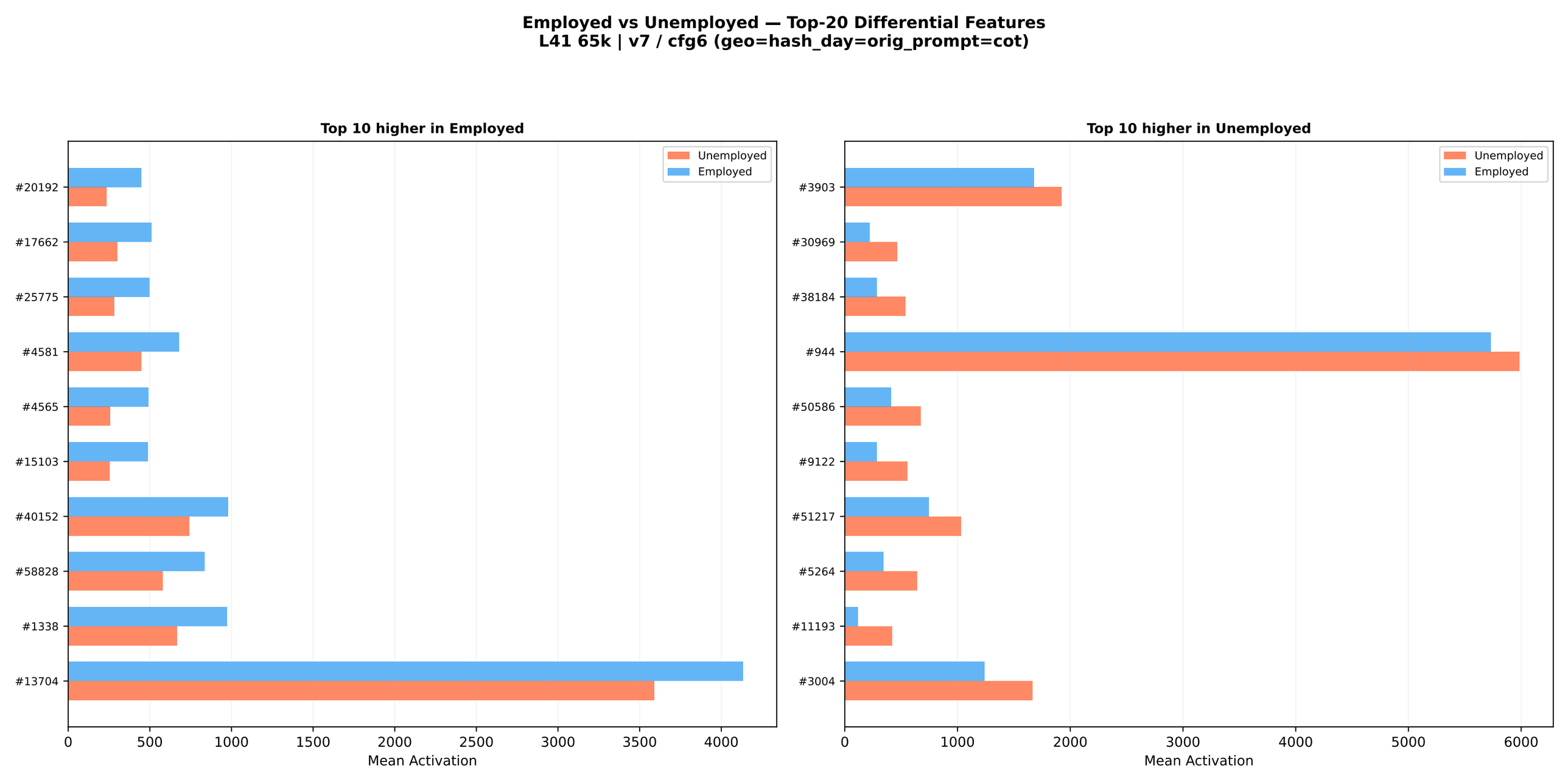

Employed vs Unemployed — Top-20 Differential Features
L12 65k | v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
#44243
#55340
#23120
#43091
#24605
#45250
#45908
#58278
#20558
#54445
Unemployed
Employed
0
50
100
150
200
Mean Activation
Top 10 higher in Unemployed
#34402
#8290
#56496
#62407
#62352
#58455
#40023
#14999
#52350
#36896
Unemployed
Employed
0
50
100
150
200
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L31 16k | v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
#348
#999
#13568
#9168
#6610
#7016
#7288
#3731
#9815
#13897
0
200
400
600
800
1000
1200
1400
1600
Mean Activation
Unemployed
Employed
Top 10 higher in Unemployed
#1863
#14394
#3790
#14492
#6240
#14153
#4230
#3862
#11297
#6446
0
100
200
300
400
Mean Activation
Unemployed
Employed


Employed vs Unemployed — Top-20 Differential Features
L31 65k | v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#33373
#7508
#9925
#34143
#1871
#9432
#10324
#48221
#3973
#19538
0
200
400
600
800
1000
1200
1400
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#28260
#52396
#1053
#6627
#3164
#24069
#10827
#46830
#693
#1112
0
200
400
600
800
1000
1200
1400
1600
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 16k | v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#15721
#9034
#5055
#6034
#2926
#697
#510
#15754
#13732
#15262
0
2000
4000
6000
8000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#4044
#15915
#3443
#9303
#441
#3977
#4929
#2108
#11033
#679
0
2000
4000
6000
8000
10000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 262k | v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#98621
#239365
#76814
#97647
#197266
#90736
#175277
#4521
#118778
#5344
0
2000
4000
6000
8000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#163588
#140356
#202686
#10839
#81340
#29180
#48059
#33775
#166156
#7047
0
200
400
600
800
1000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 65k | v7 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#20192
#17662
#25775
#4581
#4565
#15103
#40152
#58828
#1338
#13704
0
500
1000
1500
2000
2500
3000
3500
4000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#3903
#30969
#38184
#944
#50586
#9122
#51217
#5264
#11193
#3004
0
1000
2000
3000
4000
5000
6000
Mean Activation

Summary

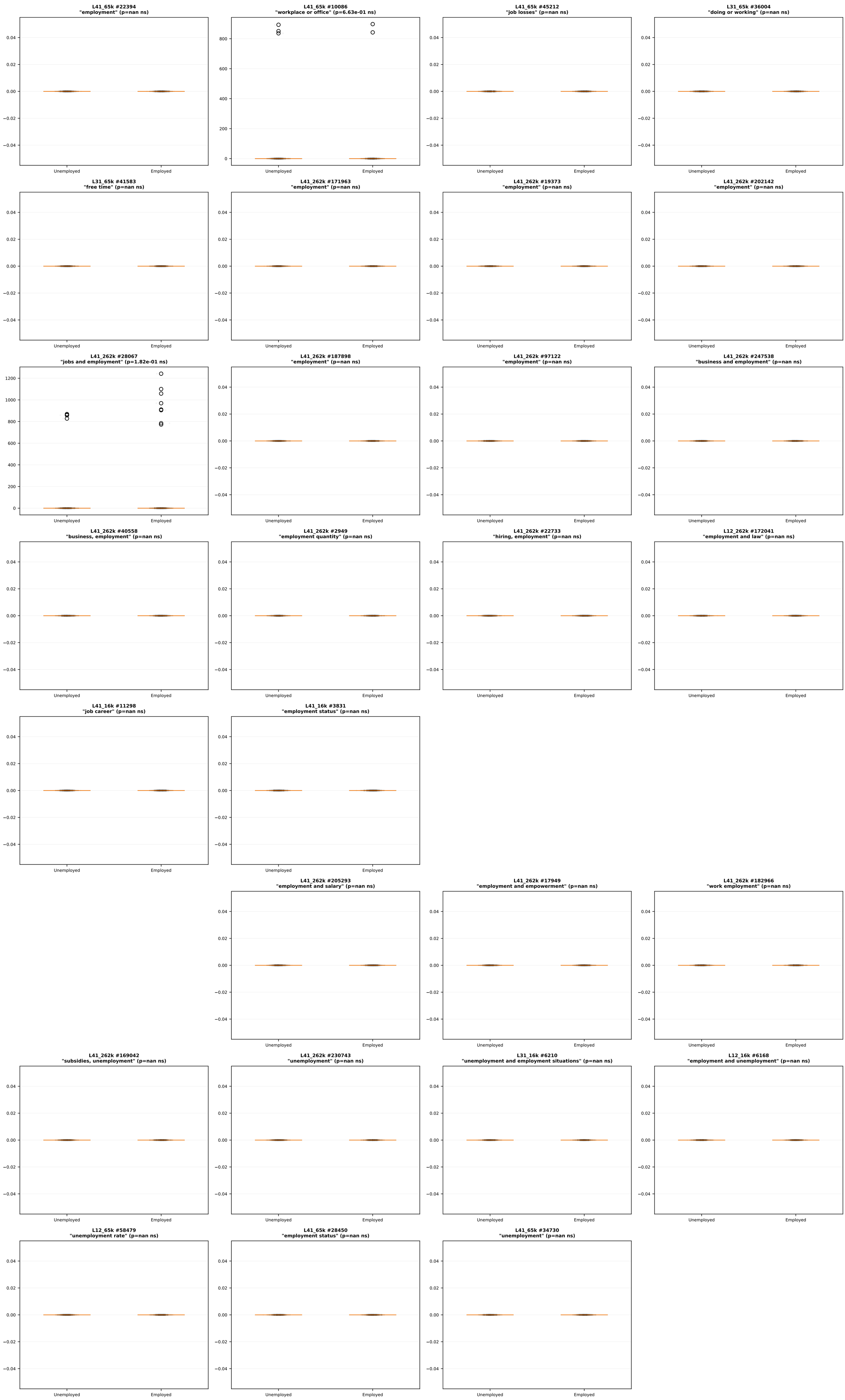
Employed vs Unemployed — SAE Feature Summary | v7 / cfg6 (geo=hash_day=orig_prompt=cot)

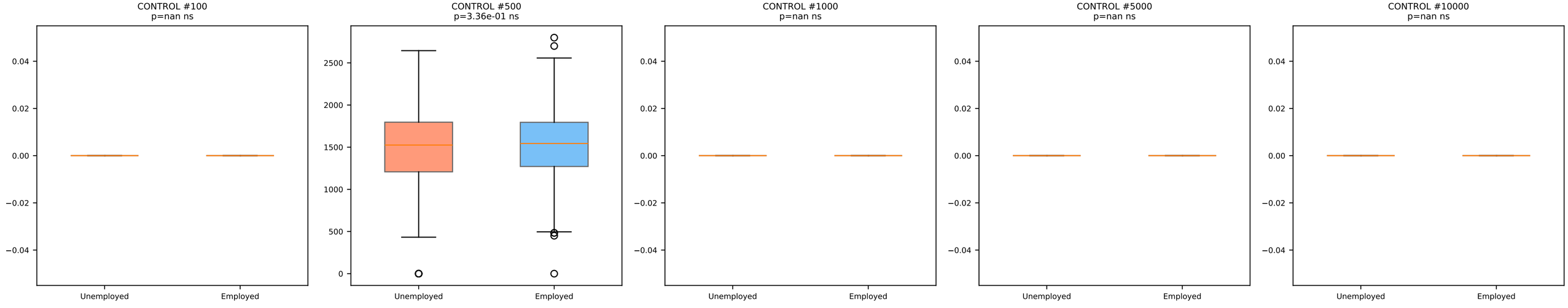
Layer	Width	Feat#	Label	mean_c0	mean_c1	d	p-val	sig	act%_c0	act%_c1
L 41	65k	22394	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10086	workplace or office	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	45212	job losses	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	36004	doing or working	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	41583	free time	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	171963	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	19373	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	202142	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	28067	jobs and employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	187898	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	97122	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	247538	business and employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	40558	business, employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	2949	employment quantity	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	22733	hiring, employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	262k	172041	employment and law	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	16k	11298	job career	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	16k	3831	employment status	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	205293	employment and salary	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	17949	employment and empowerment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	182966	work employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	169042	subsidies, unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	230743	unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	16k	6210	unemployment and employment situations	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	16k	6168	employment and unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	65k	58479	unemployment rate	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	28450	employment status	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	34730	unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	100	CONTROL_100	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	500	CONTROL_500	1761.5598	1712.2932	-0.228	4.40e-04	***	100.0%	100.0%
L 41	65k	1000	CONTROL_1000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	5000	CONTROL_5000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10000	CONTROL_10000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%

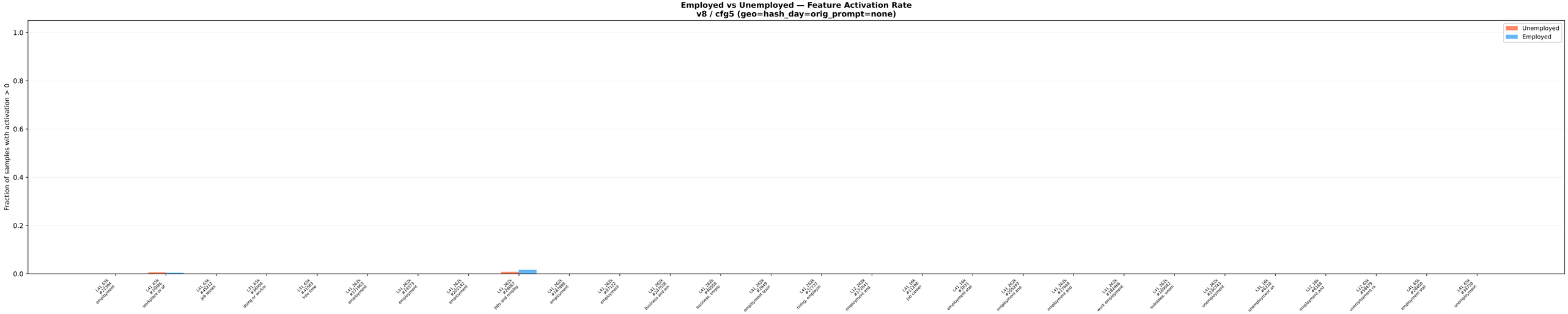
c0 = Unemployed, c1 = Employed
Cohen's d: |d|>0.2 small, |d|>0.5 medium, |d|>0.8 large
act% = fraction of samples where feature activation > 0

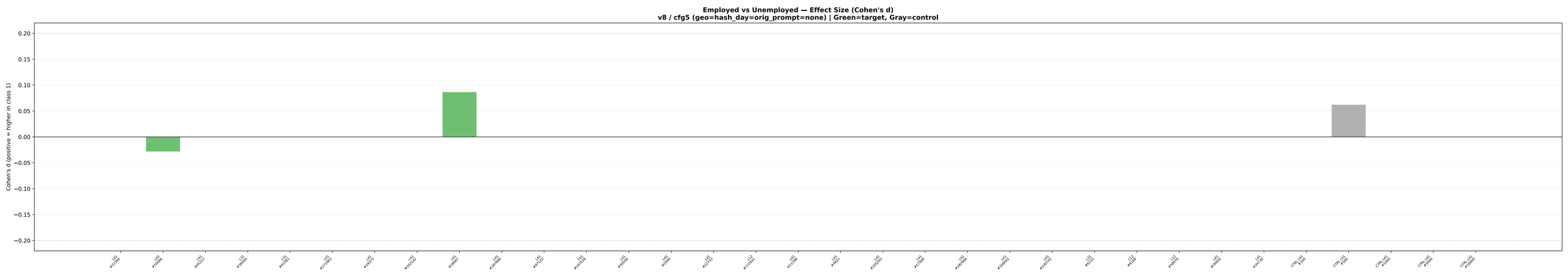
Significant target features: 0/28
Significant control features: 1/5 (should be ~0)

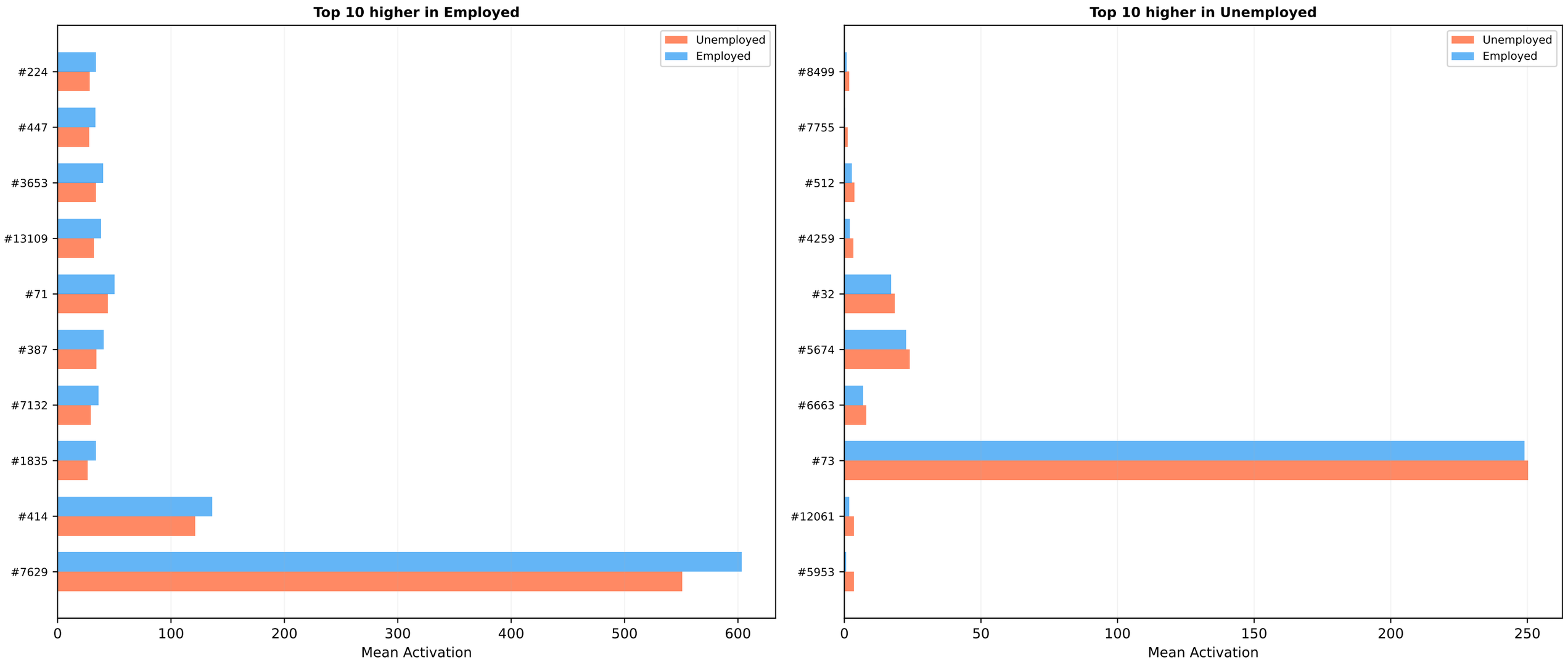
Employed vs Unemployed — SAE Feature Activations
v8 / cfg5 (geo=hash_day=orig_prompt=none) | Target features (NeuronPedia)

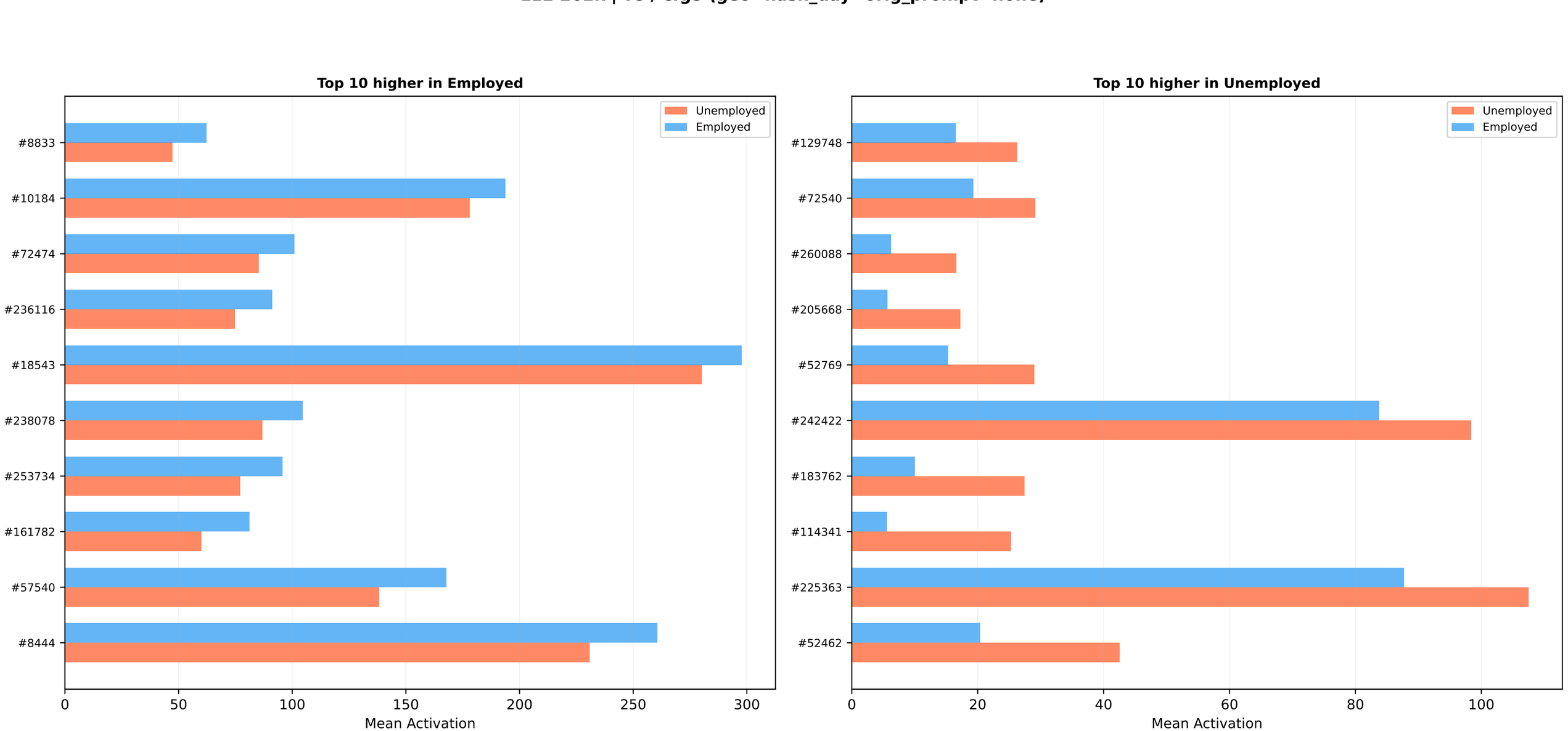


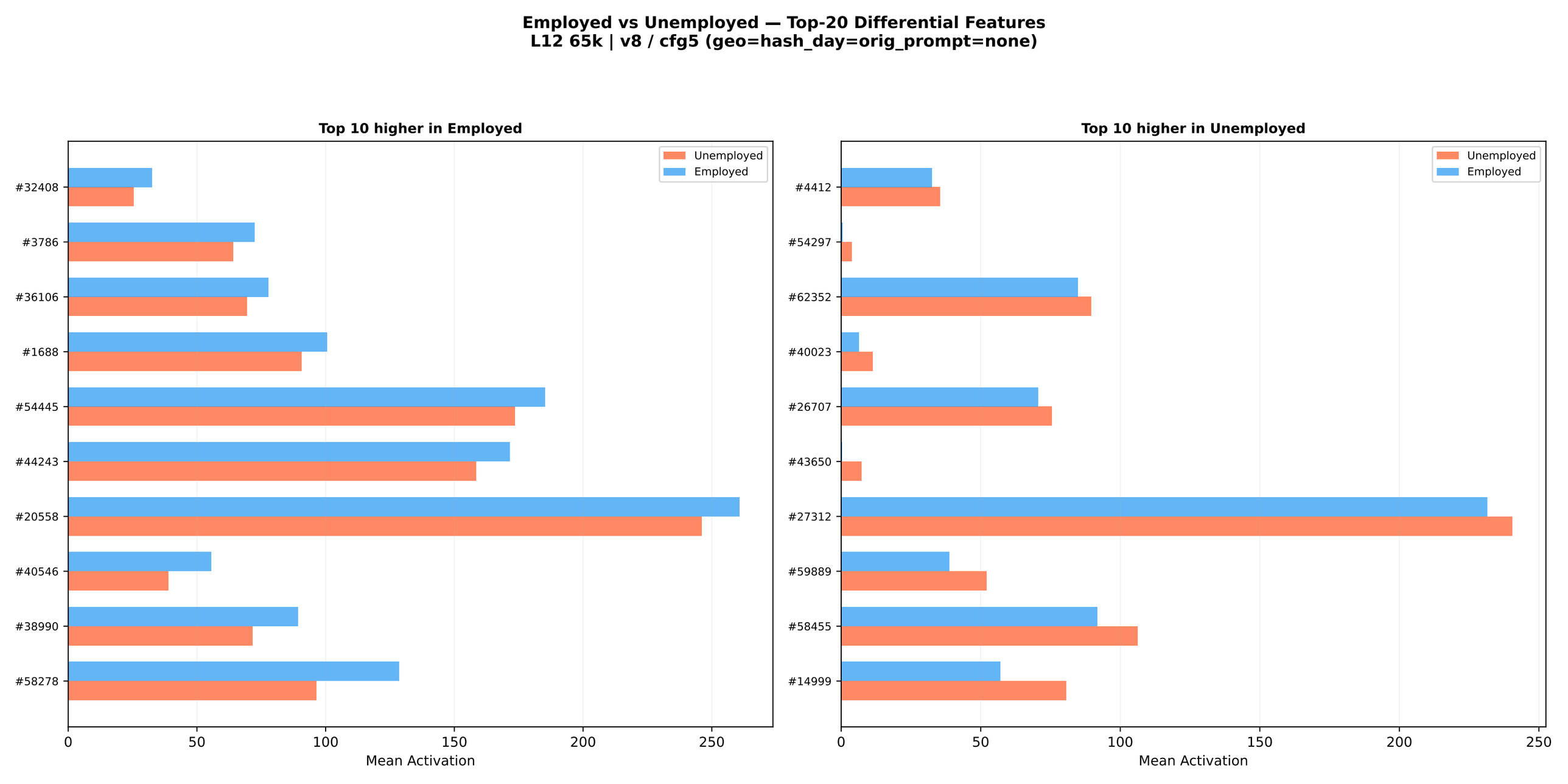

Employed vs Unemployed — Control Features (should be ns)
v8 / cfg5 (geo=hash_day=orig_prompt=none)
CONTROL #100
p=nan ns
CONTROL #500
p=3.36e-01 ns
CONTROL #1000
p=nan ns
CONTROL #5000
p=nan ns
CONTROL #10000
p=nan ns
Unemployed
Employed

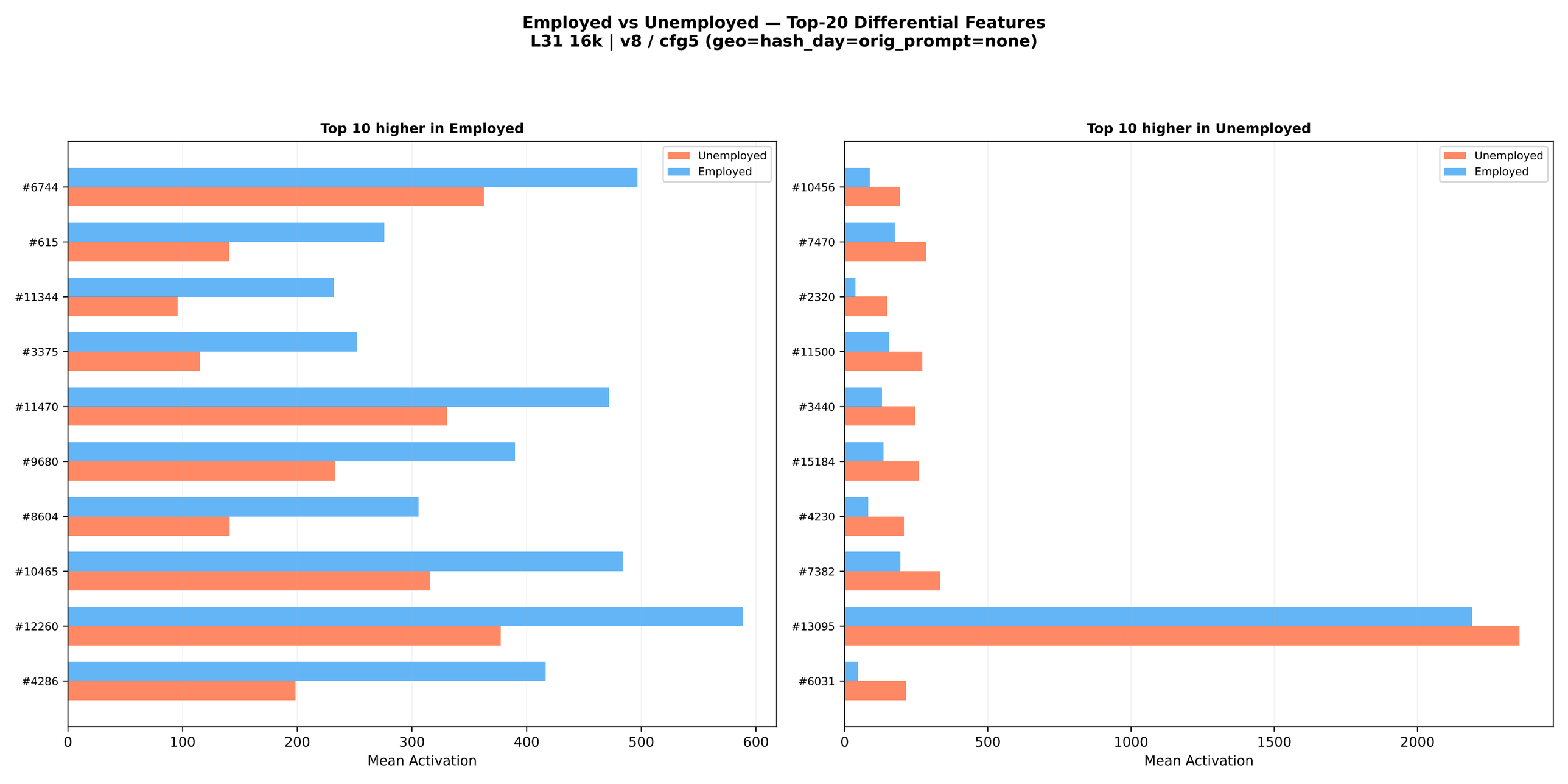

Employed vs Unemployed — Feature Activation Rate
v8 / cfg5 (geo=hash_day=orig_prompt=none)
Fraction of samples with activation > 0
1.0
0.8
0.6
0.4
0.2
0.0
Unemployed
Employed
L41 65k #22384 employment
L41 65k #10088 workplace or of
L41 65k #4523 job dates
L31 65k #26004 doing, or workin
L31 65k #1589 free time
L41 -60% #17063 employment
L41 -60% #4573 employment
L41 -60% #2014 employment
L41 -60% #24867 jobs and employ
L41 -50% #16188 employment
L41 -50% #9123 employment
L41 -50% #27538 business asia em
L41 -50% #4658 business, emplo
L41 -40% #2349 employment sitan
L41 -40% #2723 hiring, employm
L7 -40% #17043 employment asia
L41 -16% #1348 job centre
L41 -16% #1861 employment stat
L41 -50% #30503 employment and
L41 -50% #2769 employment and
L41 -50% #15086 work employment
L41 -30% #10643 subsidiaries, employ
L41 -40% #30729 unemployment
L31 -16% #4019 unemployment sh
L12 -16% #3148 employment asia
L12 -65% #8478 unemployment ra
L41 -65% #4503 employment stat
L41 -65% #14710 unemployment

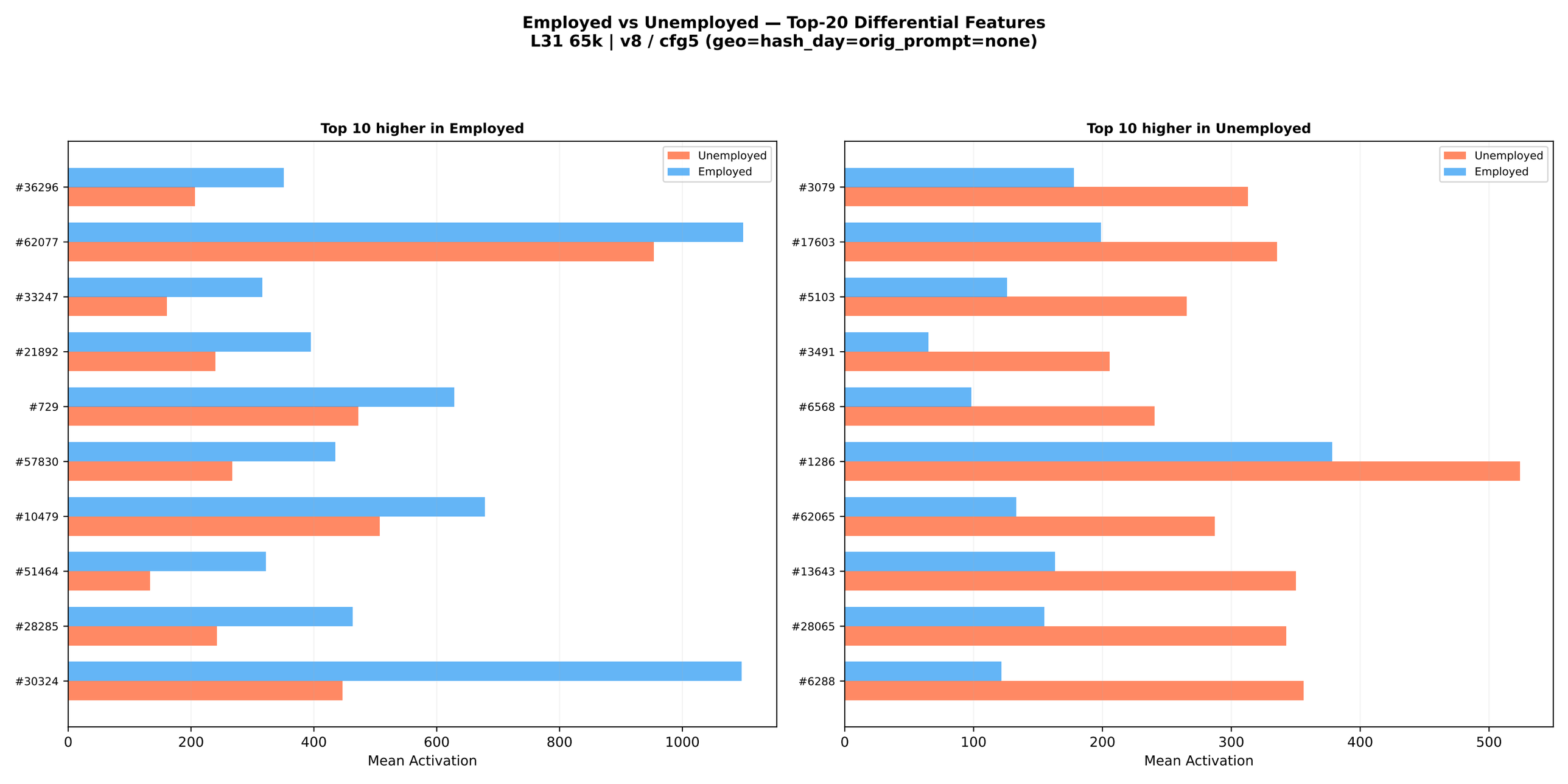

Employed vs Unemployed — Effect Size (Cohen's d)
v8 / cfg5 (geo=hash_day=orig_prompt=none) | Green=target, Gray=control
Cohen's d (positive = higher in class 1)
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
L49 #2584
L49 #1088
L45 #4513
L53 #3859
L45 #1183
L51 #7143
L51 #1673
L51 #2014
L45 #3867
L45 #1988
L45 #7122
L45 #2458
L45 #4058
L45 #848
L52 #2733
L52 #7262
L45 #1108
L45 #831
L45 #2093
L45 #1589
L45 #3088
L45 #1982
L45 #2583
L53 #610
L52 #638
L52 #6879
L45 #2980
L45 #3430
CTRL L45 #30
CTRL L45 #50
CTRL L45 #80
CTRL L45 #90
CTRL L45 #2080

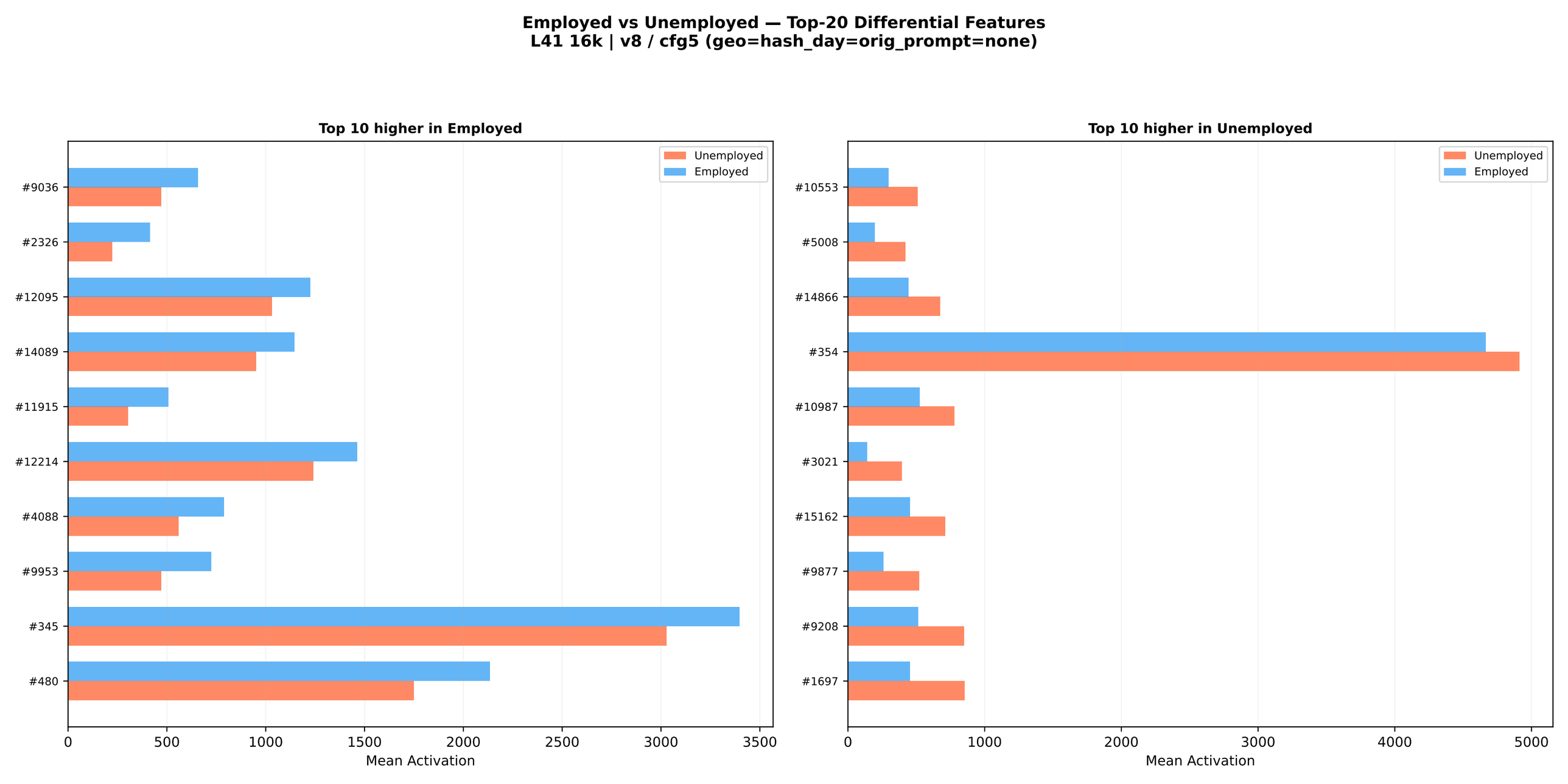

Employed vs Unemployed — Top-20 Differential Features
L12 16k | v8 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#224
#447
#3653
#13109
#71
#387
#7132
#1835
#414
#7629
0
100
200
300
400
500
600
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#8499
#7755
#512
#4259
#32
#5674
#6663
#73
#12061
#5953
0
50
100
150
200
250
Mean Activation

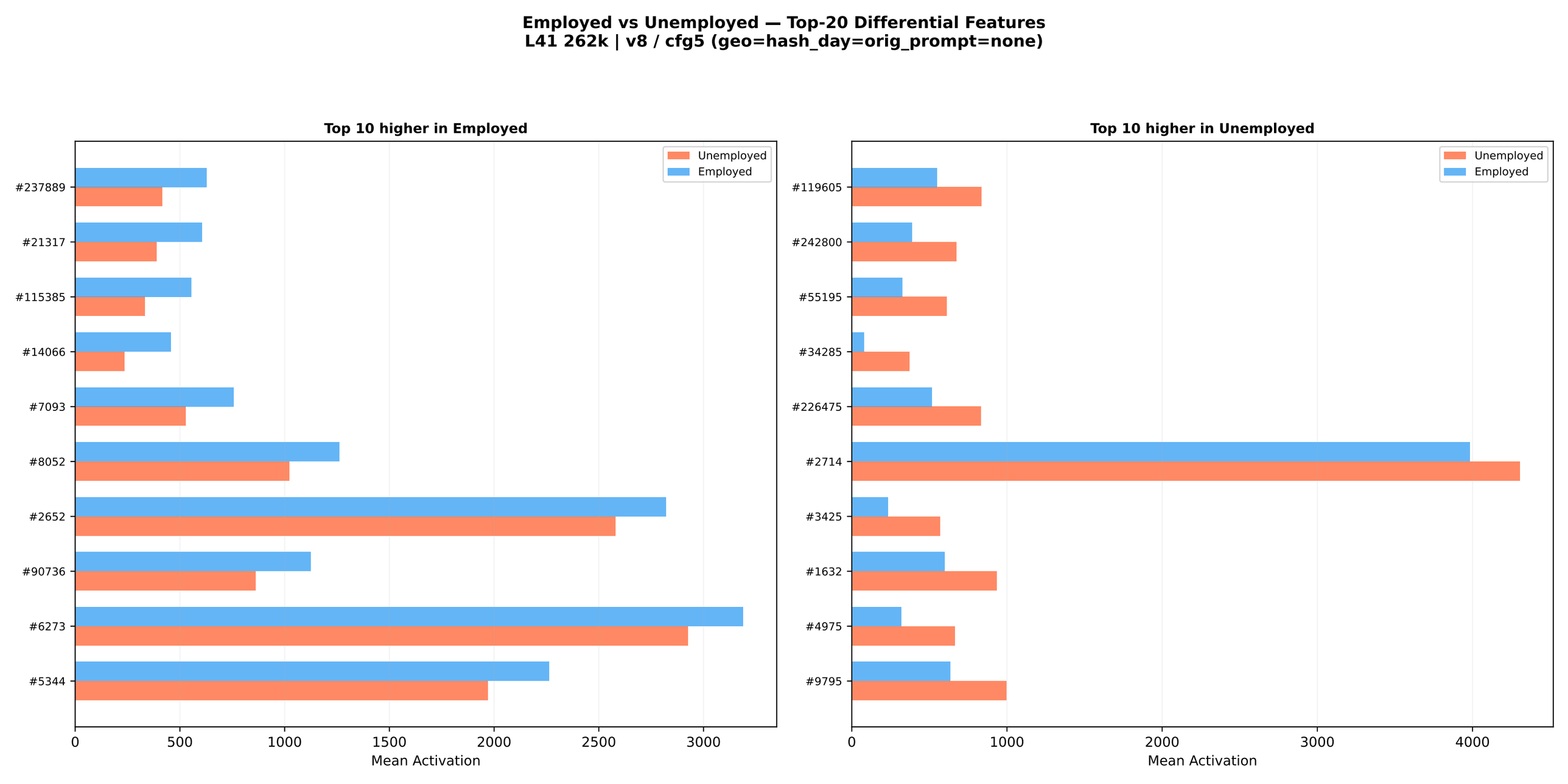

Employed vs Unemployed — Top-20 Differential Features
L12 262k | v8 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#8833
#10184
#72474
#236116
#18543
#238078
#253734
#161782
#57540
#8444
0
50
100
150
200
250
300
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#129748
#72540
#260088
#205668
#52769
#242422
#183762
#114341
#225363
#52462
0
20
40
60
80
100
Mean Activation

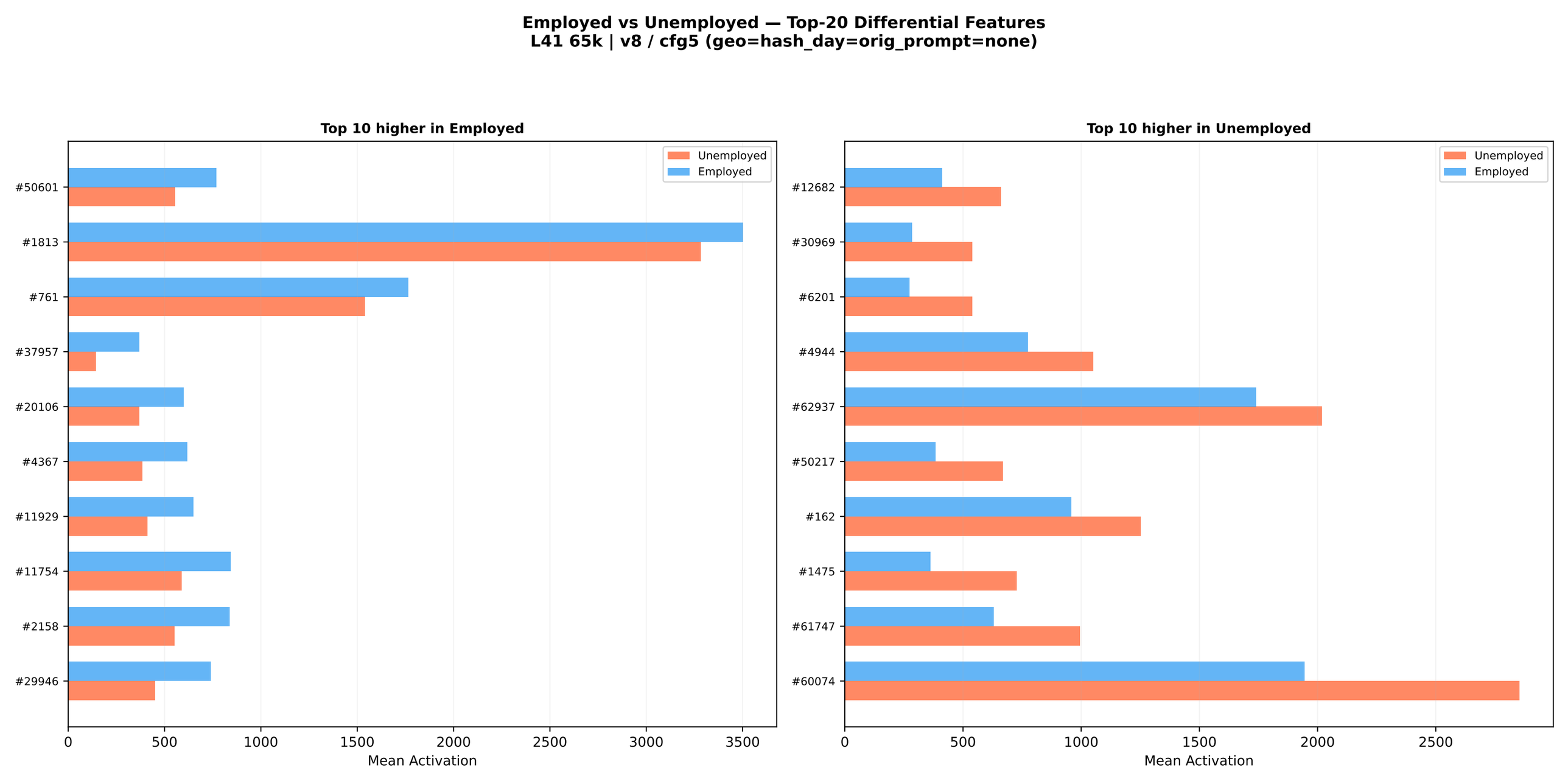

Employed vs Unemployed — Top-20 Differential Features
L12 65k | v8 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#32408
#3786
#36106
#1688
#54445
#44243
#20558
#40546
#38990
#58278
0
50
100
150
200
250
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#4412
#54297
#62352
#40023
#26707
#43650
#27312
#59889
#58455
#14999
0
50
100
150
200
250
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L31 16k | v8 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#6744
#615
#11344
#3375
#11470
#9680
#8604
#10465
#12260
#4286
0
100
200
300
400
500
600
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#10456
#7470
#2320
#11500
#3440
#15184
#4230
#7382
#13095
#6031
0
500
1000
1500
2000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L31 65k | v8 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#36296
#62077
#33247
#21892
#729
#57830
#10479
#51464
#28285
#30324
0
200
400
600
800
1000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#3079
#17603
#5103
#3491
#6568
#1286
#62065
#13643
#28065
#6288
0
100
200
300
400
500
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 16k | v8 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#9036
#2326
#12095
#14089
#11915
#12214
#4088
#9953
#345
#480
0
500
1000
1500
2000
2500
3000
3500
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#10553
#5008
#14866
#354
#10987
#3021
#15162
#9877
#9208
#1697
0
1000
2000
3000
4000
5000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 262k | v8 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#237889
#21317
#115385
#14066
#7093
#8052
#2652
#90736
#6273
#5344
0
500
1000
1500
2000
2500
3000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#119605
#242800
#55195
#34285
#226475
#2714
#3425
#1632
#4975
#9795
0
1000
2000
3000
4000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 65k | v8 / cfg5 (geo=hash_day=orig_prompt=none)
Top 10 higher in Employed
Unemployed
Employed
#50601
#1813
#761
#37957
#20106
#4367
#11929
#11754
#2158
#29946
0
500
1000
1500
2000
2500
3000
3500
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#12682
#30969
#6201
#4944
#62937
#50217
#162
#1475
#61747
#60074
0
500
1000
1500
2000
2500
Mean Activation

Summary

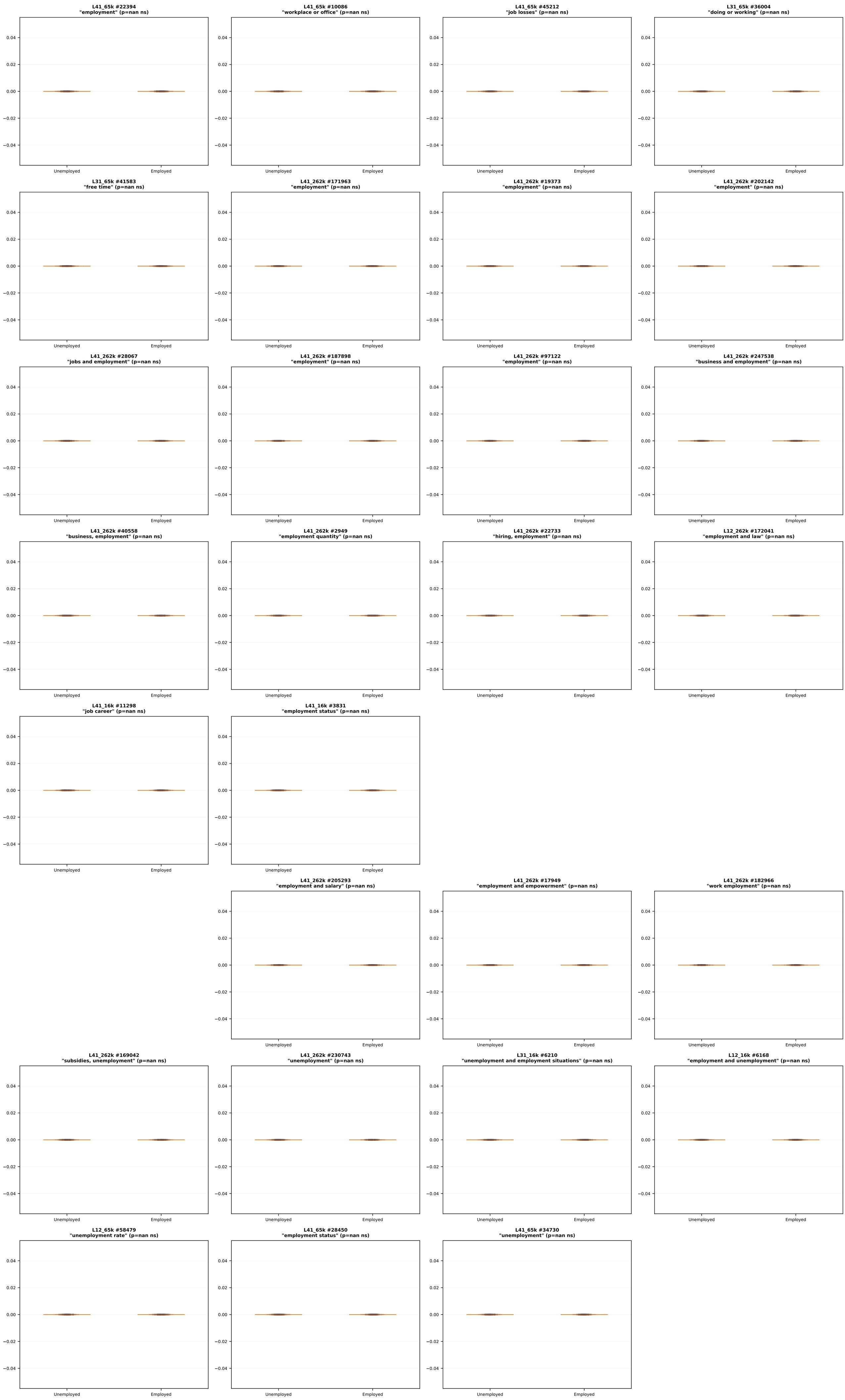
Employed vs Unemployed — SAE Feature Summary | v8 / cfg5 (geo=hash_day=orig_prompt=none)

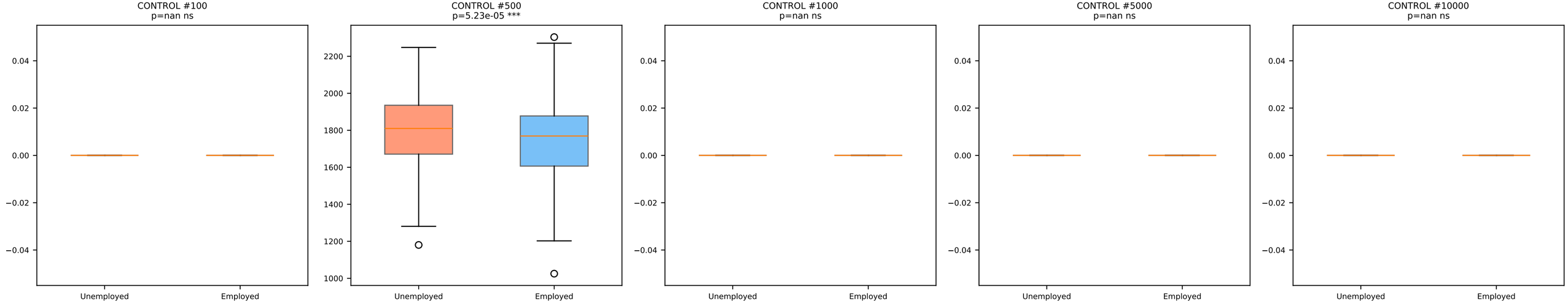
Layer	Width	Feat#	Label	mean_c0	mean_c1	d	p-val	sig	act%_c0	act%_c1
L 41	65k	22394	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10086	workplace or office	5.3758	3.6250	-0.028	6.63e-01	ns	0.6%	0.4%
L 41	65k	45212	job losses	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	36004	doing or working	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	41583	free time	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	171963	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	19373	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	202142	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	28067	jobs and employment	7.1186	16.1308	+0.086	1.82e-01	ns	0.8%	1.7%
L 41	262k	187898	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	97122	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	247538	business and employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	40558	business, employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	2949	employment quantity	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	22733	hiring, employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	262k	172041	employment and law	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	16k	11298	job career	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	16k	3831	employment status	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	205293	employment and salary	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	17949	employment and empowerment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	182966	work employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	169042	subsidies, unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	230743	unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	16k	6210	unemployment and employment situations	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	16k	6168	employment and unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	65k	58479	unemployment rate	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	28450	employment status	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	34730	unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	100	CONTROL_100	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	500	CONTROL_500	1494.8481	1520.8123	+0.062	3.36e-01	ns	99.2%	99.8%
L 41	65k	1000	CONTROL_1000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	5000	CONTROL_5000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10000	CONTROL_10000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%

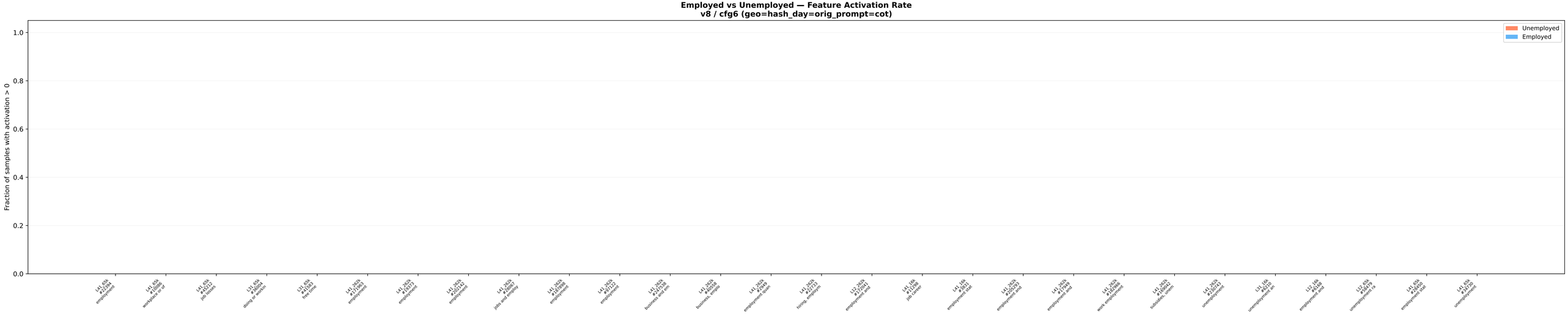
c0 = Unemployed, c1 = Employed
Cohen's d: |d|>0.2 small, |d|>0.5 medium, |d|>0.8 large
act% = fraction of samples where feature activation > 0

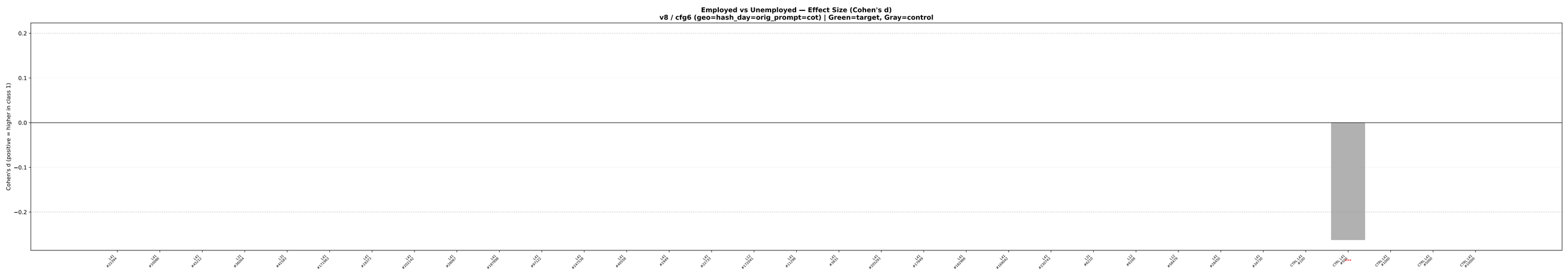
Significant target features: 0/28
Significant control features: 0/5 (should be ~0)

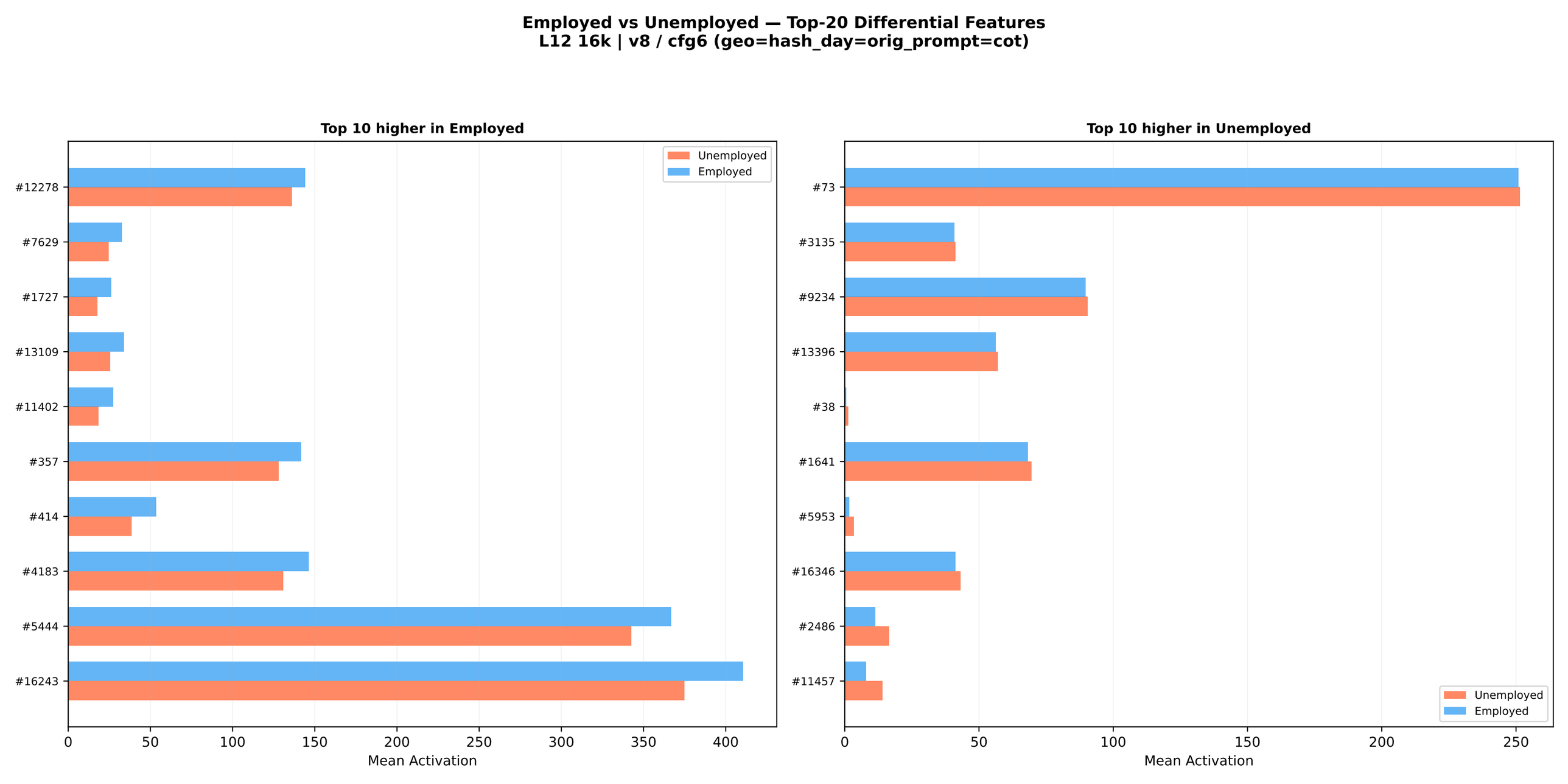
Employed vs Unemployed — SAE Feature Activations
v8 / cfg6 (geo=hash_day=orig_prompt=cot) | Target features (NeuronPedia)

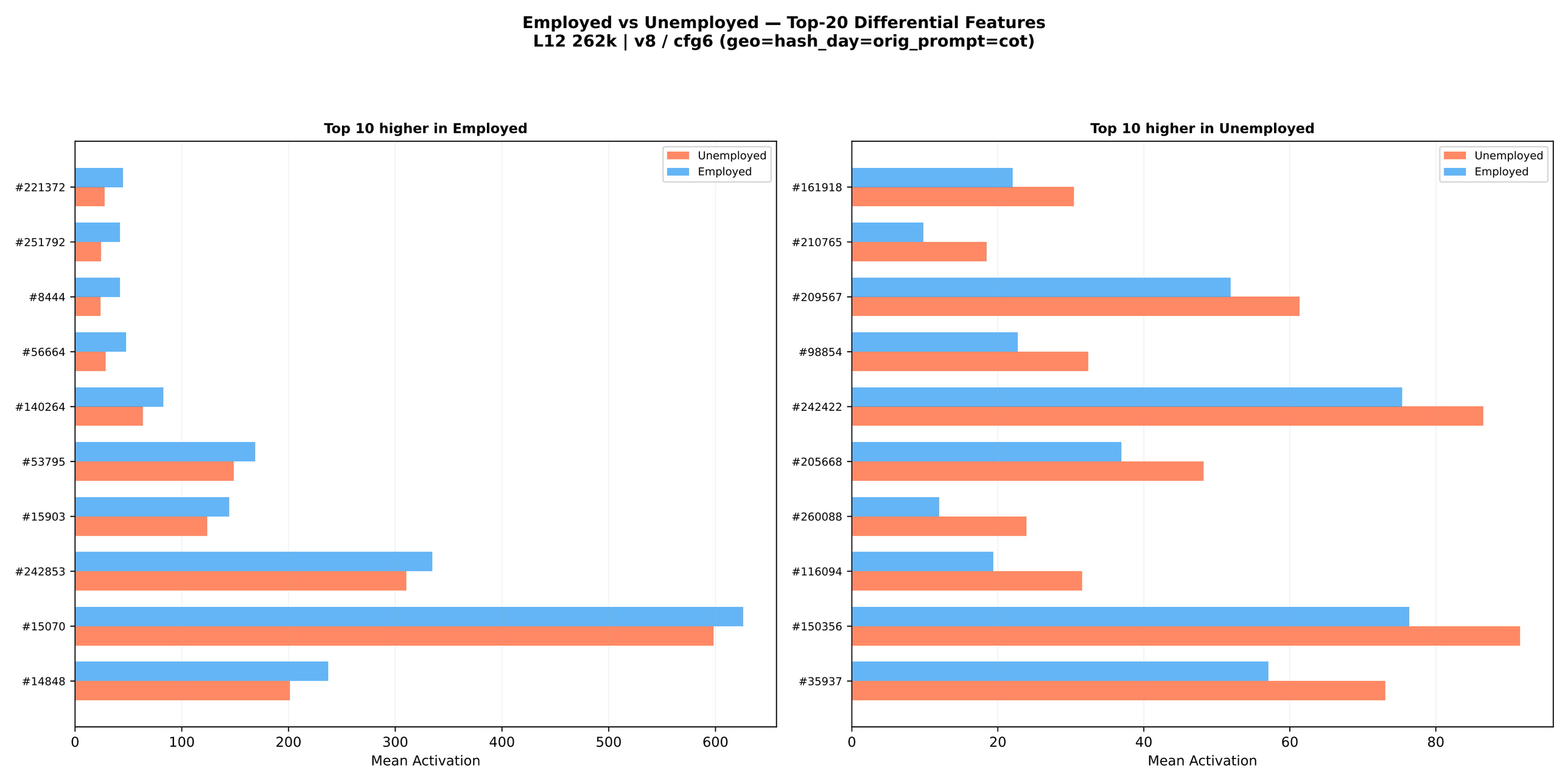


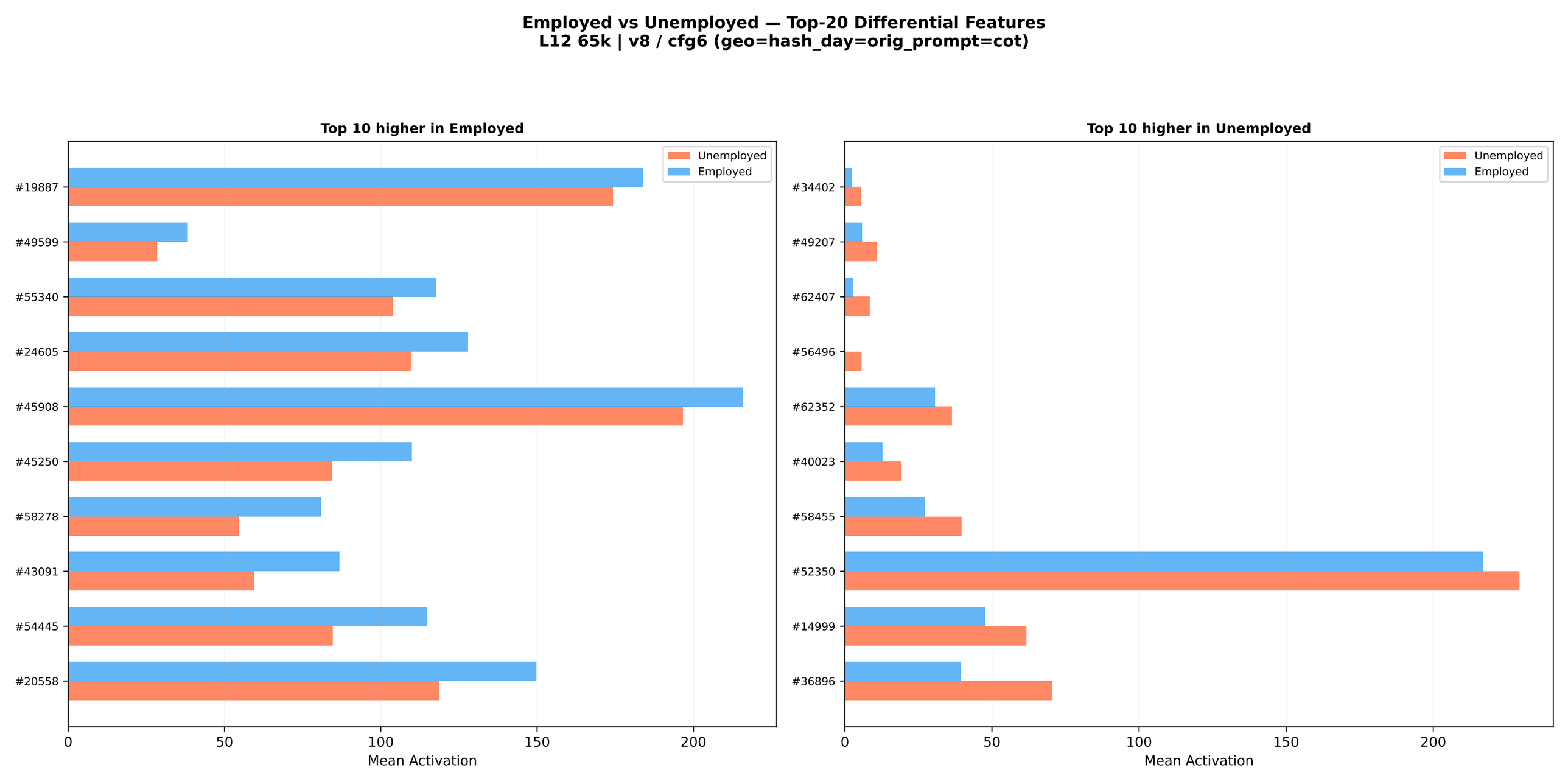

Employed vs Unemployed — Control Features (should be ns)
v8 / cfg6 (geo=hash_day=orig_prompt=cot)
CONTROL #100
p=nan ns
CONTROL #500
p=5.23e-05 ***
CONTROL #1000
p=nan ns
CONTROL #5000
p=nan ns
CONTROL #10000
p=nan ns
Unemployed
Employed

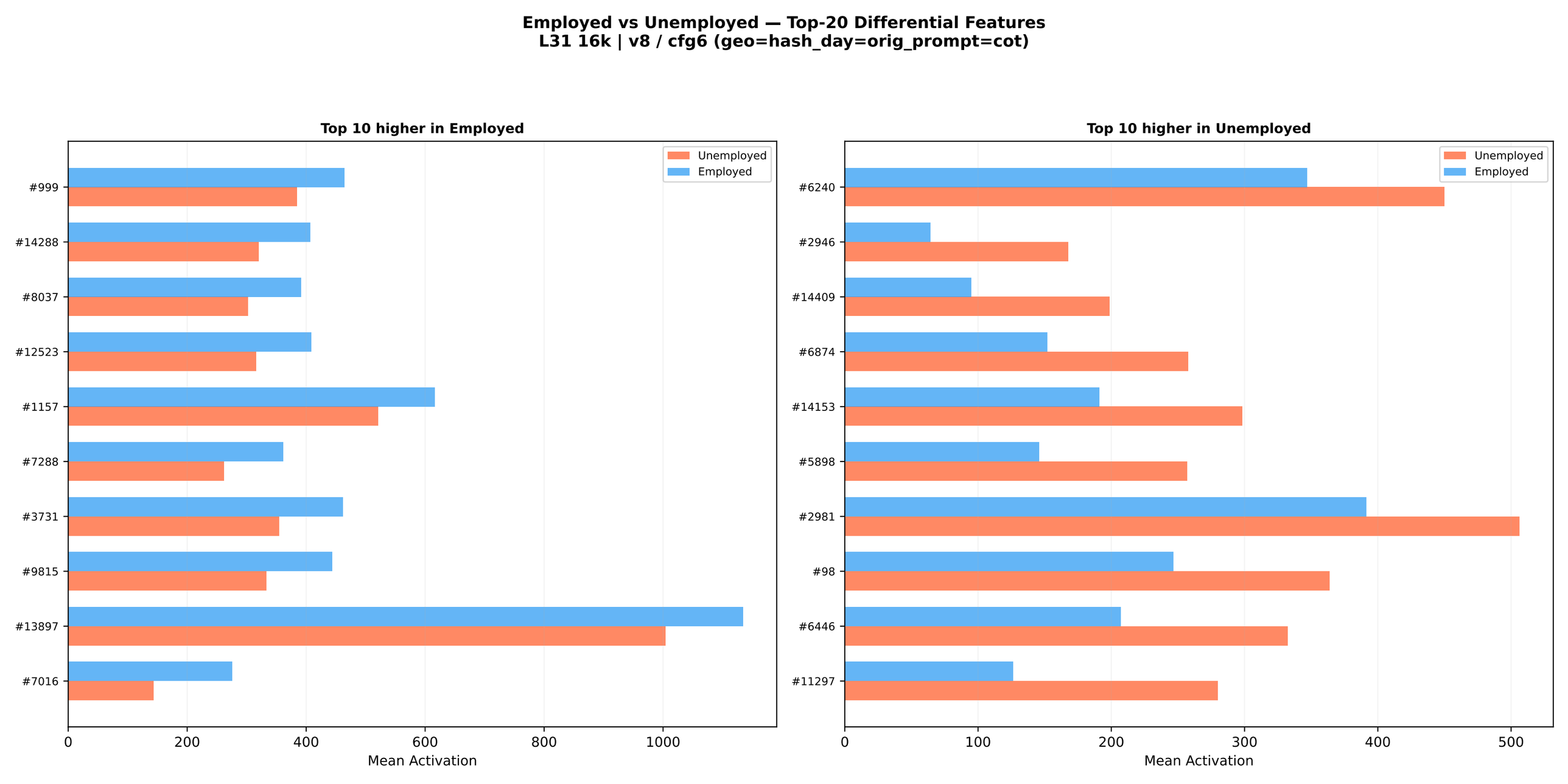

Employed vs Unemployed — Feature Activation Rate
v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Fraction of samples with activation > 0
0.0
0.2
0.4
0.6
0.8
1.0
Unemployed
Employed
L41 65k #22384 employment
L41 65k #10088 workplace or of
L41 65k #4523 job dates
L31 65k #4604 doing_or workin
L31 65k #1589 free-time
L41 -60% #17063 employment
L41 -60% #4373 employment
L41 -60% #2673 employment
L41 -60% #2867 jobs and employ
L41 -60% #18188 employment
L41 -60% #97123 employment
L41 -95% #12538 business asia em
L41 -95% #4658 business, emplo
L41 -60% #2548 employment stain
L41 -60% #2723 hiring, employm
L17 -60% #17043 employment asia
L41 18k #1048 job centre
L41 18k #1861 employment stat
L41 -60% #30503 employment and
L41 -95% #2769 employment and
L41 -95% #18088 work employment
L41 -95% #3466 subsidiaries, employ
L41 -60% #30729 unemployment
L31 18k #4019 unemployment in
L12 18k #3148 employment asia
L12 65k #8478 unemployment in
L41 65k #4503 employment stat
L41 65k #14710 unemployment

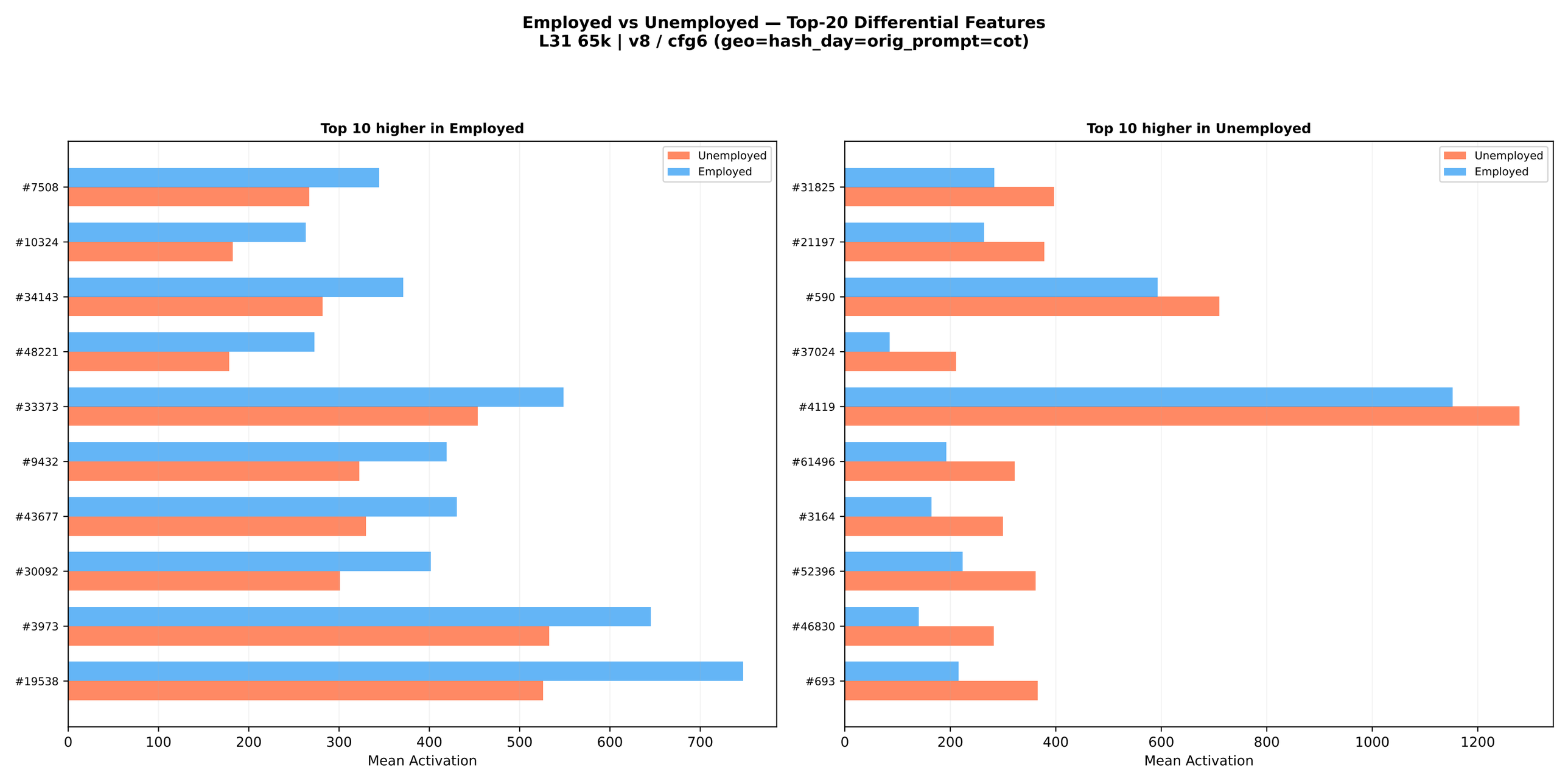

Employed vs Unemployed — Effect Size (Cohen's d)
v8 / cfg6 (geo=hash_day=orig_prompt=cot) | Green=target, Gray=control
Cohen's d (positive = higher in class 1)
0.2
0.1
0.0
−0.1
−0.2
L0 #2084
L0 #3988
L0 #4512
L0 #3804
L0 #4583
L0 #11963
L0 #9573
L0 #2042
L0 #9867
L0 #10988
L0 #9122
L0 #2438
L0 #4858
L0 #349
L0 #2133
L12 #12041
L0 #1198
L0 #11
L0 #2093
L0 #1989
L0 #10648
L0 #10982
L0 #2093
L0 #630
L0 #638
L0 #8679
L0 #2040
L0 #630
CTRL L0 #10
CTRL L0 #10 ***
CTRL L0 #1989
CTRL L0 #2089
CTRL L0 #2089

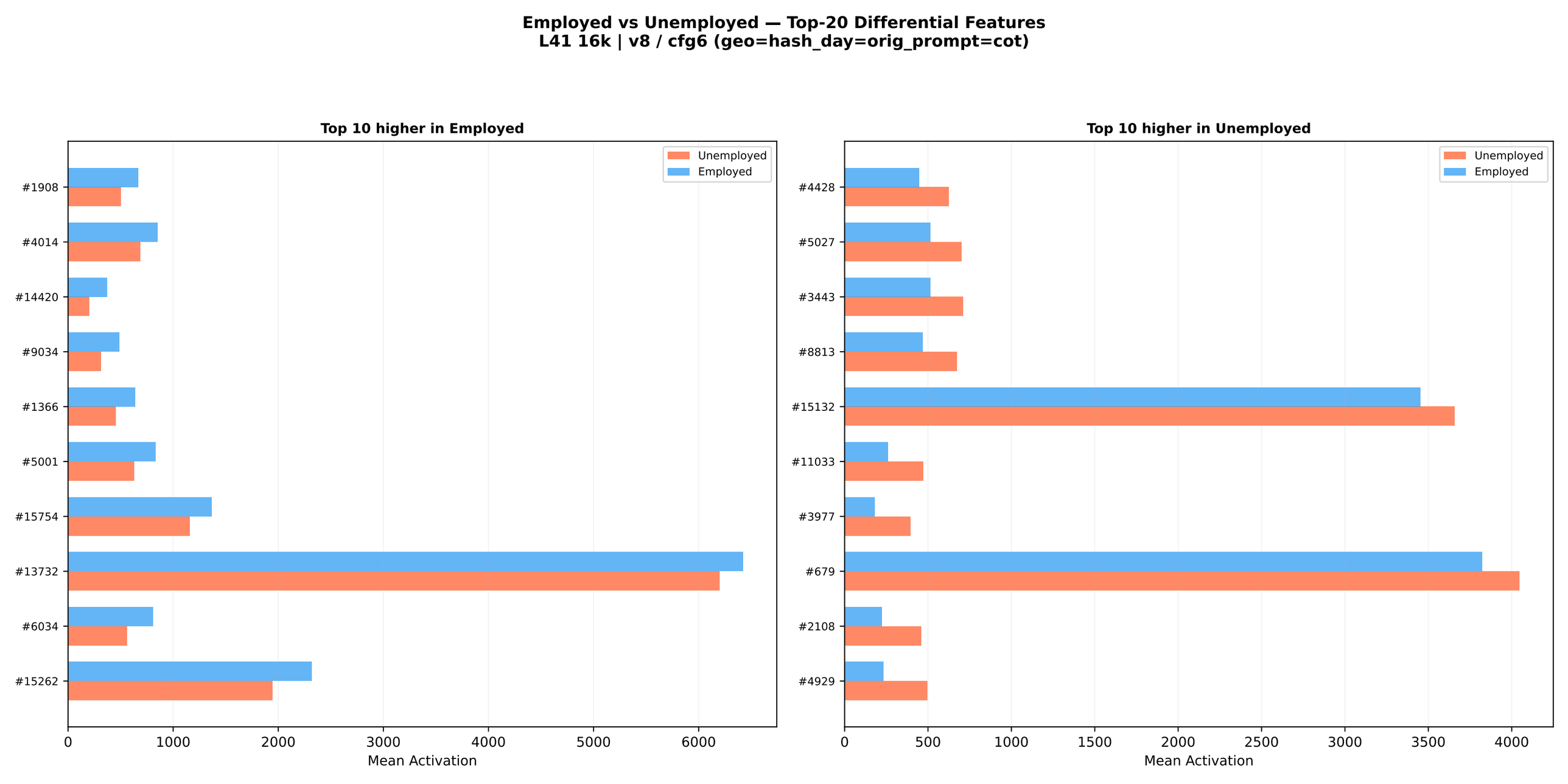

Employed vs Unemployed — Top-20 Differential Features
L12 16k | v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#12278
#7629
#1727
#13109
#11402
#357
#414
#4183
#5444
#16243
0
50
100
150
200
250
300
350
400
Mean Activation
Top 10 higher in Unemployed
#73
#3135
#9234
#13396
#38
#1641
#5953
#16346
#2486
#11457
0
50
100
150
200
250
Mean Activation

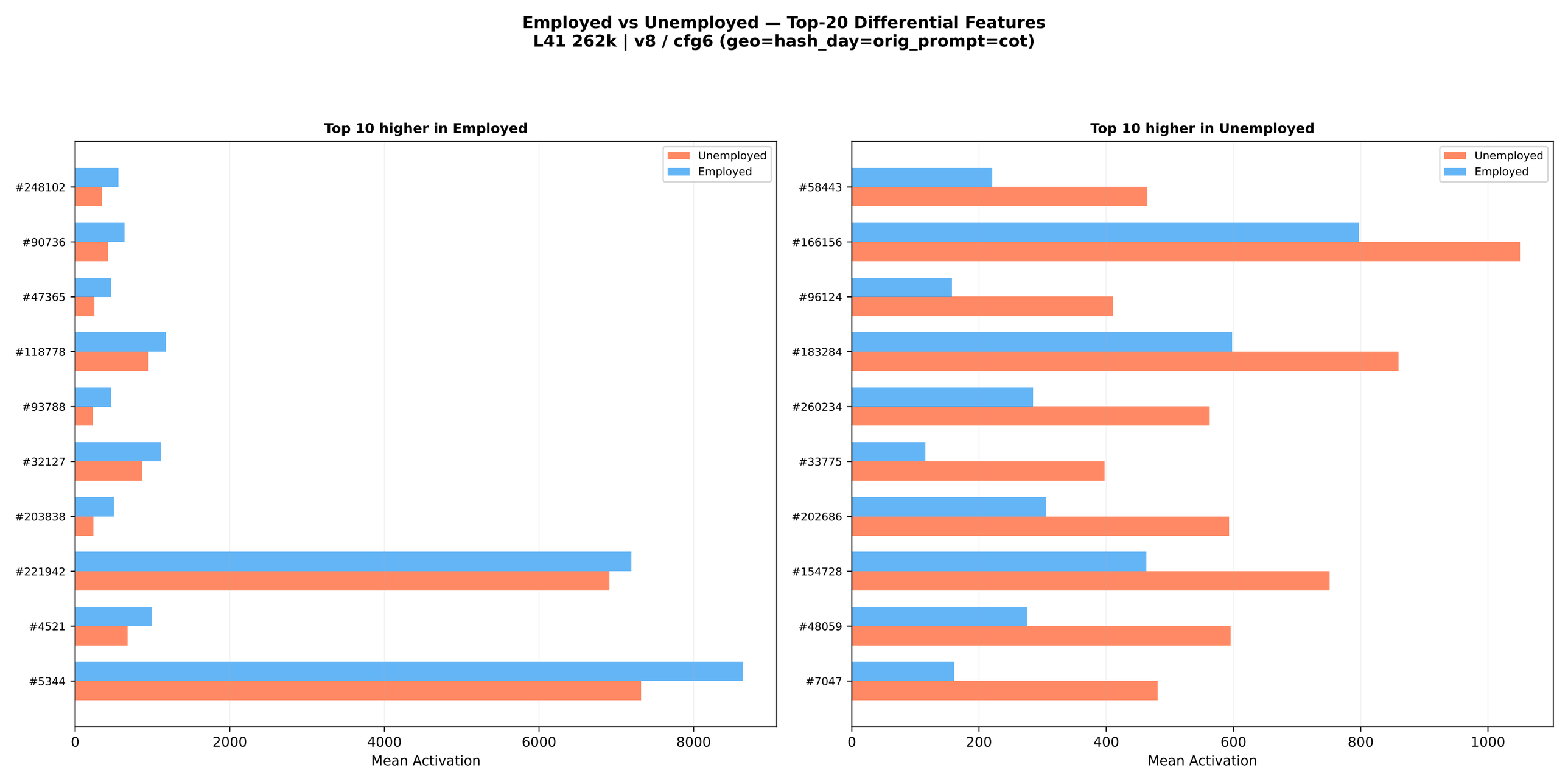

Employed vs Unemployed — Top-20 Differential Features
L12 262k | v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#221372
#251792
#8444
#56664
#140264
#53795
#15903
#242853
#15070
#14848
0
100
200
300
400
500
600
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#161918
#210765
#209567
#98854
#242422
#205668
#260088
#116094
#150356
#35937
0
20
40
60
80
Mean Activation

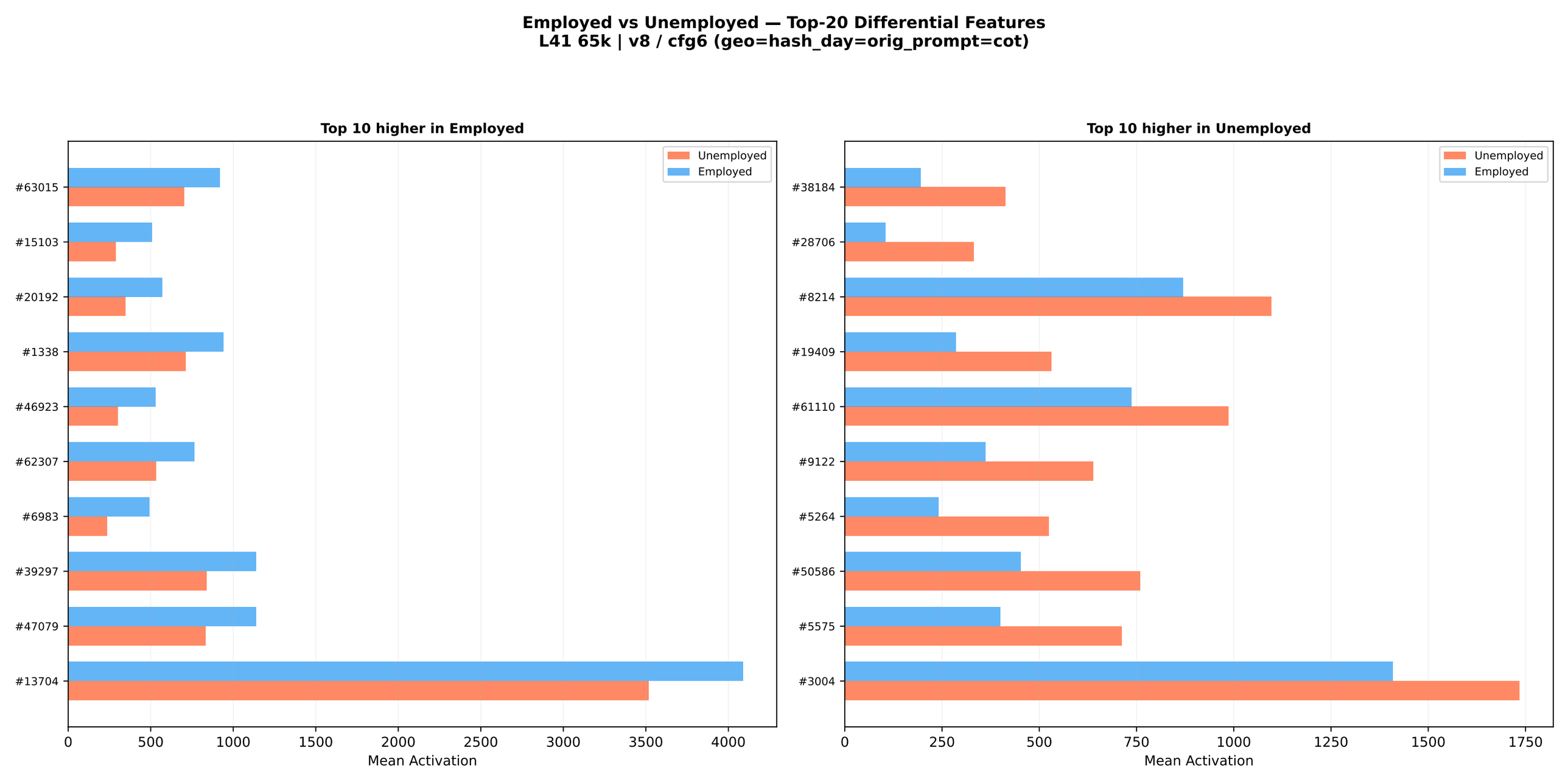

Employed vs Unemployed — Top-20 Differential Features
L12 65k | v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#19887
#49599
#55340
#24605
#45908
#45250
#58278
#43091
#54445
#20558
0
50
100
150
200
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#34402
#49207
#62407
#56496
#62352
#40023
#58455
#52350
#14999
#36896
0
50
100
150
200
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L31 16k | v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#999
#14288
#8037
#12523
#1157
#7288
#3731
#9815
#13897
#7016
0
200
400
600
800
1000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#6240
#2946
#14409
#6874
#14153
#5898
#2981
#98
#6446
#11297
0
100
200
300
400
500
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L31 65k | v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#7508
#10324
#34143
#48221
#33373
#9432
#43677
#30092
#3973
#19538
0
100
200
300
400
500
600
700
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#31825
#21197
#590
#37024
#4119
#61496
#3164
#52396
#46830
#693
0
200
400
600
800
1000
1200
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 16k | v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#1908
#4014
#14420
#9034
#1366
#5001
#15754
#13732
#6034
#15262
0
1000
2000
3000
4000
5000
6000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#4428
#5027
#3443
#8813
#15132
#11033
#3977
#679
#2108
#4929
0
500
1000
1500
2000
2500
3000
3500
4000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 262k | v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#248102
#90736
#47365
#118778
#93788
#32127
#203838
#221942
#4521
#5344
0
2000
4000
6000
8000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#58443
#166156
#96124
#183284
#260234
#33775
#202686
#154728
#48059
#7047
0
200
400
600
800
1000
Mean Activation


Employed vs Unemployed — Top-20 Differential Features
L41 65k | v8 / cfg6 (geo=hash_day=orig_prompt=cot)
Top 10 higher in Employed
Unemployed
Employed
#63015
#15103
#20192
#1338
#46923
#62307
#6983
#39297
#47079
#13704
0
500
1000
1500
2000
2500
3000
3500
4000
Mean Activation
Top 10 higher in Unemployed
Unemployed
Employed
#38184
#28706
#8214
#19409
#61110
#9122
#5264
#50586
#5575
#3004
0
250
500
750
1000
1250
1500
1750
Mean Activation

Summary

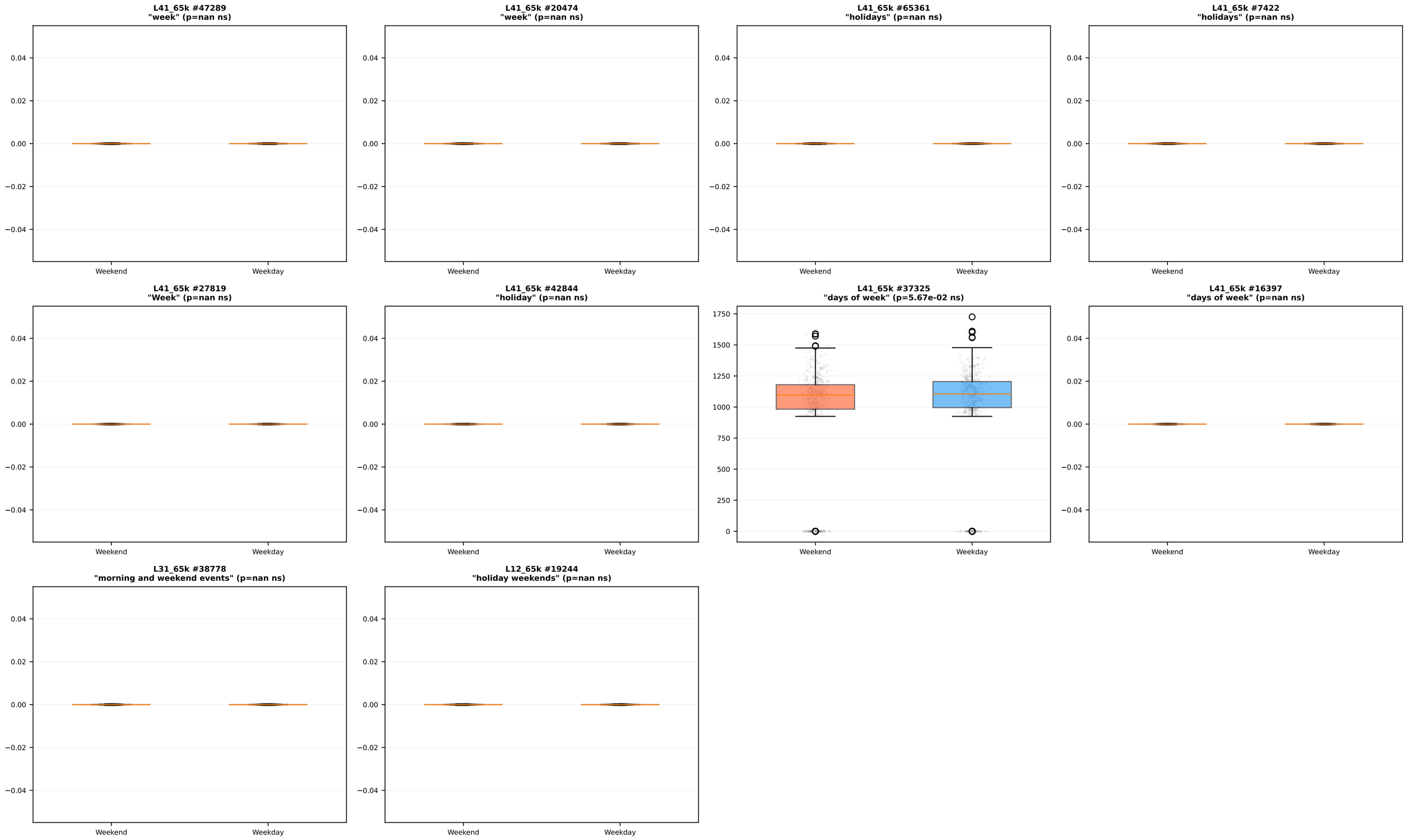
Employed vs Unemployed — SAE Feature Summary | v8 / cfg6 (geo=hash_day=orig_prompt=cot)

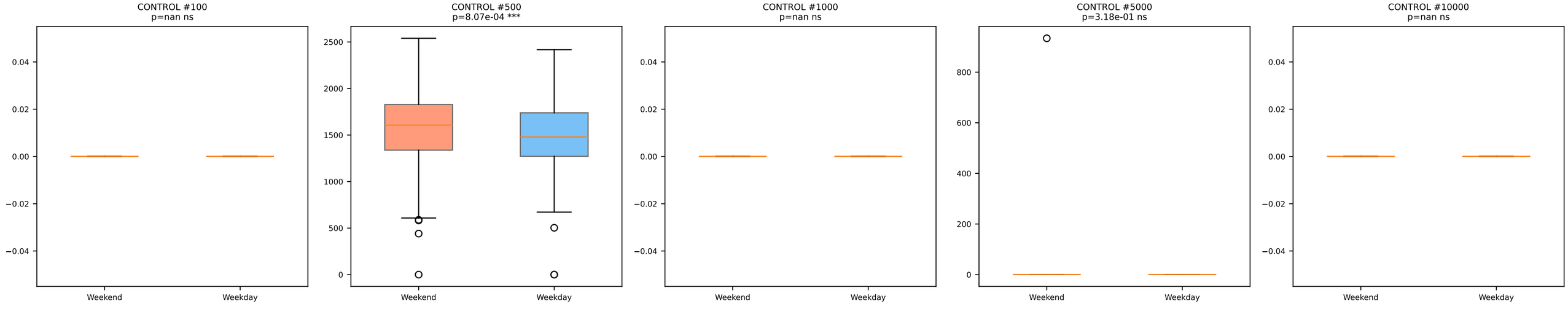
Layer	Width	Feat#	Label	mean_c0	mean_c1	d	p-val	sig	act%_c0	act%_c1
L 41	65k	22394	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10086	workplace or office	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	45212	job losses	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	36004	doing or working	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	41583	free time	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	171963	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	19373	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	202142	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	28067	jobs and employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	187898	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	97122	employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	247538	business and employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	40558	business, employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	2949	employment quantity	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	22733	hiring, employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	262k	172041	employment and law	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	16k	11298	job career	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	16k	3831	employment status	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	205293	employment and salary	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	17949	employment and empowerment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	182966	work employment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	169042	subsidies, unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	262k	230743	unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	16k	6210	unemployment and employment situations	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	16k	6168	employment and unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	65k	58479	unemployment rate	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	28450	employment status	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	34730	unemployment	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	100	CONTROL_100	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	500	CONTROL_500	1798.2026	1747.9763	-0.263	5.23e-05	***	100.0%	100.0%
L 41	65k	1000	CONTROL_1000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	5000	CONTROL_5000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10000	CONTROL_10000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%

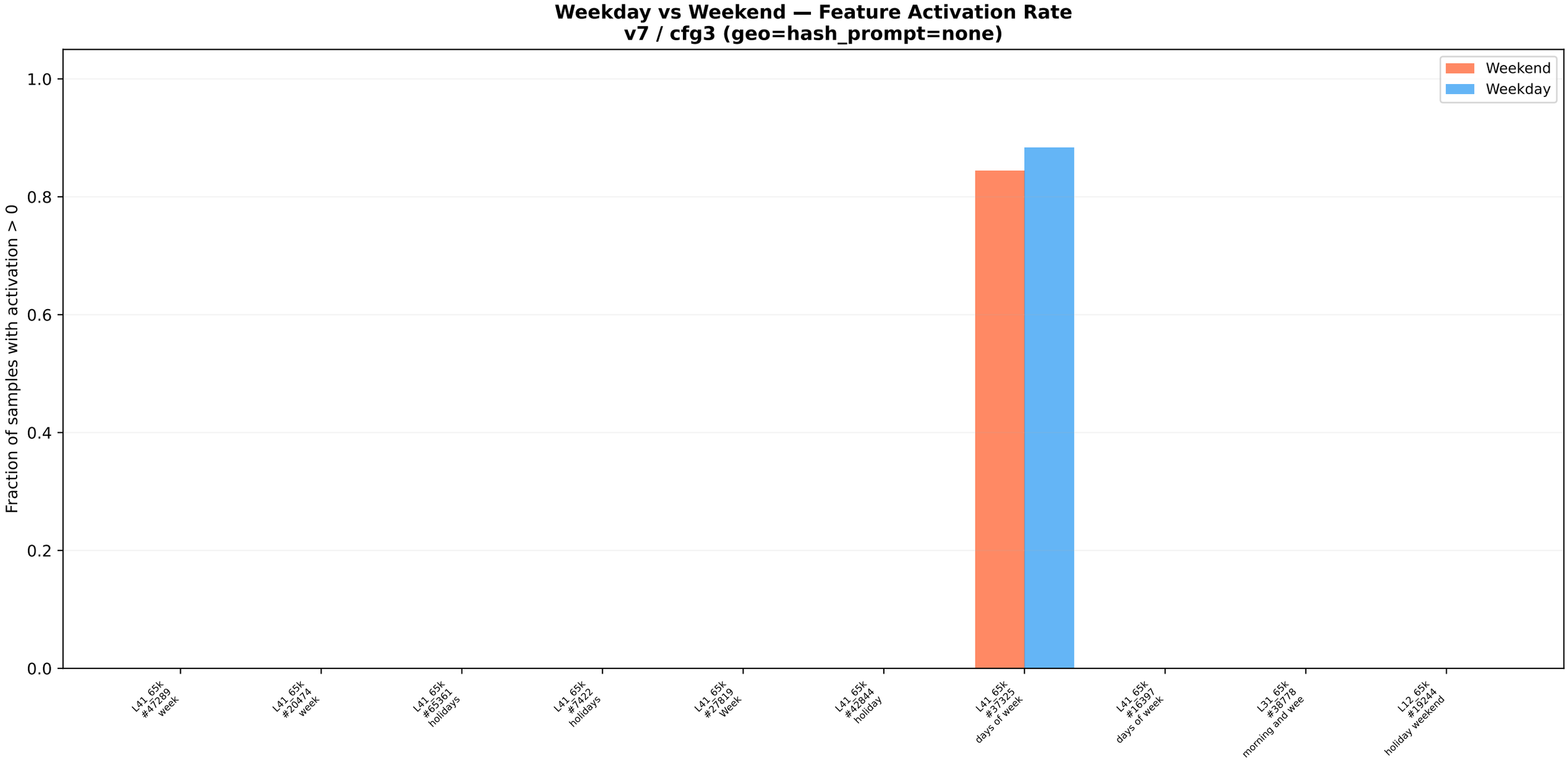
c0 = Unemployed, c1 = Employed
Cohen's d: |d|>0.2 small, |d|>0.5 medium, |d|>0.8 large
act% = fraction of samples where feature activation > 0

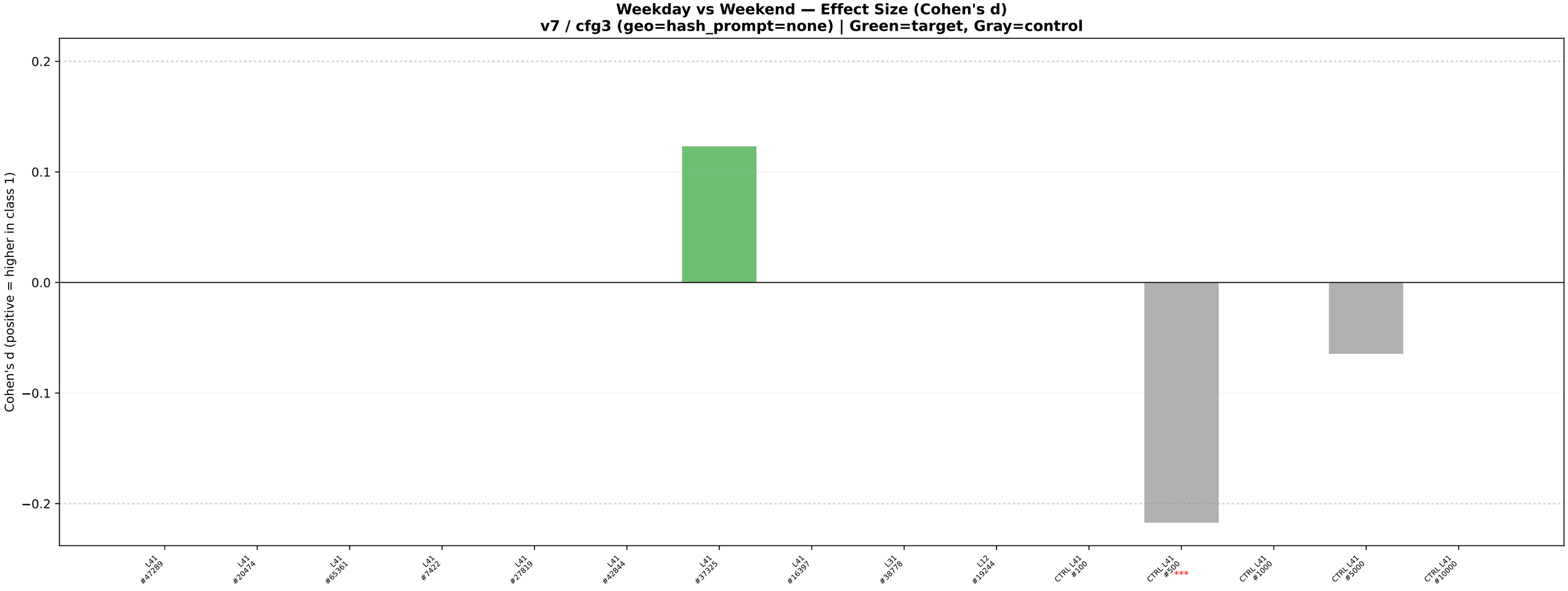
Significant target features: 0/28
Significant control features: 1/5 (should be ~0)

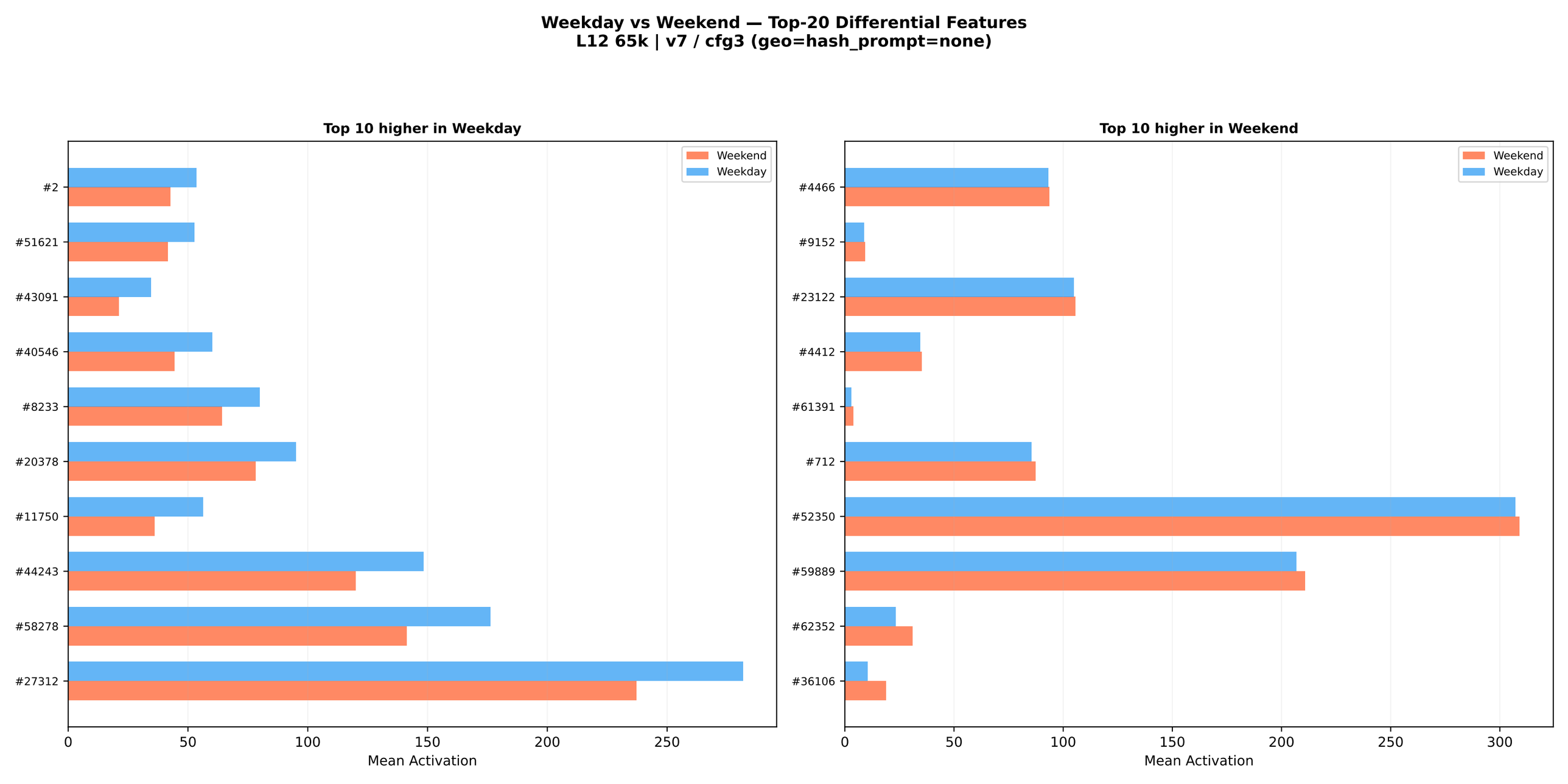
Weekday vs Weekend — SAE Feature Activations
v7 / cfg3 (geo=hash_prompt=none) | Target features (NeuronPedia)

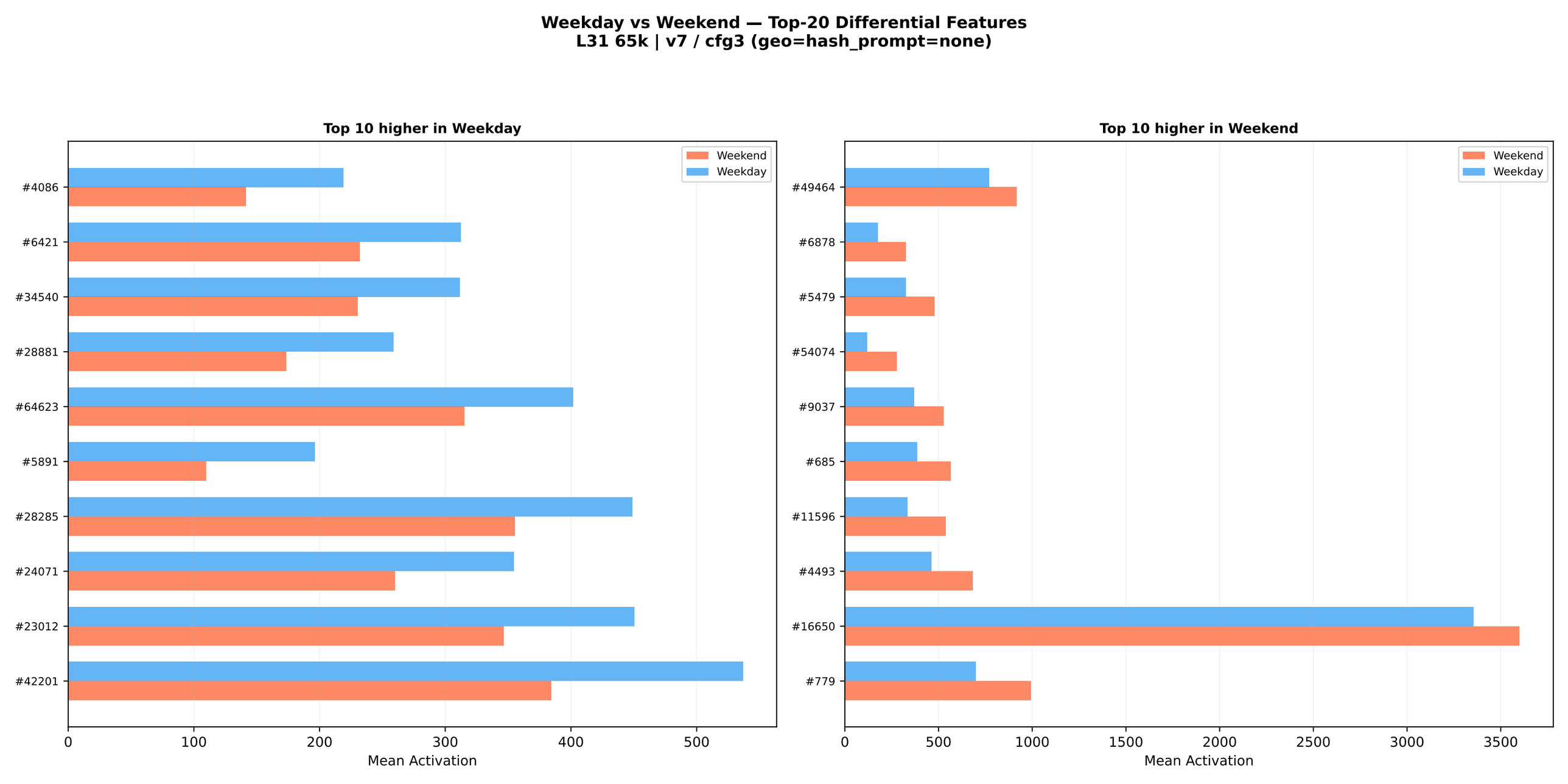


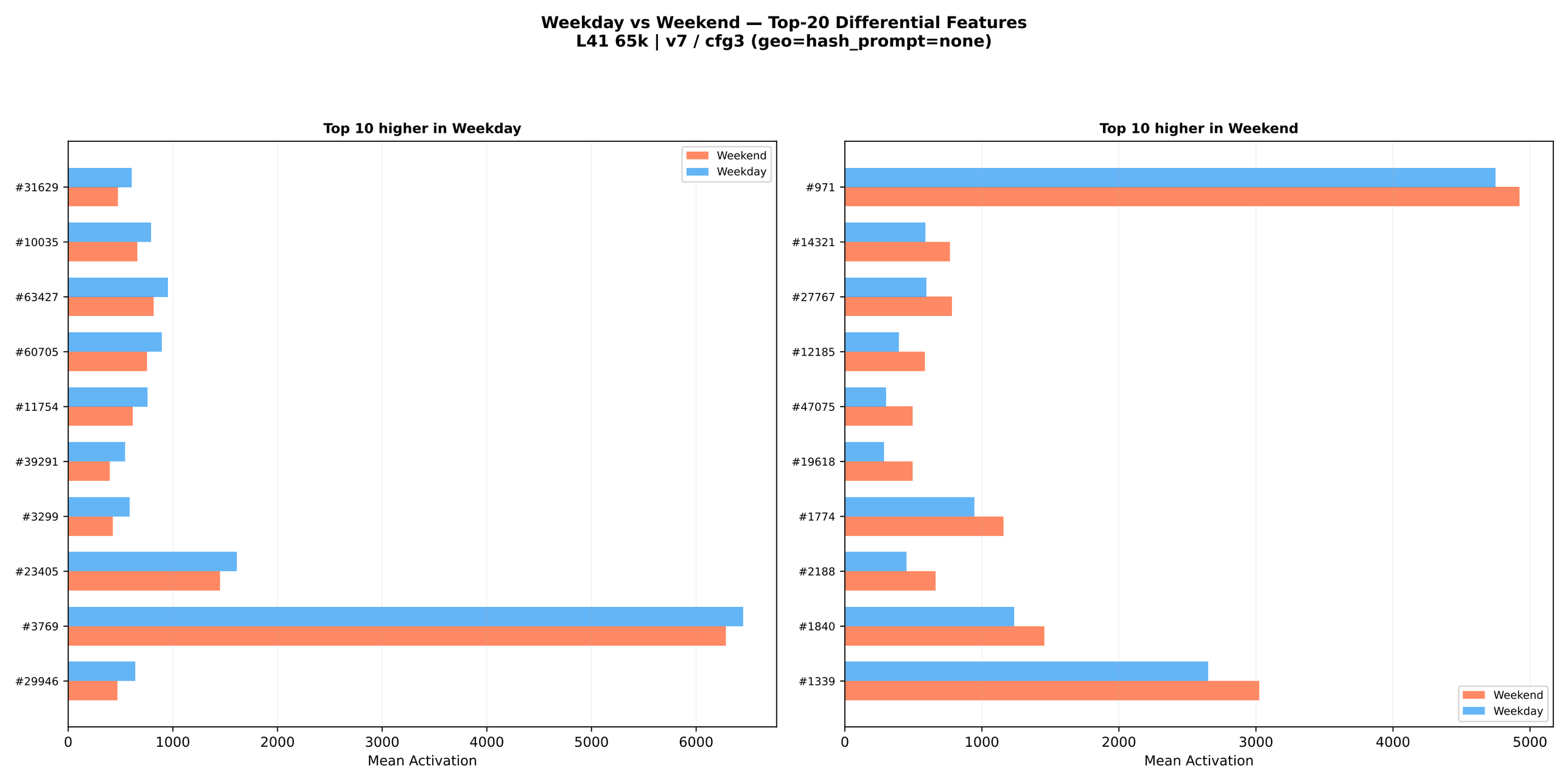

Weekday vs Weekend — Control Features (should be ns)
v7 / cfg3 (geo=hash_prompt=none)
CONTROL #100
p=nan ns
CONTROL #500
p=8.07e-04 ***
CONTROL #1000
p=nan ns
CONTROL #5000
p=3.18e-01 ns
CONTROL #10000
p=nan ns
Weekend
Weekday


Weekday vs Weekend — Feature Activation Rate
v7 / cfg3 (geo=hash_prompt=none)
Fraction of samples with activation > 0
1.0
0.8
0.6
0.4
0.2
0.0
Weekend
Weekday
L41 65k #47289 week
L41 65k #20474 week
L41 65k #65361 holidays
L41 65k #7422 holidays
L41 65k #27819 Week
L41 65k #42844 holiday
L41 65k #37325 days of week
L41 65k #16397 days of week
L31 65k #38778 morning and wee
L12 65k #19244 holiday weekend


Weekday vs Weekend — Effect Size (Cohen's d)
v7 / cfg3 (geo=hash_prompt=none) | Green=target, Gray=control
Cohen's d (positive = higher in class 1)
0.2
0.1
0.0
−0.1
−0.2
L41 #47289
L41 #20474
L41 #65361
L41 #7422
L41 #27819
L41 #42844
L41 #37325
L41 #16397
L31 #38778
L12 #19244
CTRL L41 #100
CTRL L41 #500 ***
CTRL L41 #1000
CTRL L41 #5000
CTRL L41 #10000


Weekday vs Weekend — Top-20 Differential Features
L12 65k | v7 / cfg3 (geo=hash_prompt=none)
Top 10 higher in Weekday
Weekend
Weekday
#2
#51621
#43091
#40546
#8233
#20378
#11750
#44243
#58278
#27312
0
50
100
150
200
250
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#4466
#9152
#23122
#4412
#61391
#712
#52350
#59889
#62352
#36106
0
50
100
150
200
250
300
Mean Activation


Weekday vs Weekend — Top-20 Differential Features
L31 65k | v7 / cfg3 (geo=hash_prompt=none)
Top 10 higher in Weekday
Weekend
Weekday
#4086
#6421
#34540
#28881
#64623
#5891
#28285
#24071
#23012
#42201
0
100
200
300
400
500
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#49464
#6878
#5479
#54074
#9037
#685
#11596
#4493
#16650
#779
0
500
1000
1500
2000
2500
3000
3500
Mean Activation


Weekday vs Weekend — Top-20 Differential Features
L41 65k | v7 / cfg3 (geo=hash_prompt=none)
Top 10 higher in Weekday
Weekend
Weekday
#31629
#10035
#63427
#60705
#11754
#39291
#3299
#23405
#3769
#29946
0
1000
2000
3000
4000
5000
6000
Mean Activation
Top 10 higher in Weekend
#971
#14321
#27767
#12185
#47075
#19618
#1774
#2188
#1840
#1339
0
1000
2000
3000
4000
5000
Mean Activation

Summary

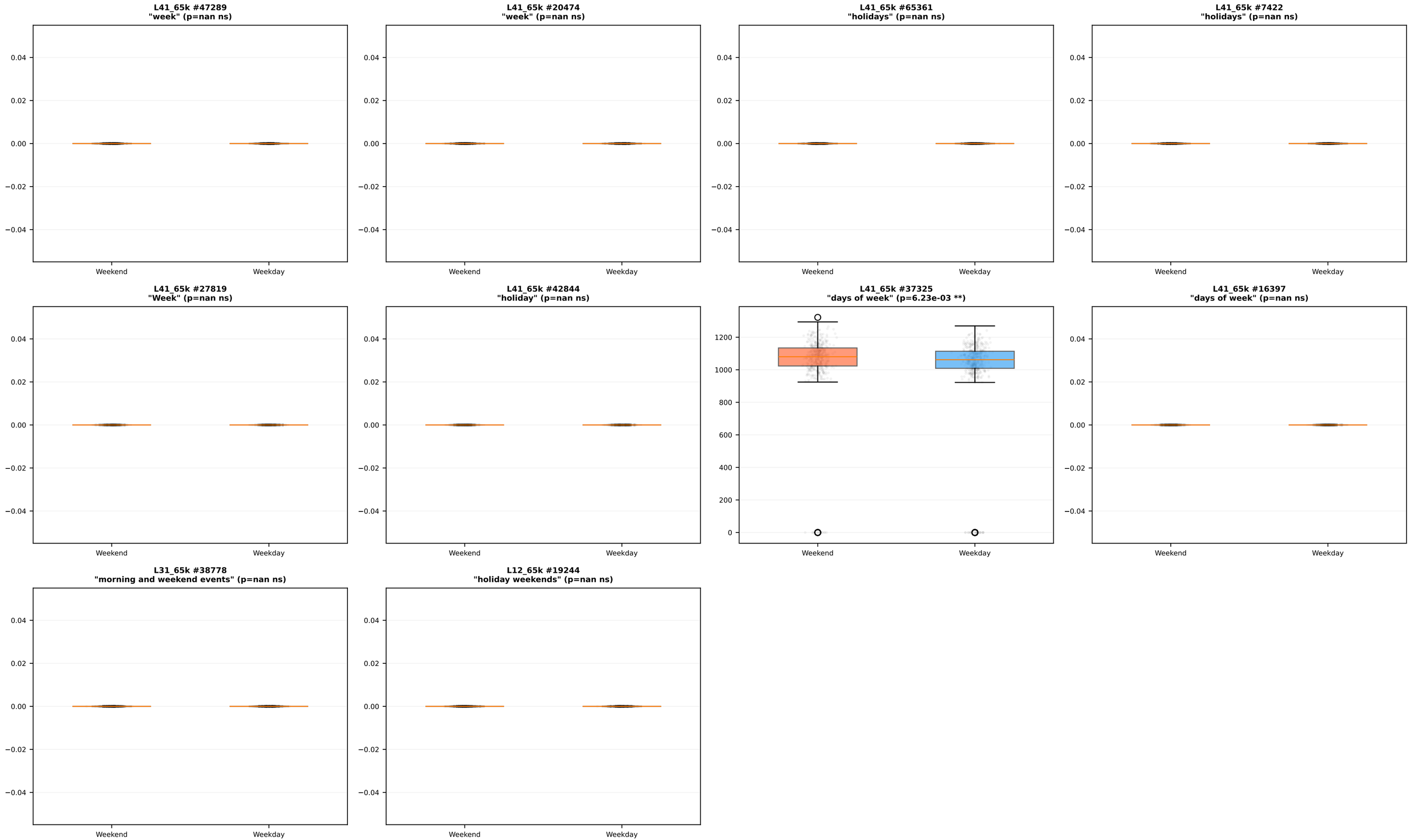
Weekday vs Weekend — SAE Feature Summary | v7 / cfg3 (geo=hash_prompt=none)

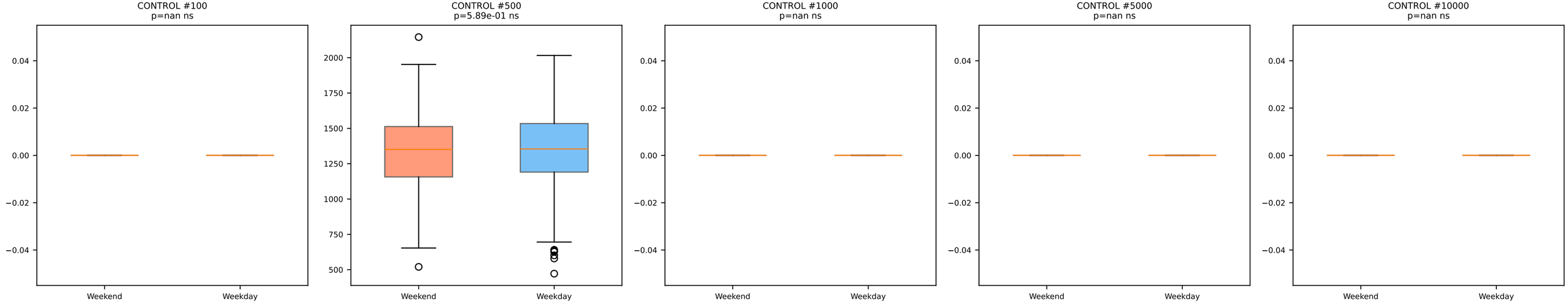
Layer	Width	Feat#	Label	mean_c0	mean_c1	d	p-val	sig	act%_c0	act%_c1
L 41	65k	47289	week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	20474	week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	65361	holidays	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	7422	holidays	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	27819	Week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	42844	holiday	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	37325	days of week	956.4201	1006.6772	+0.123	5.67e-02	ns	84.4%	88.3%
L 41	65k	16397	days of week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	38778	morning and weekend events	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	65k	19244	holiday weekends	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	100	CONTROL_100	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	500	CONTROL_500	1570.3185	1492.7239	-0.217	8.07e-04	***	99.8%	99.6%
L 41	65k	1000	CONTROL_1000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	5000	CONTROL_5000	1.9456	0.0000	-0.065	3.18e-01	ns	0.2%	0.0%
L 41	65k	10000	CONTROL_10000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%

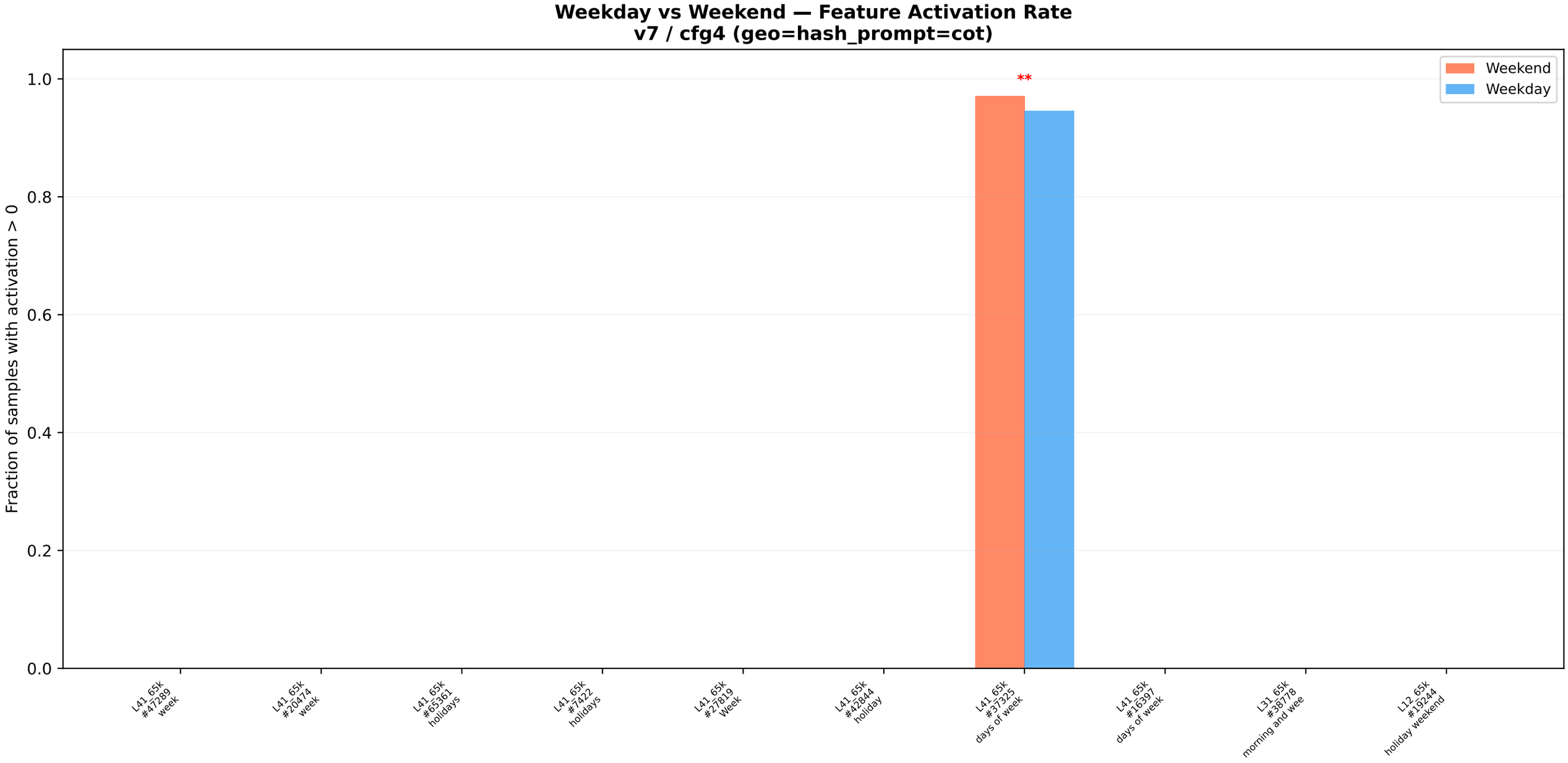
c0 = Weekend, c1 = Weekday
Cohen's d: |d|>0.2 small, |d|>0.5 medium, |d|>0.8 large
act% = fraction of samples where feature activation > 0

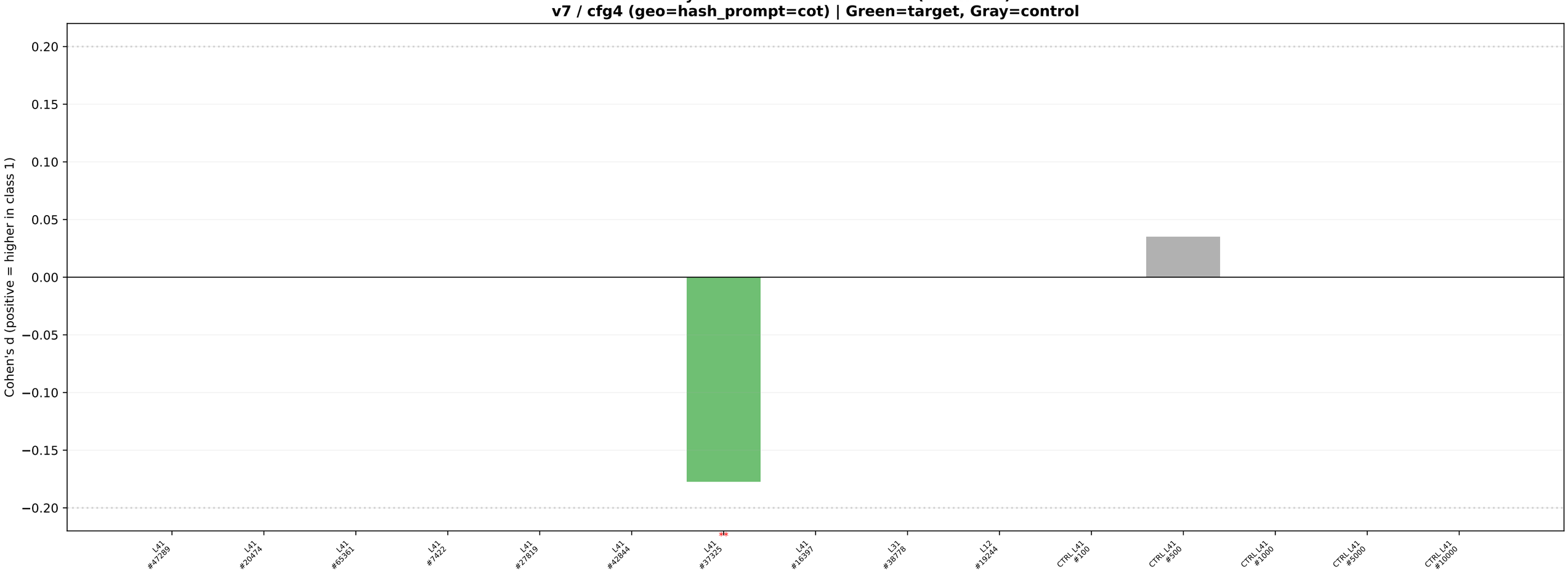
Significant target features: 0/10
Significant control features: 1/5 (should be ~0)

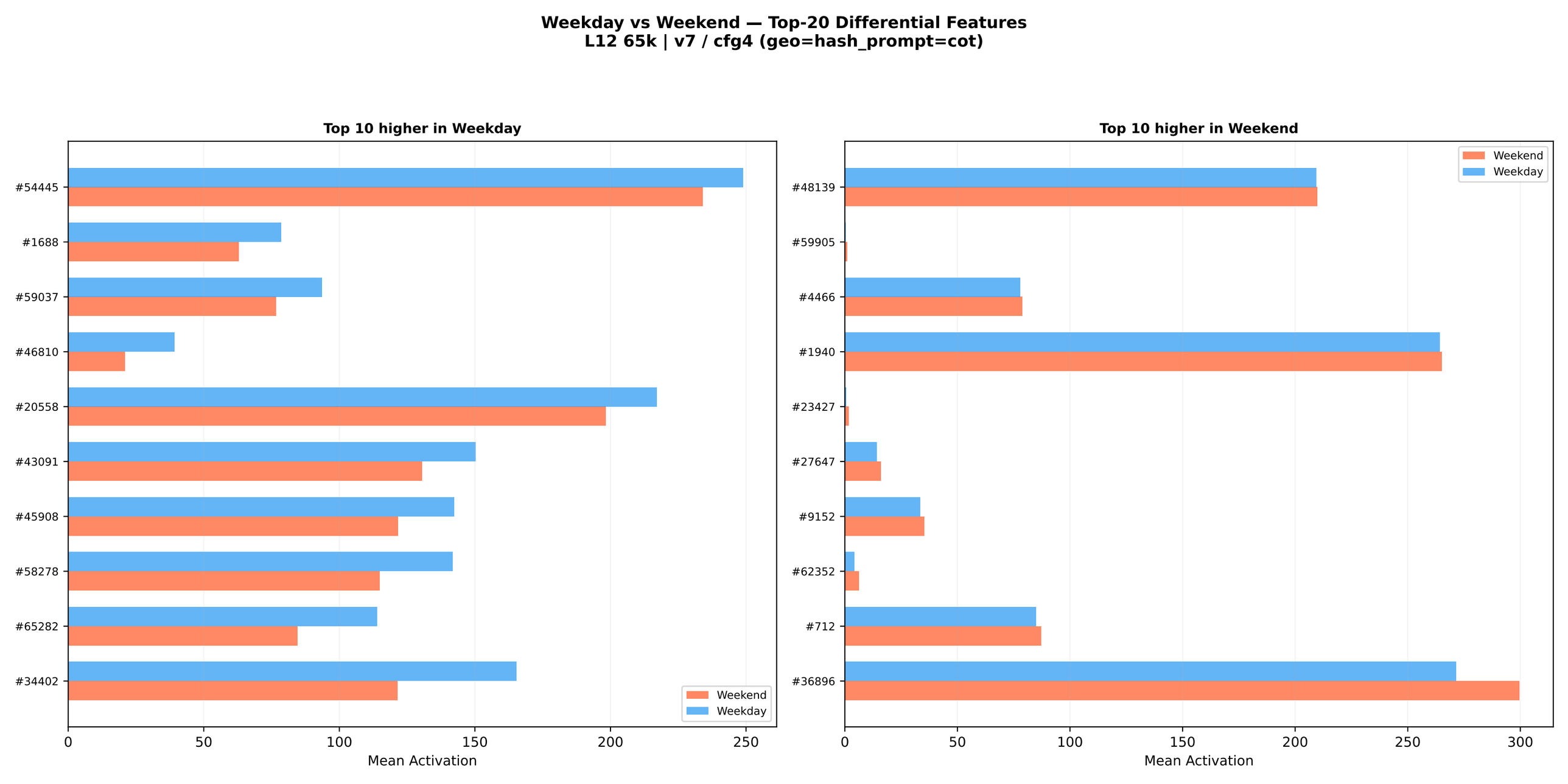
Weekday vs Weekend — SAE Feature Activations
v7 / cfg4 (geo=hash_prompt=cot) | Target features (NeuronPedia)

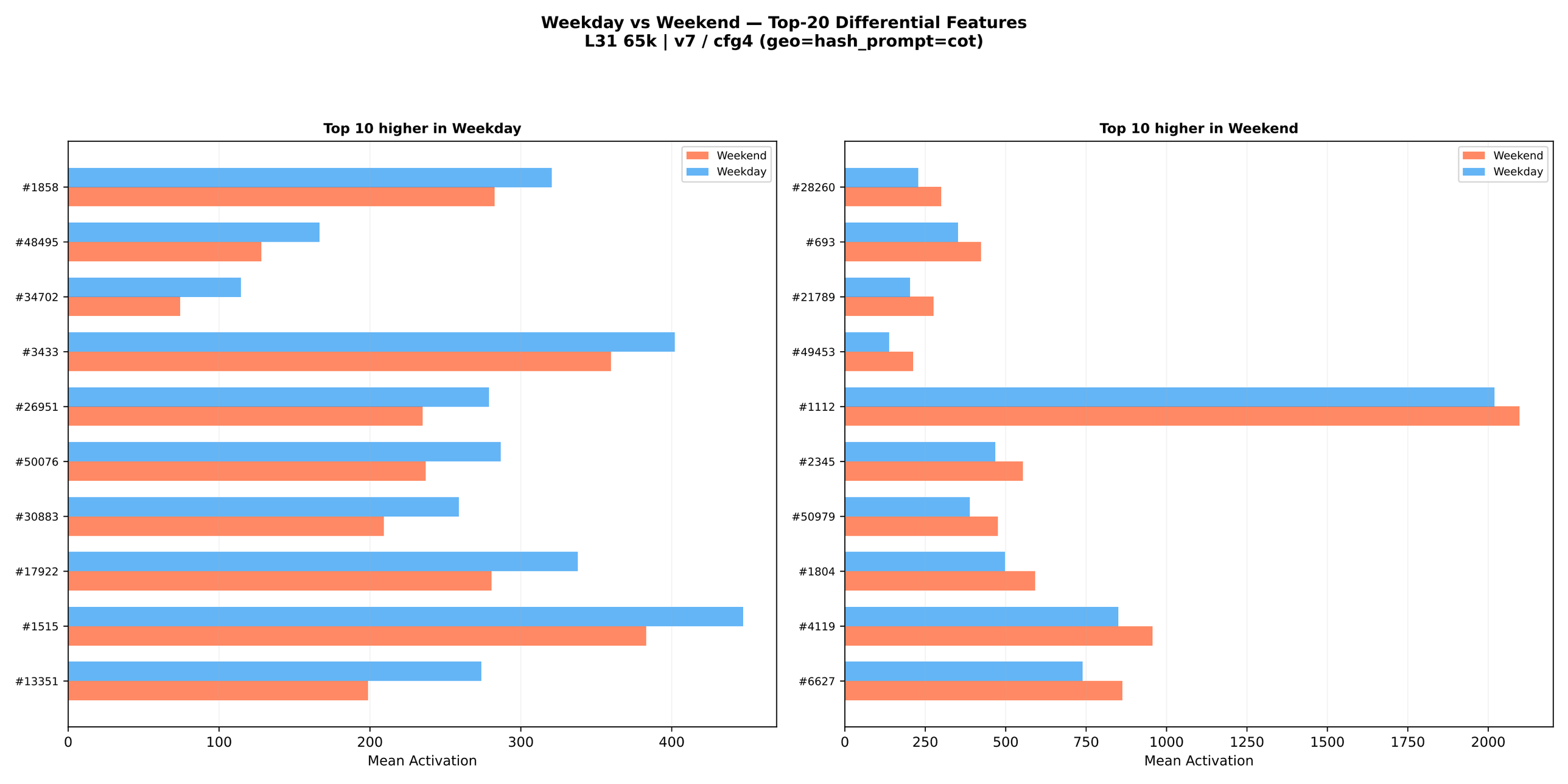


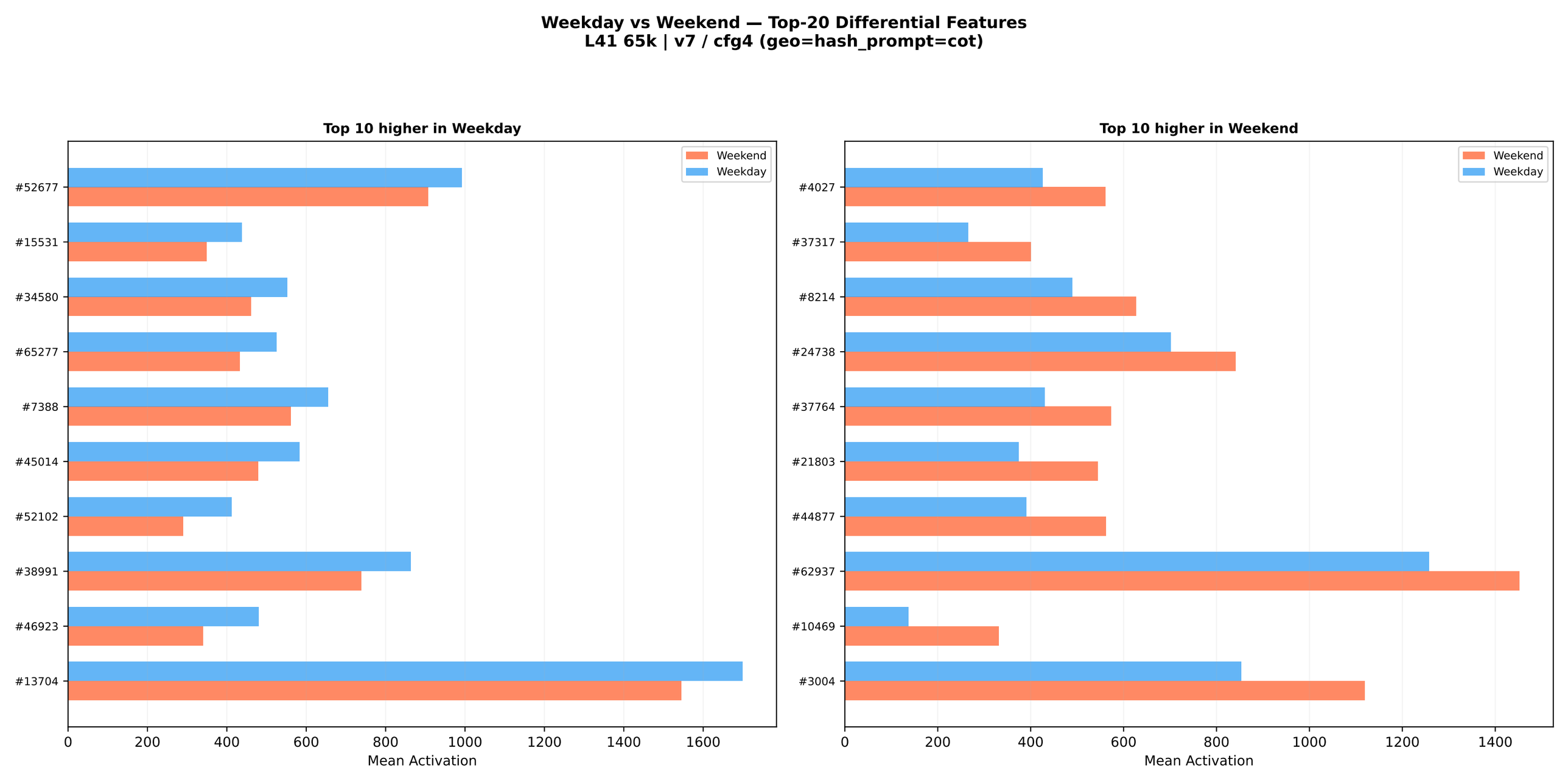

Weekday vs Weekend — Control Features (should be ns)
v7 / cfg4 (geo=hash_prompt=cot)
CONTROL #100
p=nan ns
CONTROL #500
p=5.89e-01 ns
CONTROL #1000
p=nan ns
CONTROL #5000
p=nan ns
CONTROL #10000
p=nan ns
Weekend
Weekday


Weekday vs Weekend — Feature Activation Rate
v7 / cfg4 (geo=hash_prompt=cot)
Fraction of samples with activation > 0
1.0
0.8
0.6
0.4
0.2
0.0
Weekend
Weekday
**
L41 65k #47289 week
L41 65k #20474 week
L41 65k #65361 holidays
L41 65k #7422 holidays
L41 65k #27819 Week
L41 65k #42844 holiday
L41 65k #37325 days of week
L41 65k #16397 days of week
L31 65k #38778 morning and wee
L12 65k #19244 holiday weekend


Weekday vs Weekend — Effect Size (Cohen's d)
v7 / cfg4 (geo=hash_prompt=cot) | Green=target, Gray=control
Cohen's d (positive = higher in class 1)
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
**
L41 #47289
L41 #20474
L41 #65361
L41 #7422
L41 #27819
L41 #42844
L41 #37325
L41 #16397
L31 #38778
L12 #19244
CTRL L41 #100
CTRL L41 #500
CTRL L41 #1000
CTRL L41 #5000
CTRL L41 #10000


Weekday vs Weekend — Top-20 Differential Features
L12 65k | v7 / cfg4 (geo=hash_prompt=cot)
Top 10 higher in Weekday
#54445
#1688
#59037
#46810
#20558
#43091
#45908
#58278
#65282
#34402
Weekend
Weekday
0
50
100
150
200
250
Mean Activation
Top 10 higher in Weekend
#48139
#59905
#4466
#1940
#23427
#27647
#9152
#62352
#712
#36896
Weekend
Weekday
0
50
100
150
200
250
300
Mean Activation


Weekday vs Weekend — Top-20 Differential Features
L31 65k | v7 / cfg4 (geo=hash_prompt=cot)
Top 10 higher in Weekday
Weekend
Weekday
#1858
#48495
#34702
#3433
#26951
#50076
#30883
#17922
#1515
#13351
0
100
200
300
400
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#28260
#693
#21789
#49453
#1112
#2345
#50979
#1804
#4119
#6627
0
250
500
750
1000
1250
1500
1750
2000
Mean Activation


Weekday vs Weekend — Top-20 Differential Features
L41 65k | v7 / cfg4 (geo=hash_prompt=cot)
Top 10 higher in Weekday
Weekend
Weekday
#52677
#15531
#34580
#65277
#7388
#45014
#52102
#38991
#46923
#13704
0
200
400
600
800
1000
1200
1400
1600
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#4027
#37317
#8214
#24738
#37764
#21803
#44877
#62937
#10469
#3004
0
200
400
600
800
1000
1200
1400
Mean Activation

Summary

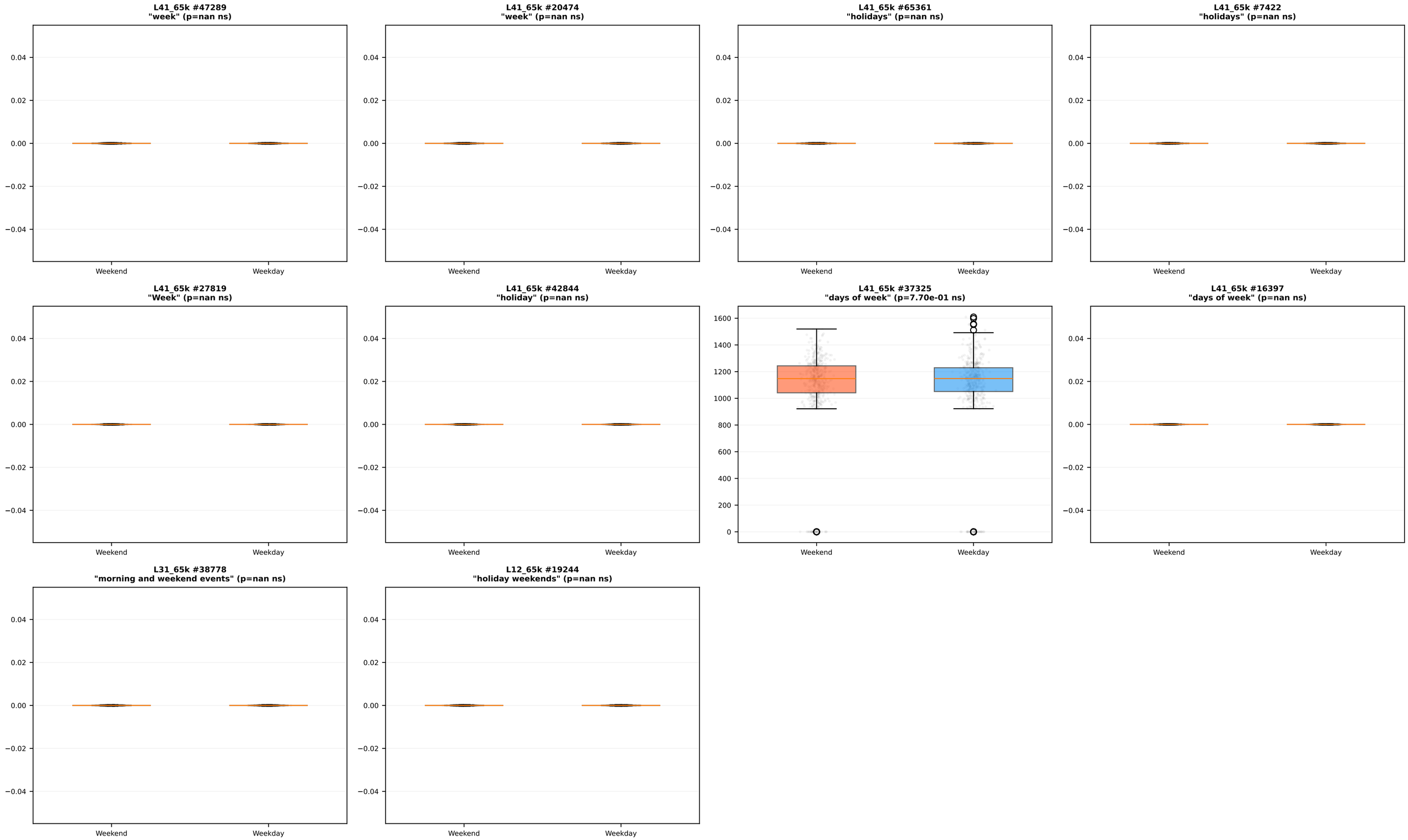
Weekday vs Weekend — SAE Feature Summary | v7 / cfg4 (geo=hash_prompt=cot)

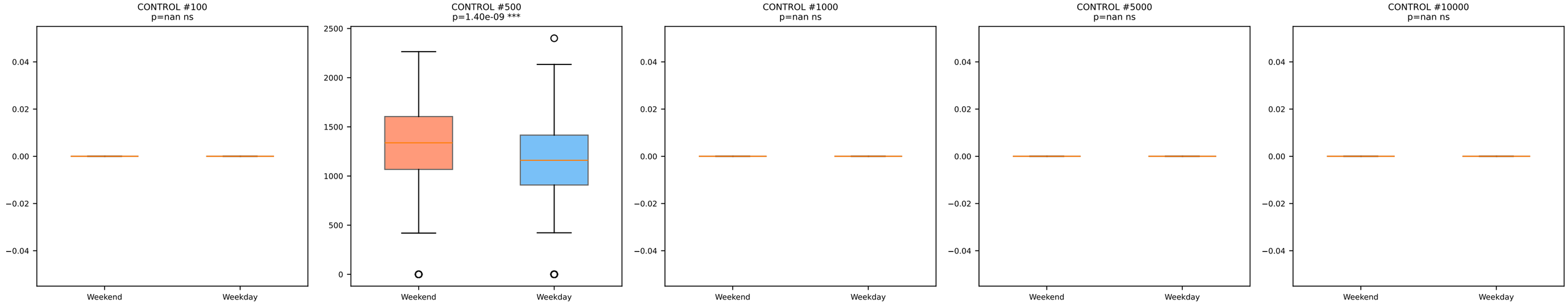
Layer	Width	Feat#	Label	mean_c0	mean_c1	d	p-val	sig	act%_c0	act%_c1
L 41	65k	47289	week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	20474	week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	65361	holidays	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	7422	holidays	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	27819	Week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	42844	holiday	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	37325	days of week	1052.2836	1012.2404	-0.177	6.23e-03	**	97.1%	94.6%
L 41	65k	16397	days of week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	38778	morning and weekend events	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	65k	19244	holiday weekends	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	100	CONTROL_100	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	500	CONTROL_500	1337.3716	1346.2238	+0.035	5.89e-01	ns	100.0%	100.0%
L 41	65k	1000	CONTROL_1000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	5000	CONTROL_5000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10000	CONTROL_10000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%

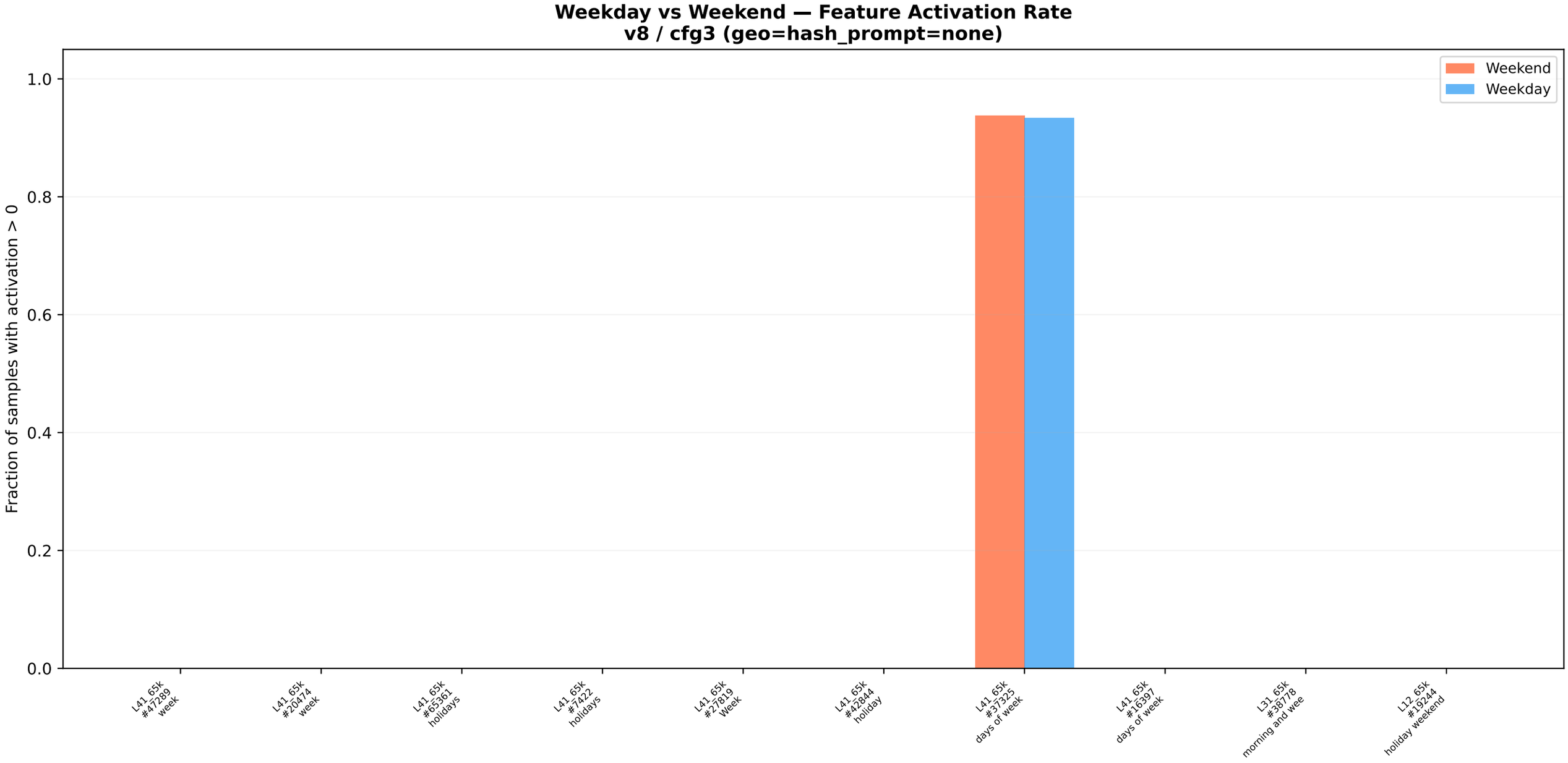
c0 = Weekend, c1 = Weekday
Cohen's d: |d|>0.2 small, |d|>0.5 medium, |d|>0.8 large
act% = fraction of samples where feature activation > 0

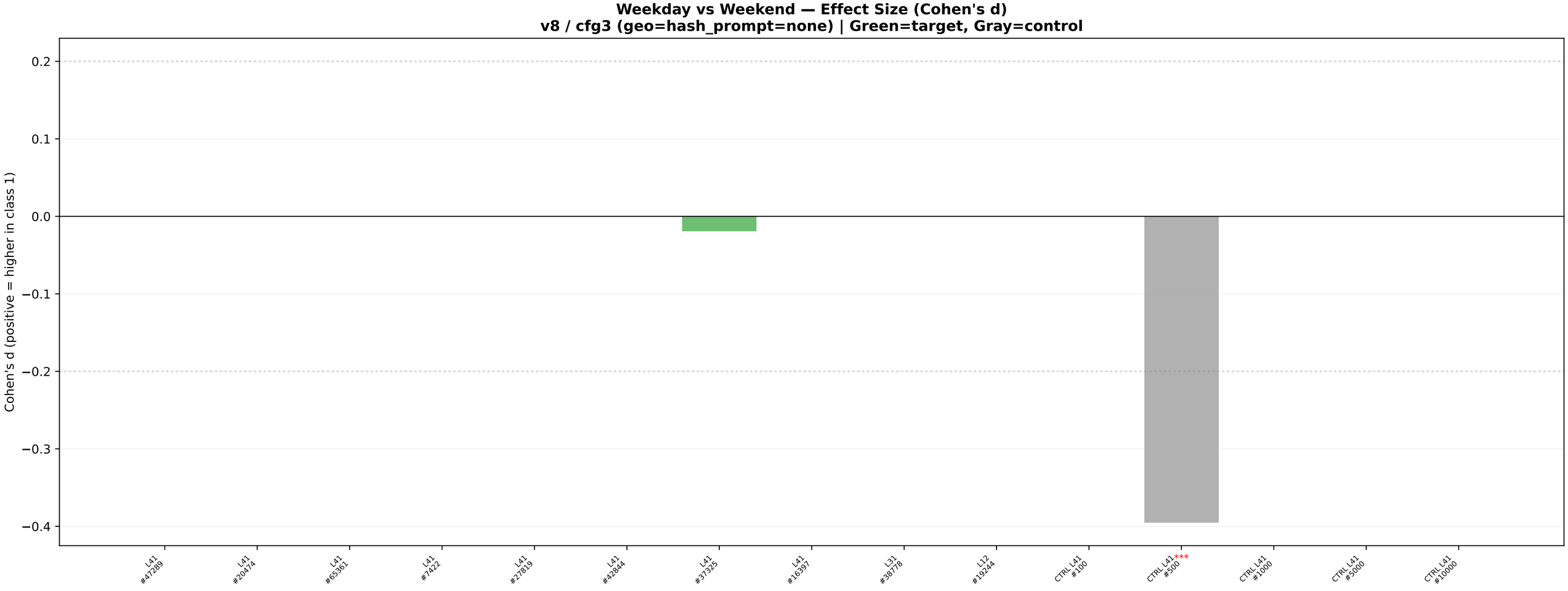
Significant target features: 1/10
Significant control features: 0/5 (should be ~0)

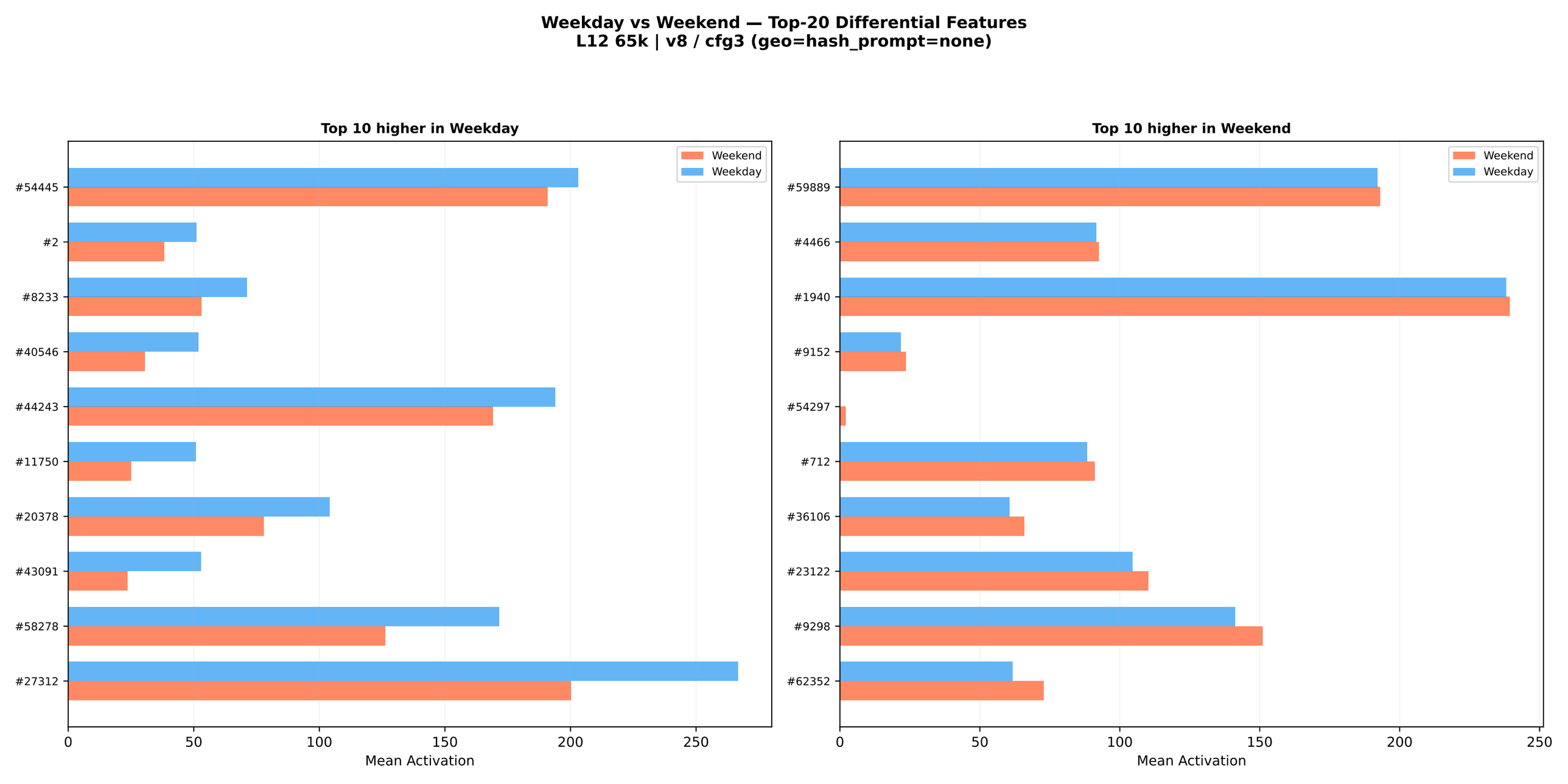
Weekday vs Weekend — SAE Feature Activations
v8 / cfg3 (geo=hash_prompt=none) | Target features (NeuronPedia)

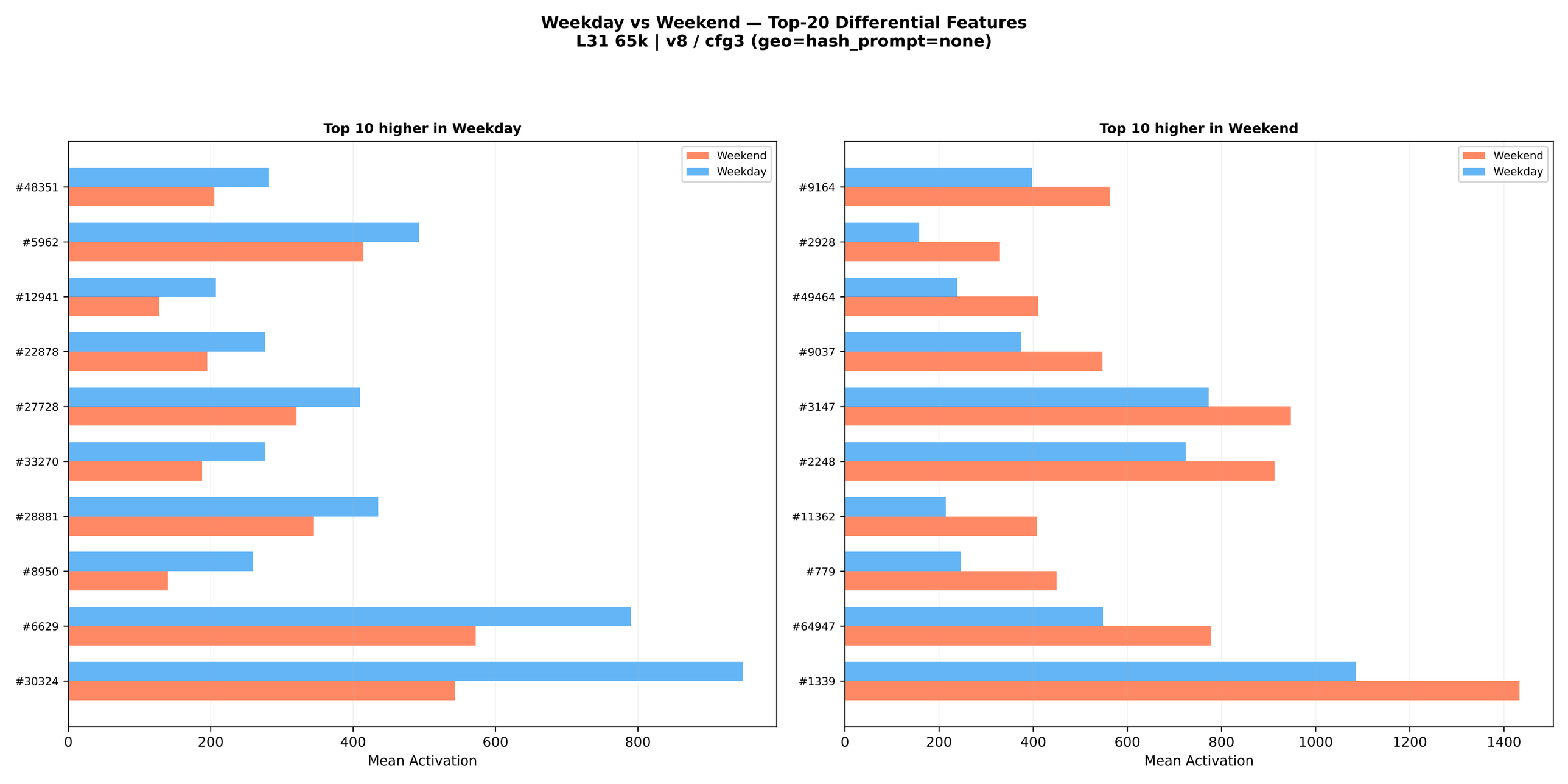


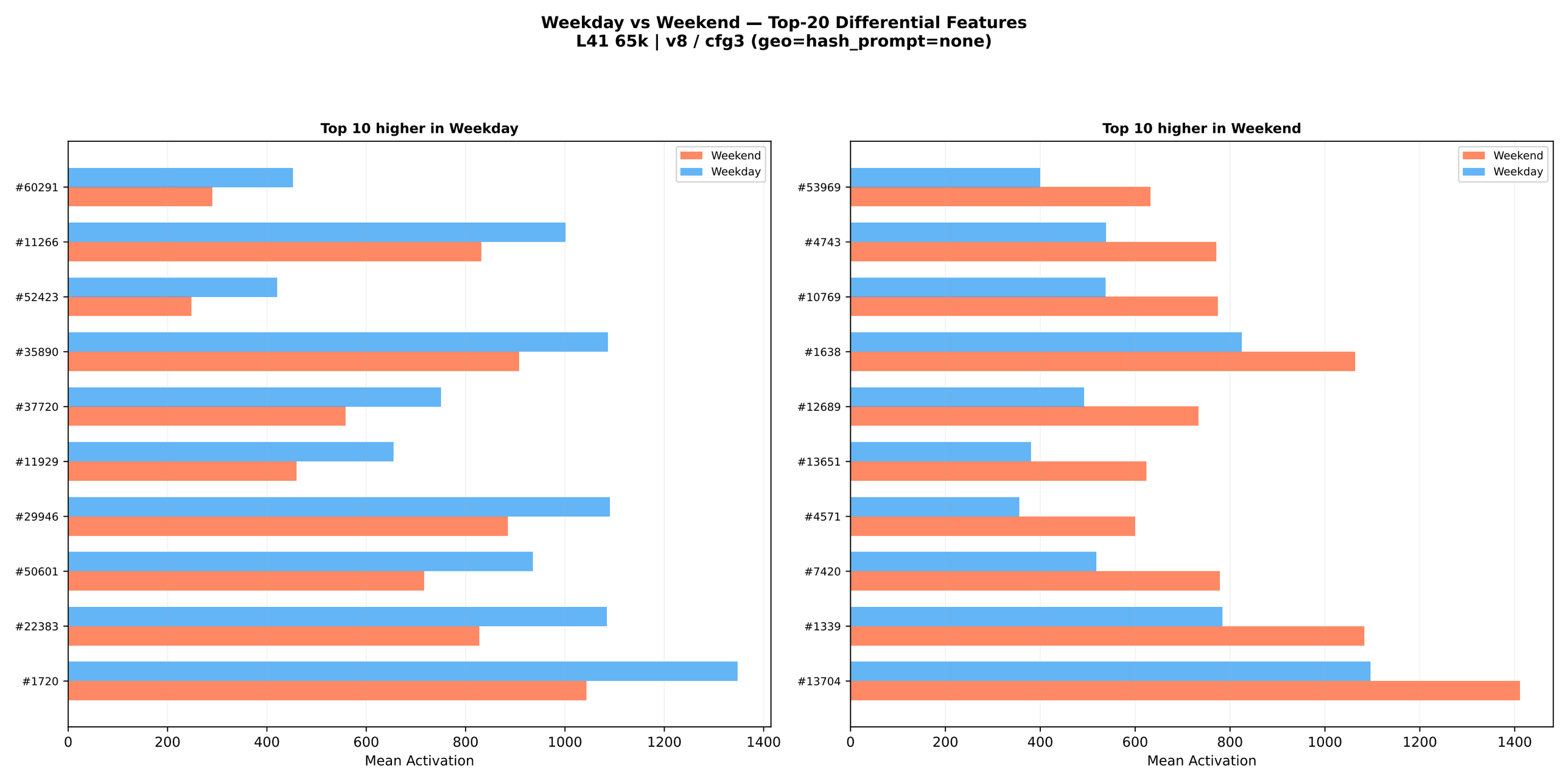

Weekday vs Weekend — Control Features (should be ns)
v8 / cfg3 (geo=hash_prompt=none)
CONTROL #100
p=nan ns
CONTROL #500
p=1.40e-09 ***
CONTROL #1000
p=nan ns
CONTROL #5000
p=nan ns
CONTROL #10000
p=nan ns
Weekend
Weekday


Weekday vs Weekend — Feature Activation Rate
v8 / cfg3 (geo=hash_prompt=none)
Fraction of samples with activation > 0
1.0
0.8
0.6
0.4
0.2
0.0
Weekend
Weekday
L41 65k #47289 week
L41 65k #20474 week
L41 65k #65361 holidays
L41 65k #7422 holidays
L41 65k #27819 Week
L41 65k #42844 holiday
L41 65k #37325 days of week
L41 65k #16397 days of week
L31 65k #38778 morning and wee
L12 65k #19244 holiday weekend


Weekday vs Weekend — Effect Size (Cohen's d)
v8 / cfg3 (geo=hash_prompt=none) | Green=target, Gray=control
Cohen's d (positive = higher in class 1)
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
L41 #47289
L41 #20474
L41 #65361
L41 #7422
L41 #27819
L41 #42844
L41 #37325
L41 #16397
L31 #38778
L12 #19244
CTRL L41 #100
CTRL L41 #500 ***
CTRL L41 #1000
CTRL L41 #5000
CTRL L41 #10000


Weekday vs Weekend — Top-20 Differential Features
L12 65k | v8 / cfg3 (geo=hash_prompt=none)
Top 10 higher in Weekday
Weekend
Weekday
#54445
#2
#8233
#40546
#44243
#11750
#20378
#43091
#58278
#27312
0
50
100
150
200
250
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#59889
#4466
#1940
#9152
#54297
#712
#36106
#23122
#9298
#62352
0
50
100
150
200
250
Mean Activation


Weekday vs Weekend — Top-20 Differential Features
L31 65k | v8 / cfg3 (geo=hash_prompt=none)
Top 10 higher in Weekday
Weekend
Weekday
#48351
#5962
#12941
#22878
#27728
#33270
#28881
#8950
#6629
#30324
0
200
400
600
800
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#9164
#2928
#49464
#9037
#3147
#2248
#11362
#779
#64947
#1339
0
200
400
600
800
1000
1200
1400
Mean Activation


Weekday vs Weekend — Top-20 Differential Features
L41 65k | v8 / cfg3 (geo=hash_prompt=none)
Top 10 higher in Weekday
Weekend
Weekday
#60291
#11266
#52423
#35890
#37720
#11929
#29946
#50601
#22383
#1720
0
200
400
600
800
1000
1200
1400
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#53969
#4743
#10769
#1638
#12689
#13651
#4571
#7420
#1339
#13704
0
200
400
600
800
1000
1200
1400
Mean Activation

Summary

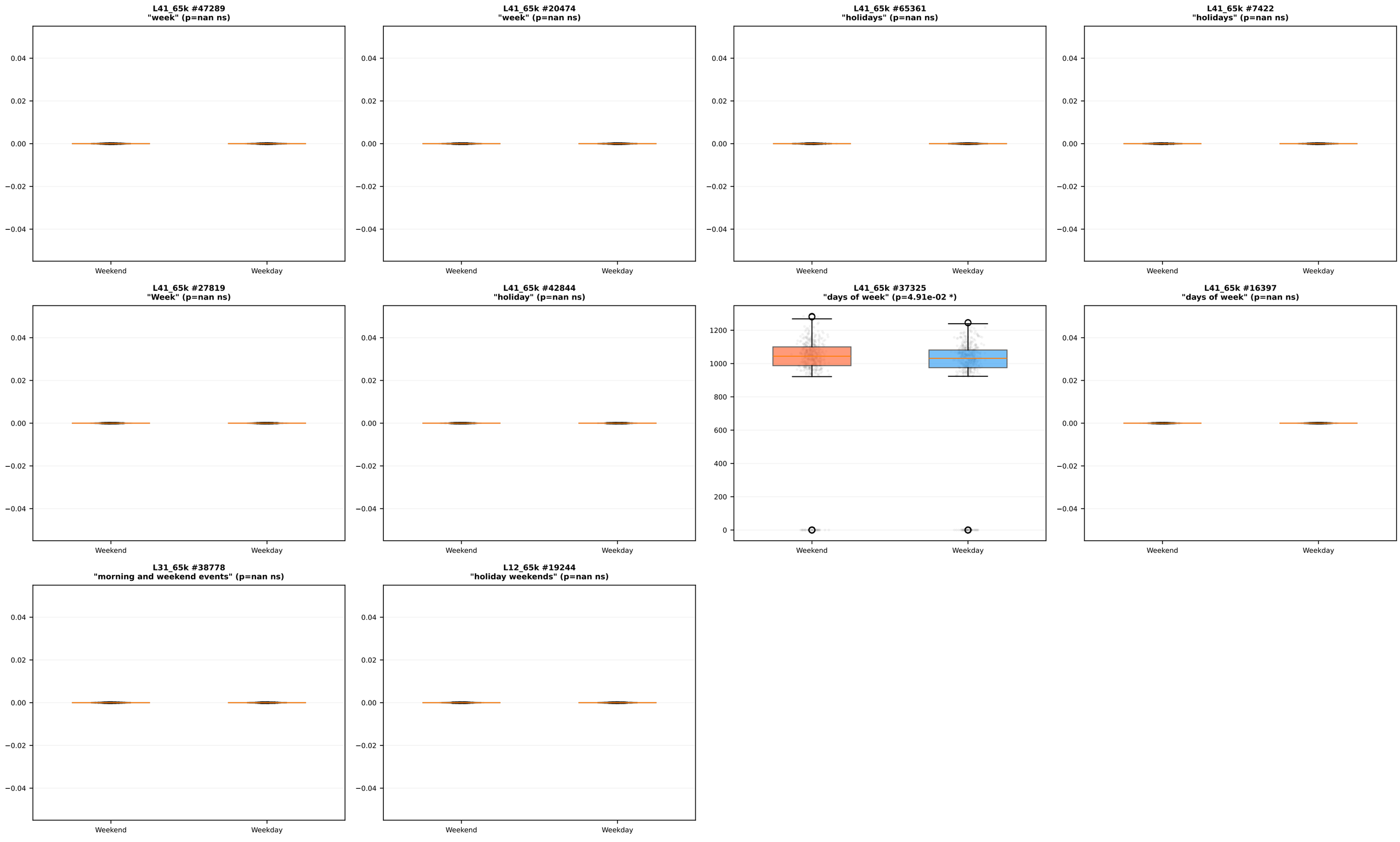
Weekday vs Weekend — SAE Feature Summary | v8 / cfg3 (geo=hash_prompt=none)

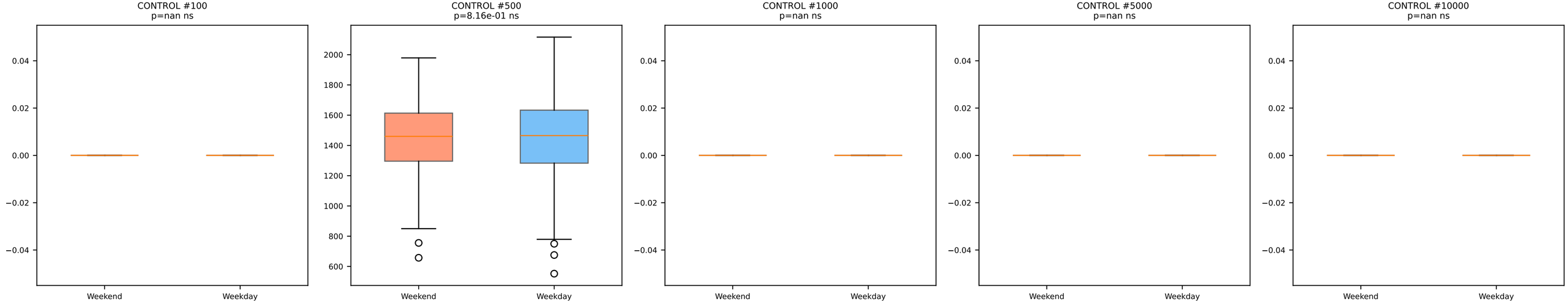
Layer	Width	Feat#	Label	mean_c0	mean_c1	d	p-val	sig	act%_c0	act%_c1
L 41	65k	47289	week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	20474	week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	65361	holidays	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	7422	holidays	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	27819	Week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	42844	holiday	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	37325	days of week	1090.4833	1084.5774	-0.019	7.70e-01	ns	93.8%	93.3%
L 41	65k	16397	days of week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	38778	morning and weekend events	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	65k	19244	holiday weekends	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	100	CONTROL_100	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	500	CONTROL_500	1310.6606	1151.5441	-0.395	1.40e-09	***	98.3%	96.9%
L 41	65k	1000	CONTROL_1000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	5000	CONTROL_5000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10000	CONTROL_10000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%

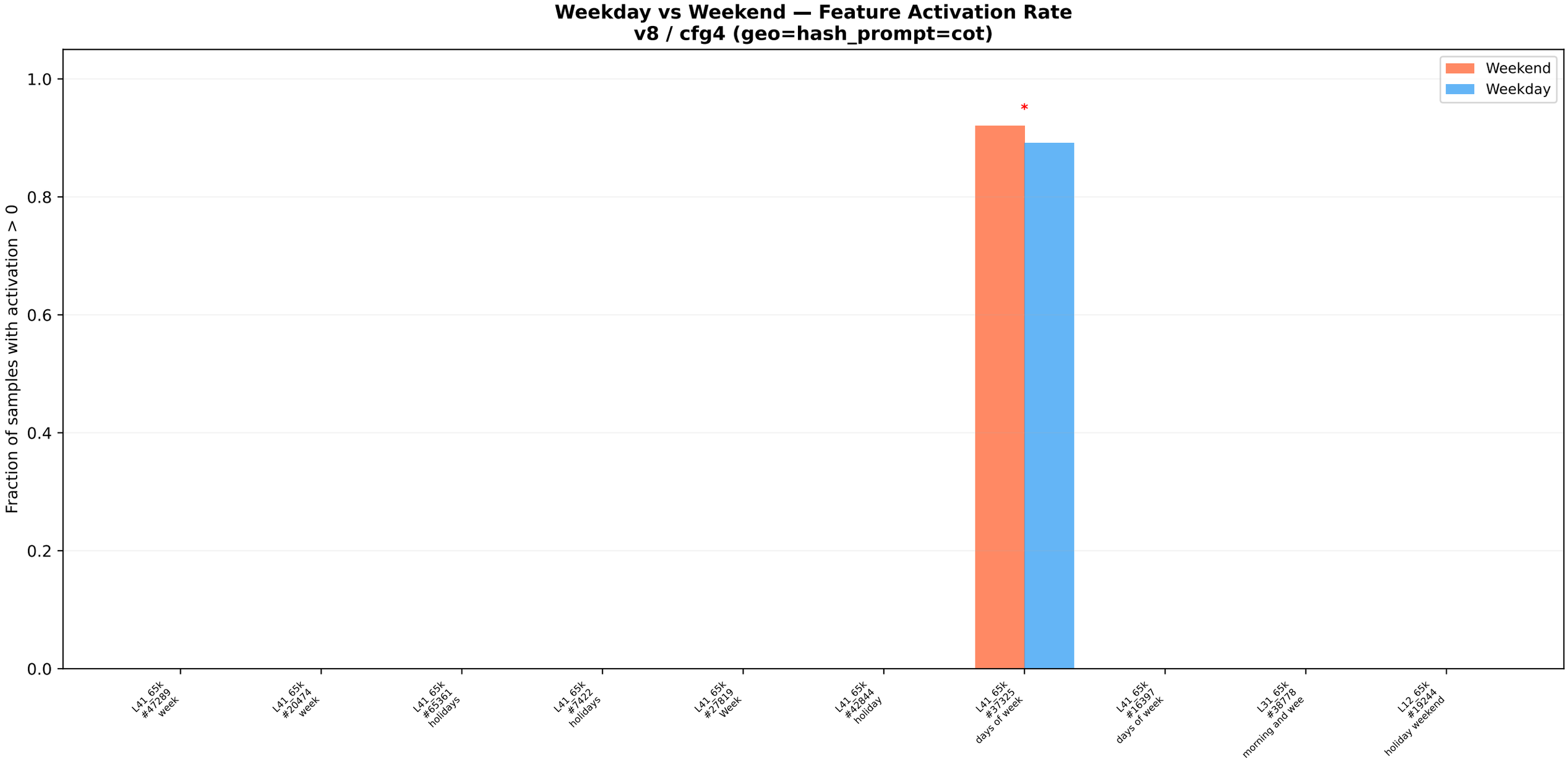
c0 = Weekend, c1 = Weekday
Cohen's d: |d|>0.2 small, |d|>0.5 medium, |d|>0.8 large
act% = fraction of samples where feature activation > 0

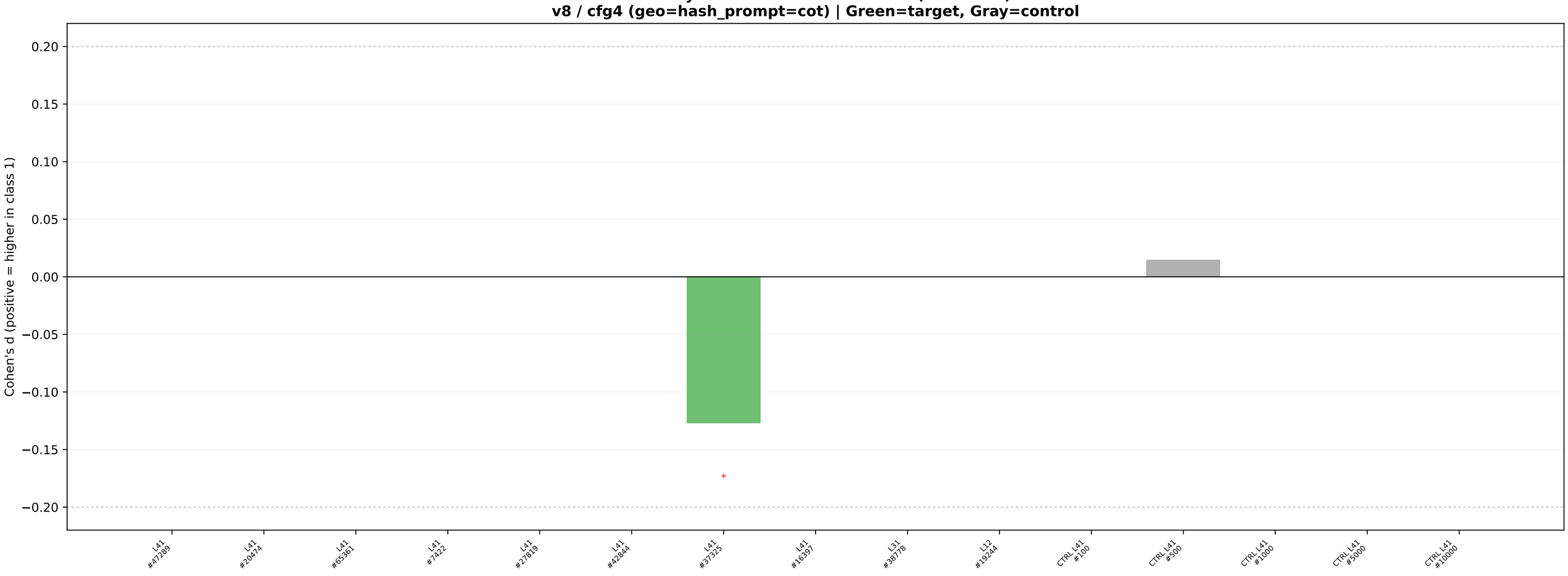
Significant target features: 0/10
Significant control features: 1/5 (should be ~0)

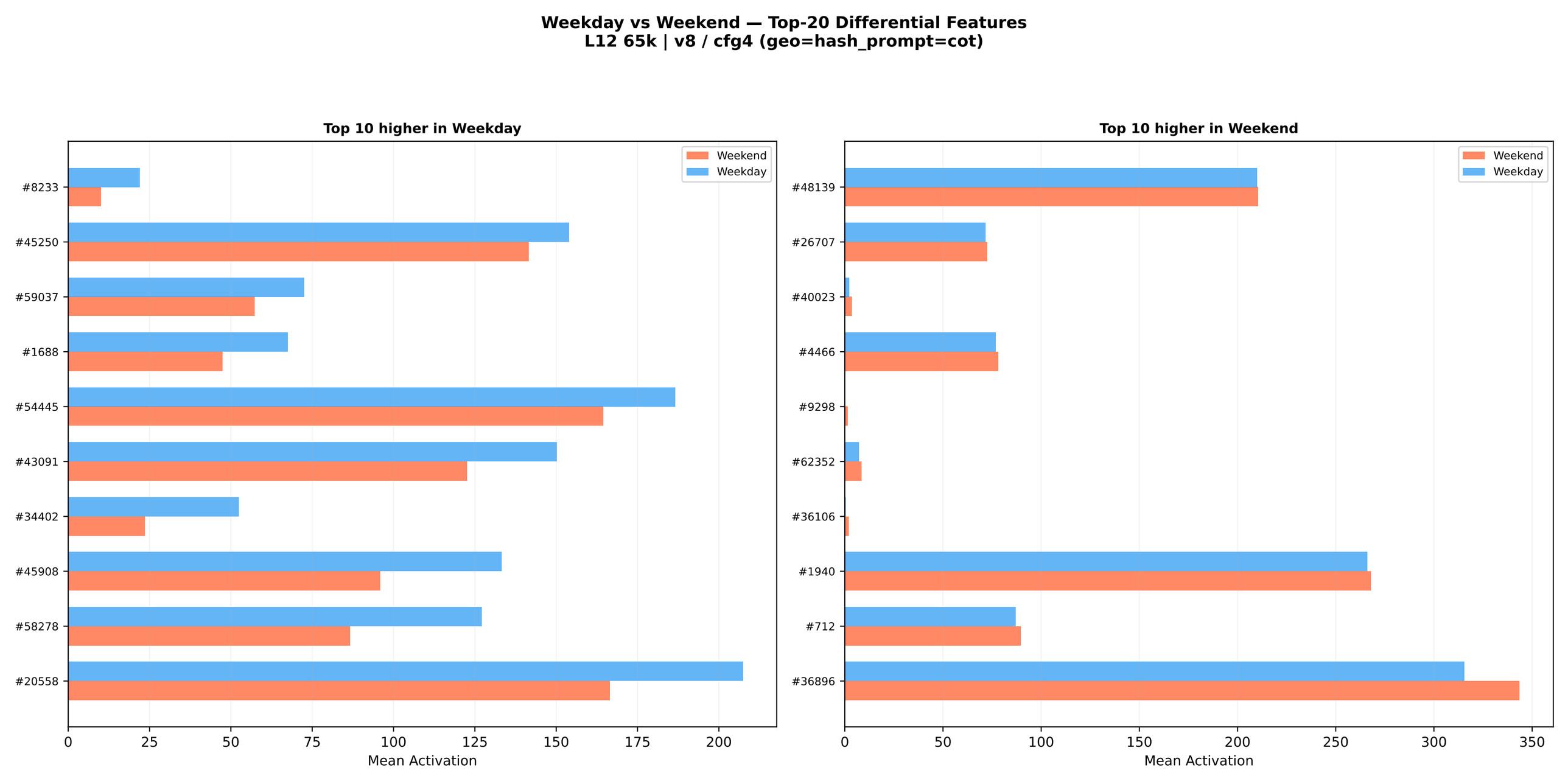
Weekday vs Weekend — SAE Feature Activations
v8 / cfg4 (geo=hash_prompt=cot) | Target features (NeuronPedia)

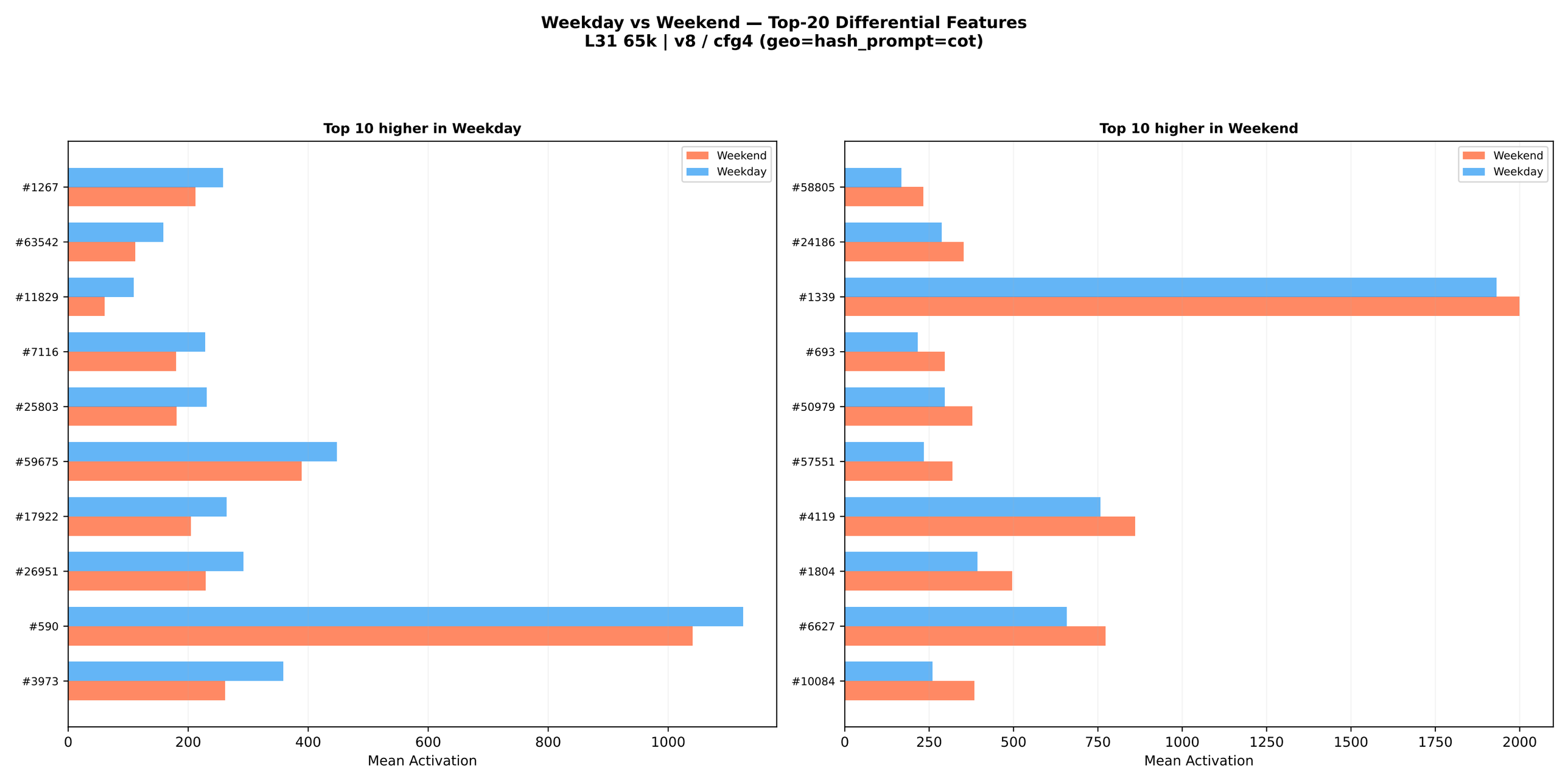


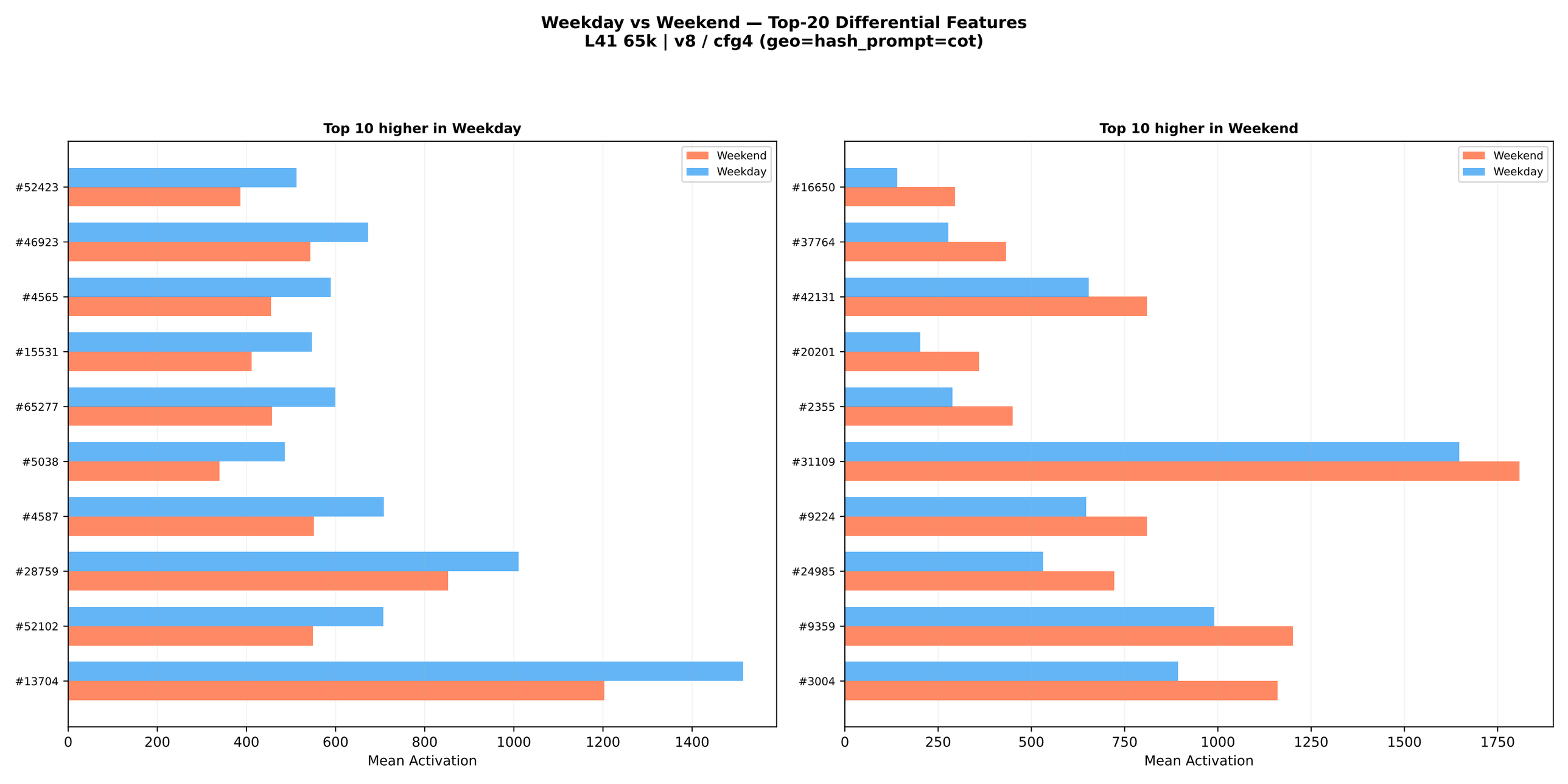

Weekday vs Weekend — Control Features (should be ns)
v8 / cfg4 (geo=hash_prompt=cot)
CONTROL #100
p=nan ns
CONTROL #500
p=8.16e-01 ns
CONTROL #1000
p=nan ns
CONTROL #5000
p=nan ns
CONTROL #10000
p=nan ns
Weekend
Weekday


Weekday vs Weekend — Feature Activation Rate
v8 / cfg4 (geo=hash_prompt=cot)
Fraction of samples with activation > 0
1.0
0.8
0.6
0.4
0.2
0.0
Weekend
Weekday
*
L41 65k #47289 week
L41 65k #20474 week
L41 65k #65361 holidays
L41 65k #7422 holidays
L41 65k #27819 Week
L41 65k #42844 holiday
L41 65k #37325 days of week
L41 65k #16397 days of week
L31 65k #38778 morning and wee
L12 65k #19244 holiday weekend


Weekday vs Weekend — Effect Size (Cohen's d)
v8 / cfg4 (geo=hash_prompt=cot) | Green=target, Gray=control
Cohen's d (positive = higher in class 1)
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
*
L41 #47289
L41 #20474
L41 #65361
L41 #7422
L41 #27819
L41 #42844
L41 #37325
L41 #16397
L31 #38778
L12 #19244
CTRL L41 #100
CTRL L41 #500
CTRL L41 #1000
CTRL L41 #5000
CTRL L41 #10000


Weekday vs Weekend — Top-20 Differential Features
L12 65k | v8 / cfg4 (geo=hash_prompt=cot)
Top 10 higher in Weekday
Weekend
Weekday
#8233
#45250
#59037
#1688
#54445
#43091
#34402
#45908
#58278
#20558
0
25
50
75
100
125
150
175
200
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#48139
#26707
#40023
#4466
#9298
#62352
#36106
#1940
#712
#36896
0
50
100
150
200
250
300
350
Mean Activation


Weekday vs Weekend — Top-20 Differential Features
L31 65k | v8 / cfg4 (geo=hash_prompt=cot)
Top 10 higher in Weekday
Weekend
Weekday
#1267
#63542
#11829
#7116
#25803
#59675
#17922
#26951
#590
#3973
0
200
400
600
800
1000
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#58805
#24186
#1339
#693
#50979
#57551
#4119
#1804
#6627
#10084
0
250
500
750
1000
1250
1500
1750
2000
Mean Activation


Weekday vs Weekend — Top-20 Differential Features
L41 65k | v8 / cfg4 (geo=hash_prompt=cot)
Top 10 higher in Weekday
Weekend
Weekday
#52423
#46923
#4565
#15531
#65277
#5038
#4587
#28759
#52102
#13704
0
200
400
600
800
1000
1200
1400
Mean Activation
Top 10 higher in Weekend
Weekend
Weekday
#16650
#37764
#42131
#20201
#2355
#31109
#9224
#24985
#9359
#3004
0
250
500
750
1000
1250
1500
1750
Mean Activation

Summary

Weekday vs Weekend — SAE Feature Summary | v8 / cfg4 (geo=hash_prompt=cot)

Layer	Width	Feat#	Label	mean_c0	mean_c1	d	p-val	sig	act%_c0	act%_c1
L 41	65k	47289	week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	20474	week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	65361	holidays	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	7422	holidays	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	27819	Week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	42844	holiday	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	37325	days of week	973.3784	933.5079	-0.127	4.91e-02	*	92.1%	89.2%
L 41	65k	16397	days of week	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 31	65k	38778	morning and weekend events	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 12	65k	19244	holiday weekends	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	100	CONTROL_100	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	500	CONTROL_500	1443.0209	1446.5419	+0.015	8.16e-01	ns	100.0%	100.0%
L 41	65k	1000	CONTROL_1000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	5000	CONTROL_5000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%
L 41	65k	10000	CONTROL_10000	0.0000	0.0000	+0.000	nan	ns	0.0%	0.0%

c0 = Weekend, c1 = Weekday
Cohen's d: |d|>0.2 small, |d|>0.5 medium, |d|>0.8 large
act% = fraction of samples where feature activation > 0

Significant target features: 1/10
Significant control features: 0/5 (should be ~0)